Mediastinum

Dr.m.almusaibli

Mediastinum

***Definition:-** It is a thick mass of tissues occupying the median part of the thoracic cavity forming a median septum between the 2 pleural cavities, extending from the sternum anteriorly to the vertebral column posteriorly.

***Structure:** it includes a large number of thoracic structures:

(a) the heart & the large vessels of the thorax.

(b) the tubes entering the thorax (trachea & esophagus).

(c) the nerves, lymph nodes & lymph vessels of thorax.

*Boundaries:

-**Anteriorly:** sternum.

-**Posteriorly:** vertebral column.

-**Superiorly**: the plane of thoracic inlet.

-**Inferiorly**: the diaphragm.

-**On each side:** the lung & mediastinal pleura.

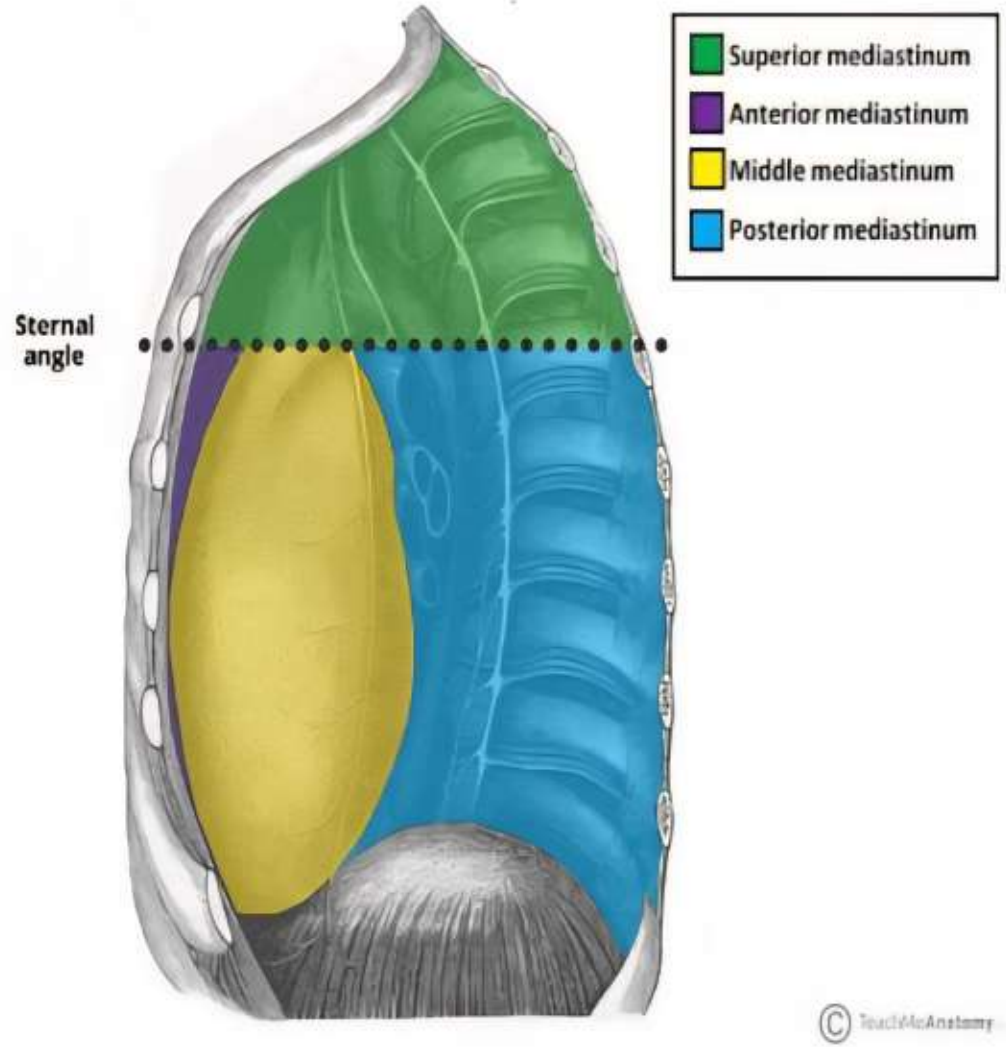
*Divisions:

-For descriptive purposes, the mediastinum is divided by an imaginary plane extending from the sternal angle (anteriorly) to the lower border of the body of T4 vertebra (posteriorly) into 2 parts:

(a) Superior mediastinum: above the imaginary plane.

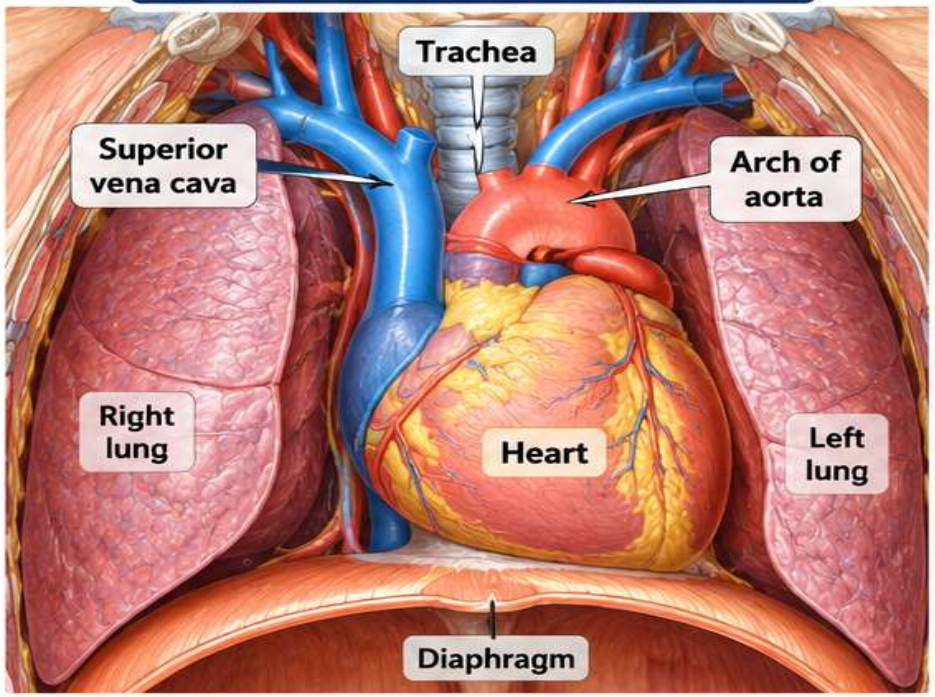
(B) Inferior mediastinum: below the imaginary plane.

The inf. mediastinum is further subdivided into 3 parts: (anterior, middle & posterior)

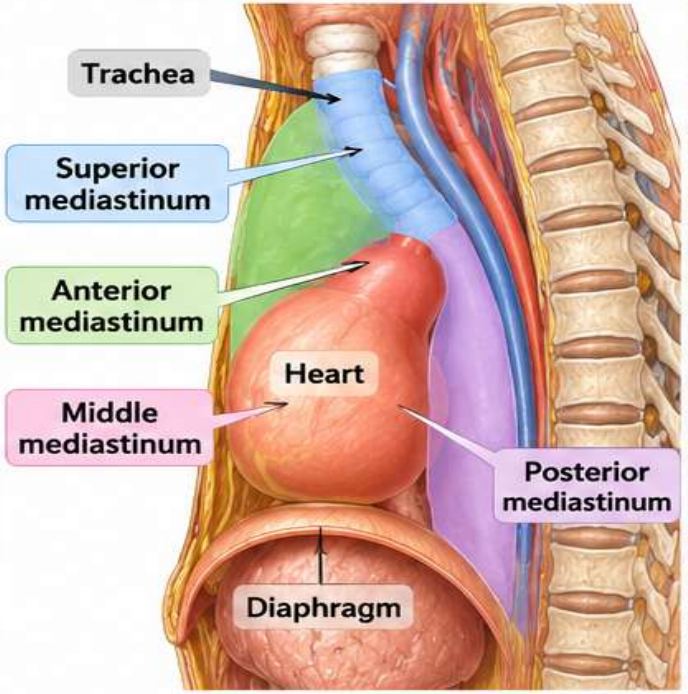


*The 4 subdivision of mediastinum:

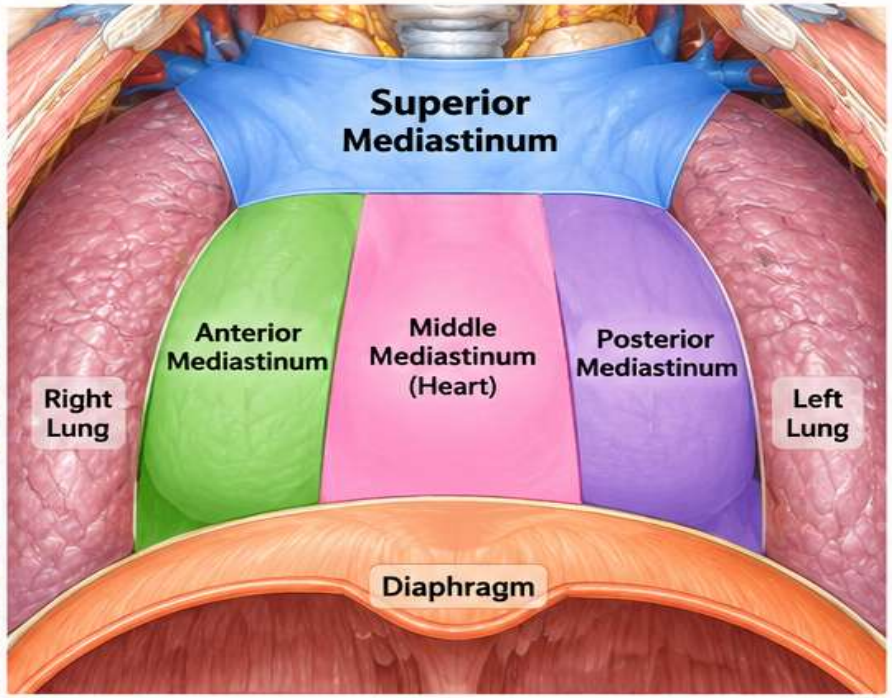
The Mediastinum



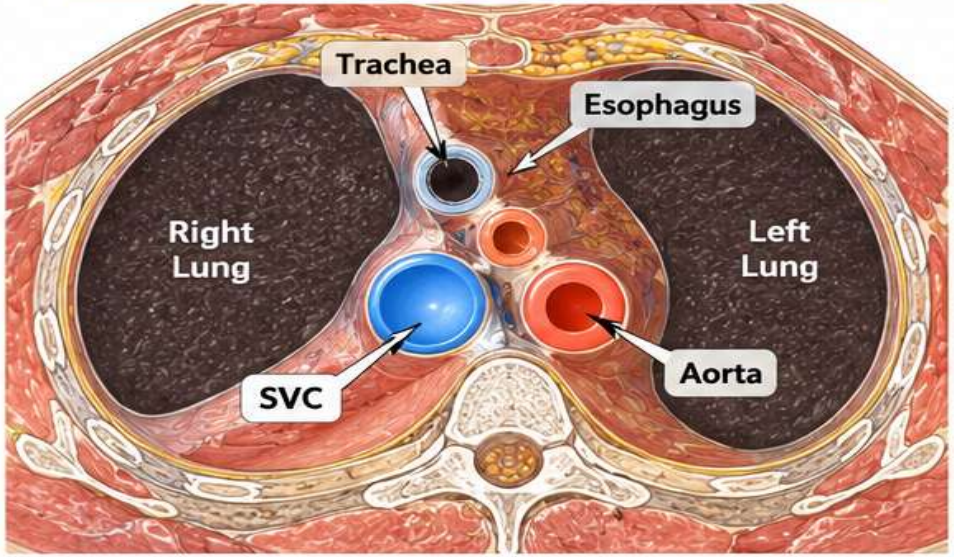
Sagittal View



Compartments of the Mediastinum



Transverse Section (at T4 level)



Boundaries of the Mediastinum

Superior Boundary:
- Thoracic inlet (T1 vertebra, 1st ribs, superior border of manubrium)

Inferior Boundary:
- Diaphragm

Anterior Boundary:
- Sternum and costal cartilages

Posterior Boundary:
- Vertebral bodies (T5 – T12)

Lateral Boundaries:
- Pleura

Contents of Each Compartment

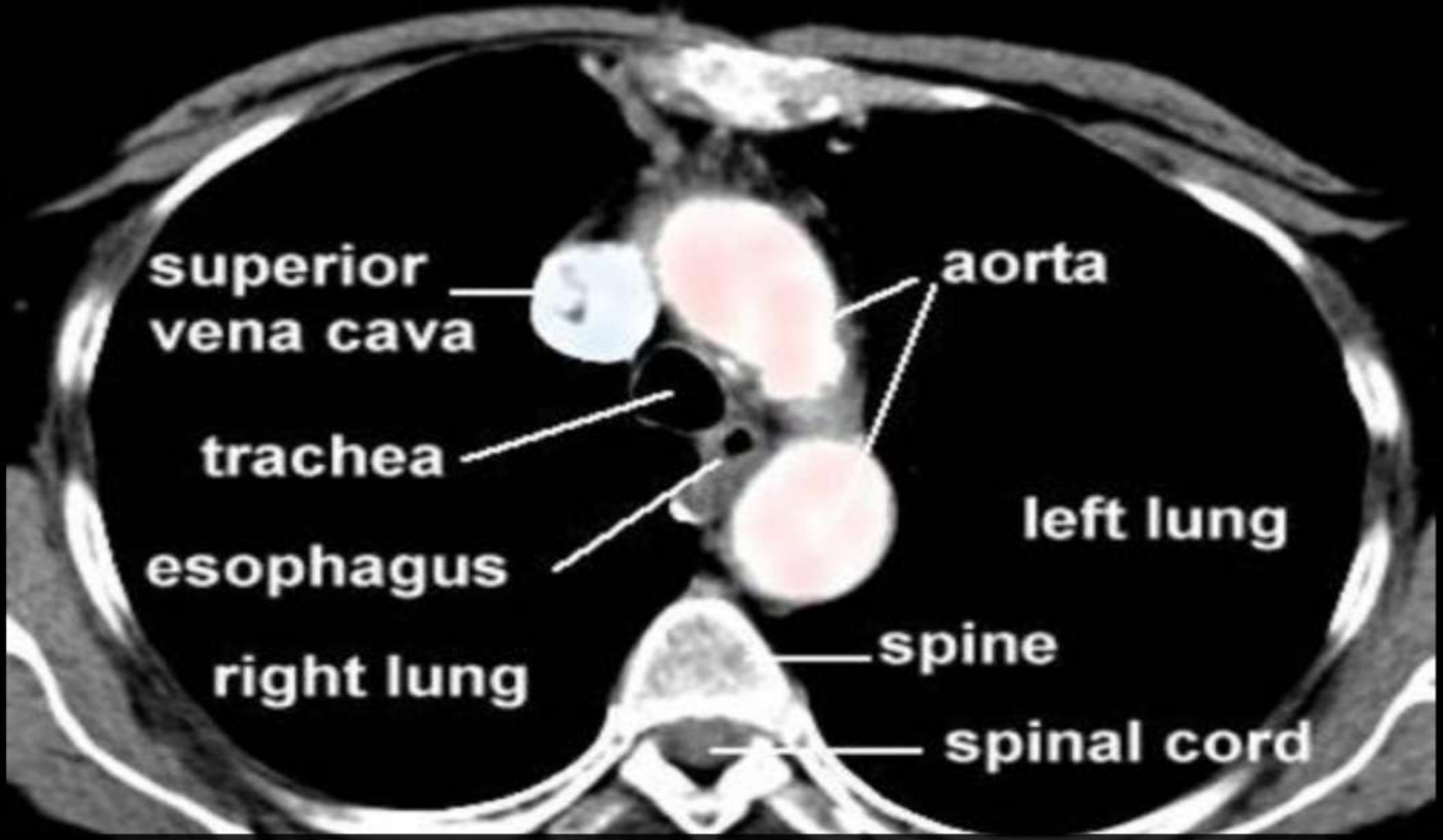
Superior Mediastinum
- Thymus
- Trachea
- Esophagus
- Arterial branches (brachiocephalic aa., left common carotid, left subclavian aa.)
- Veins (SVC, brachiocephalic veins)
- Thoracic duct
- Nerves (vagus, phrenic)

Anterior Mediastinum
- Fat and lymph nodes
- Remnants of thymus

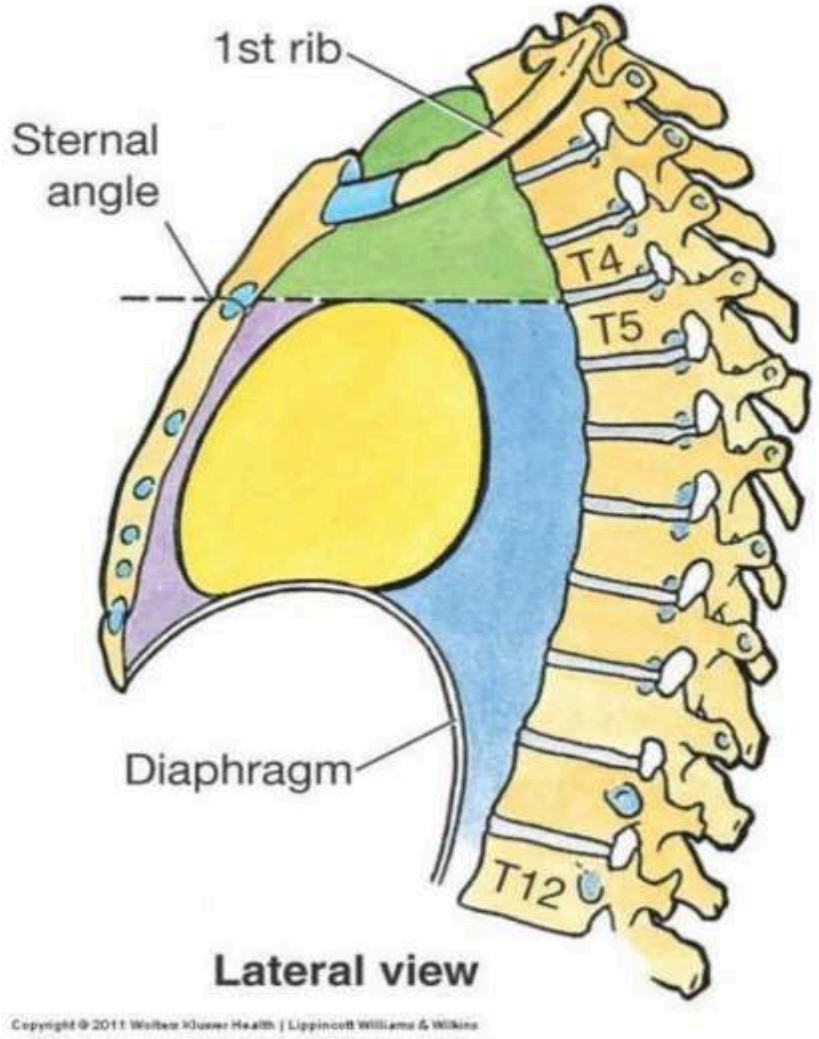
Middle Mediastinum
- Heart
- Ascending aorta
- Pulmonary trunk
- Pulmonary veins
- Pericardium

Posterior Mediastinum
- Descending aorta
- Esophagus
- Thoracic duct
- Azygos system
- Sympathetic trunks

Normal CT Mediastinum



الثلاثاء، ٣ نوفمبر



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Mediastinum

- **Superior**
- **Inferior:**
  - Anterior
  - Middle
  - Posterior

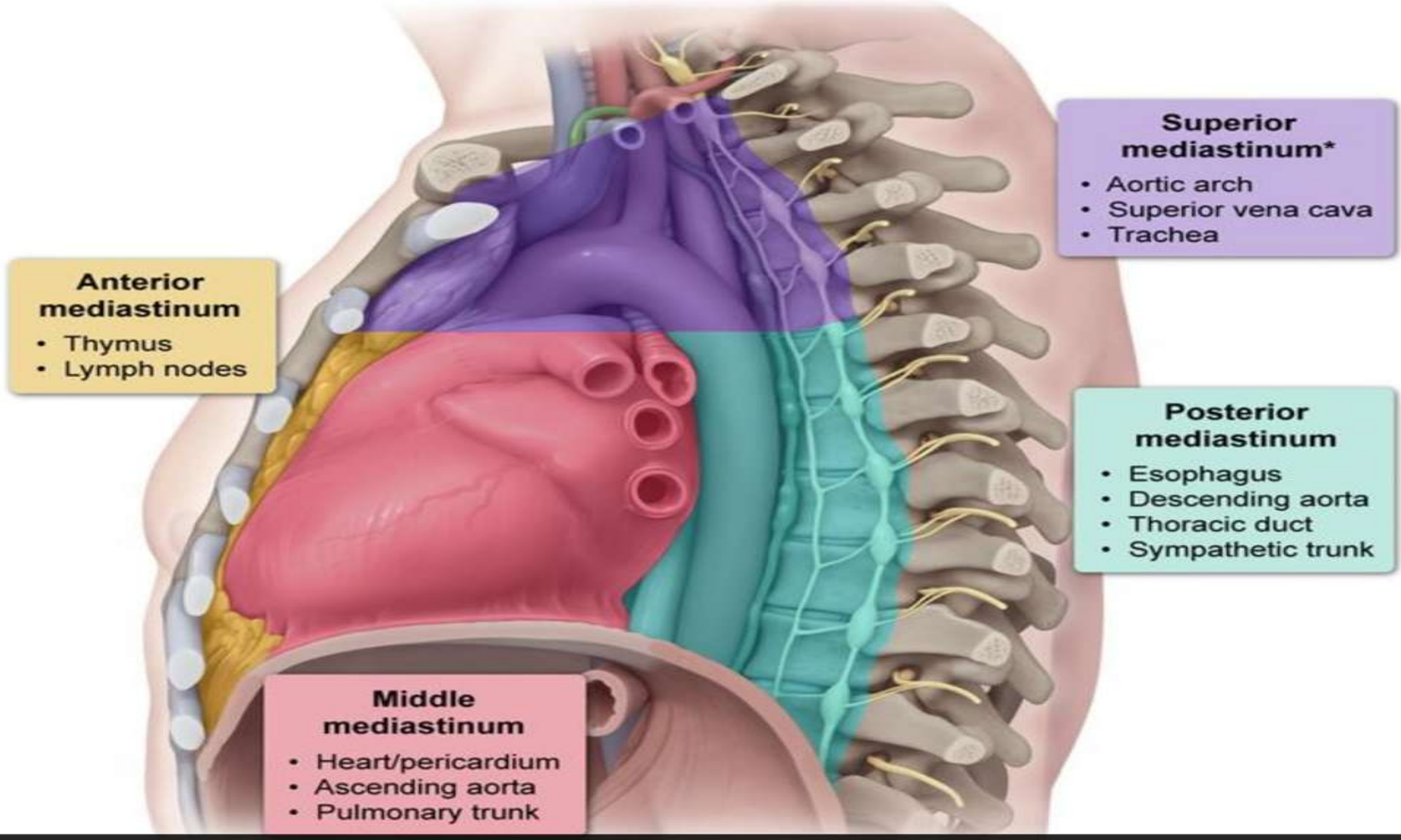
Superior : trachea, thymus, brachiocephalic vein, aortic arch, esophagus thoracic duct

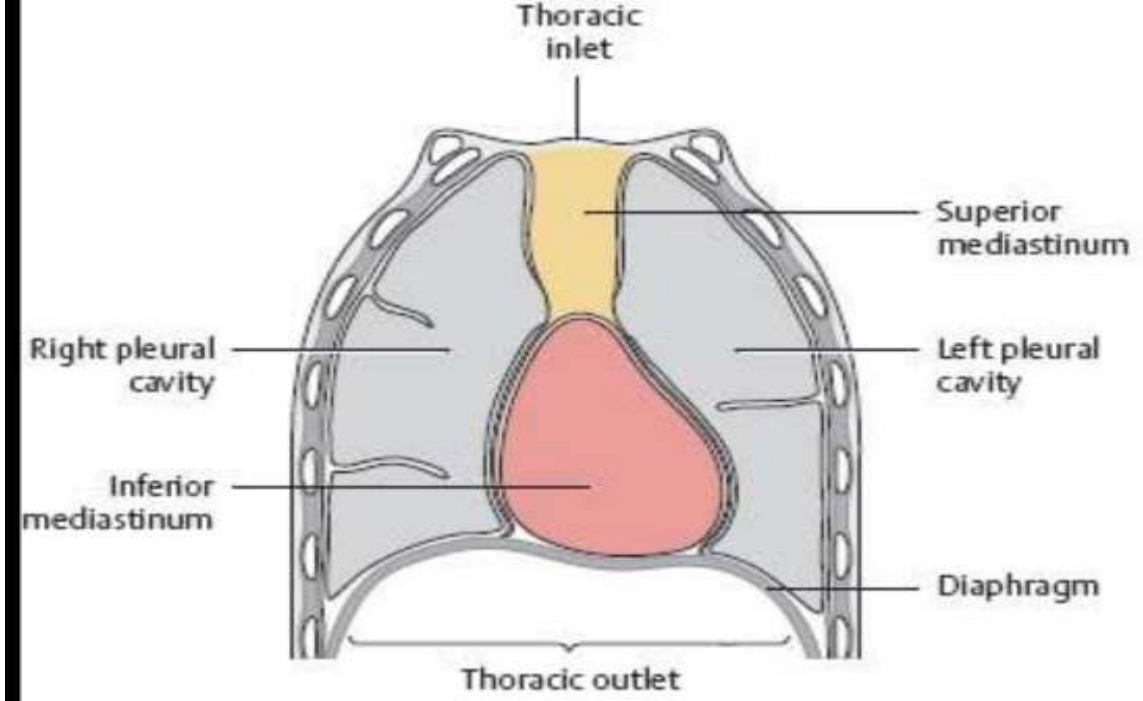
Middle: heart, ascending aorta, pulmonary trunk and veins, phrenic nerves

Posterior: esopahagus vagus nerves, descending aorta, thoracic duct, sympathetic trunks

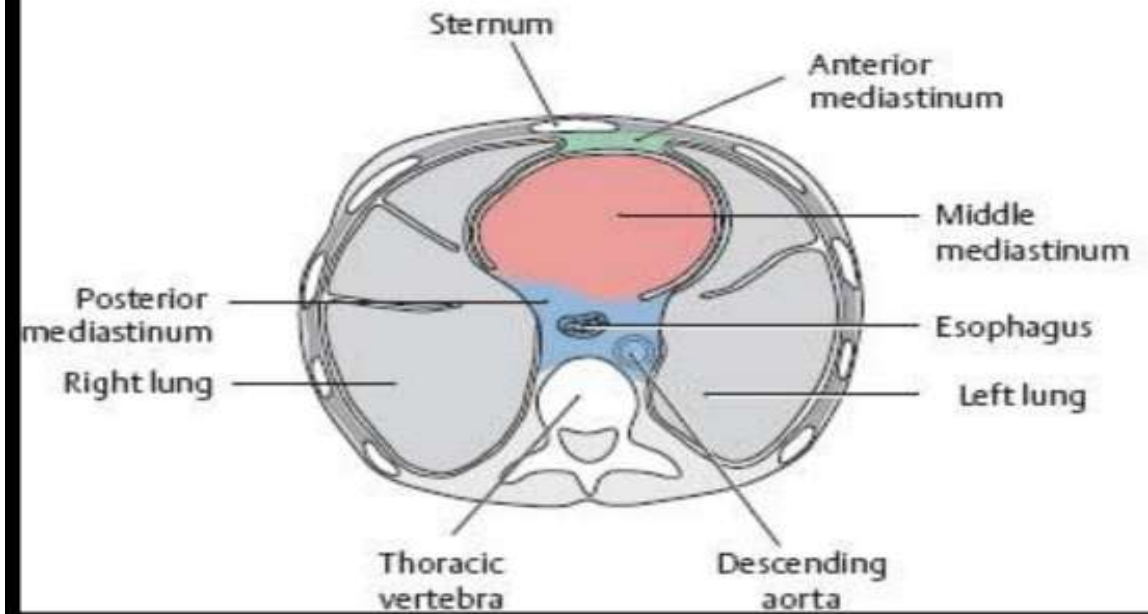
Anterior: fat, connective tissue, *thymus in child*

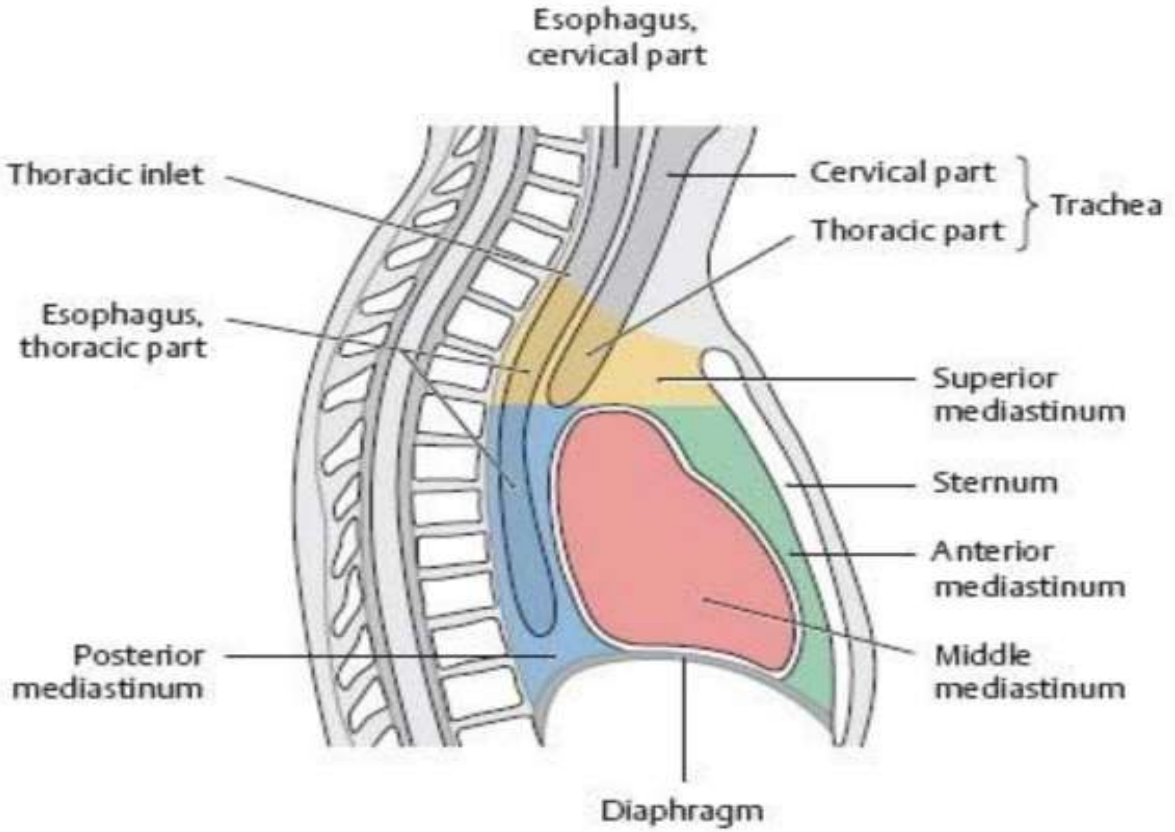
Mediastinal compartments



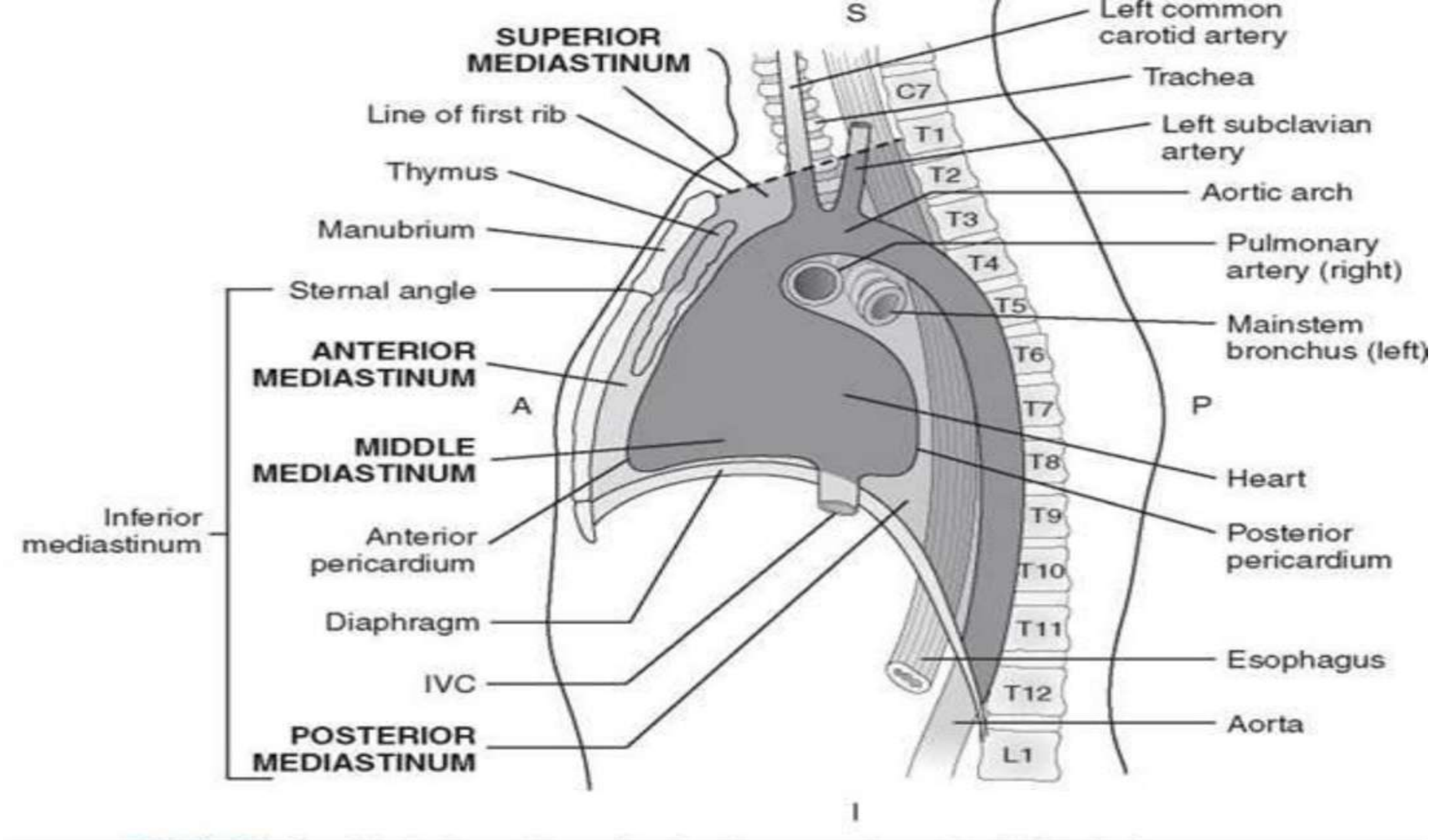


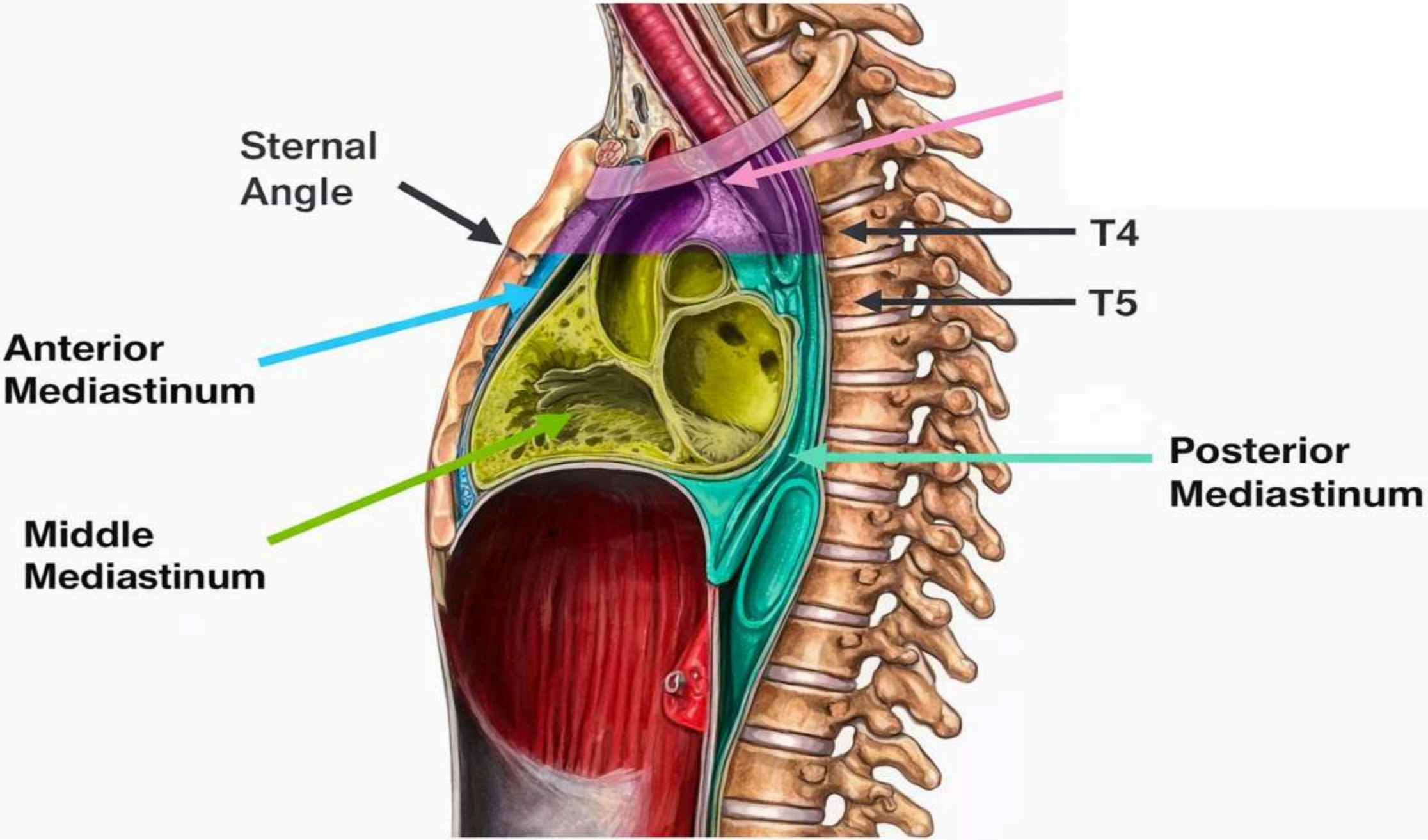
A Anterior view (coronal section).





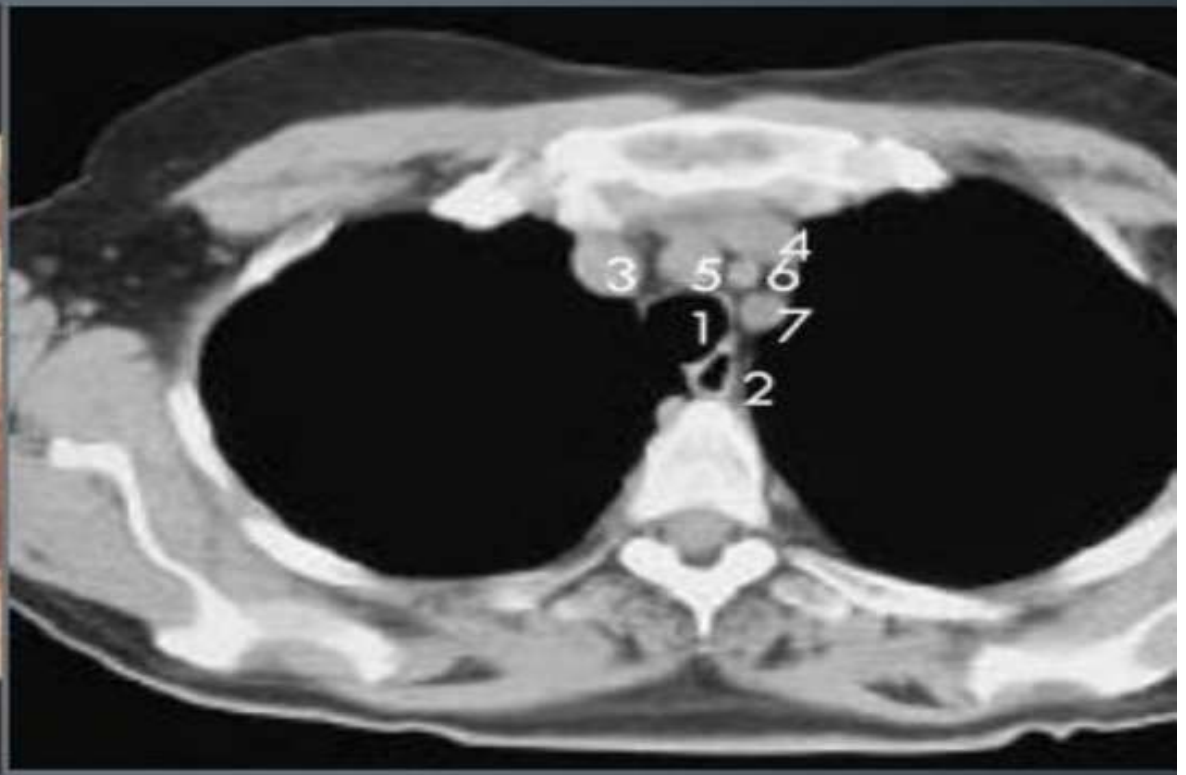
B Lateral view (midsagittal section).


S
SUPERIOR MEDIASTINUM
Line of first rib
Thymus
Manubrium
Sternal angle
ANTERIOR MEDIASTINUM
MIDDLE MEDIASTINUM
Inferior mediastinum
Anterior pericardium
Diaphragm
IVC
POSTERIOR MEDIASTINUM
A
P
I
C7
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
L1
Left common carotid artery
Trachea
Left subclavian artery
Aortic arch
Pulmonary artery (right)
Mainstem bronchus (left)
Heart
Posterior pericardium
Esophagus
Aorta

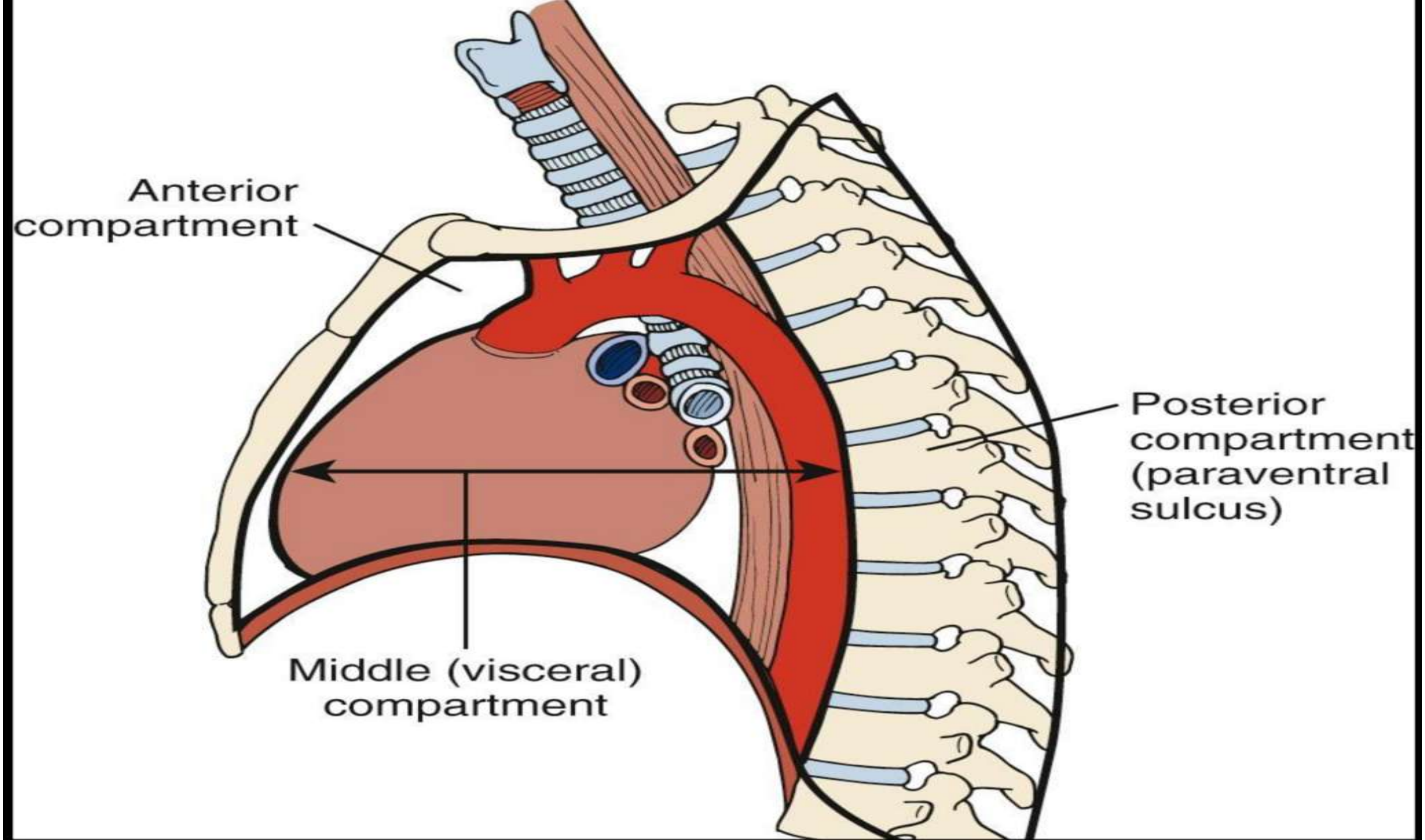

Sternal
Angle
T4
T5
Anterior
Mediastinum
Posterior
Mediastinum
Middle
Mediastinum

Cross sectional Anatomy on CT
T3 level

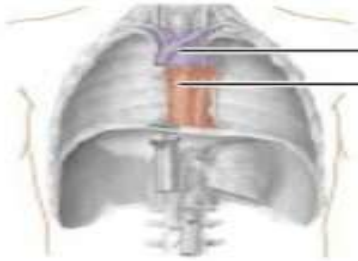




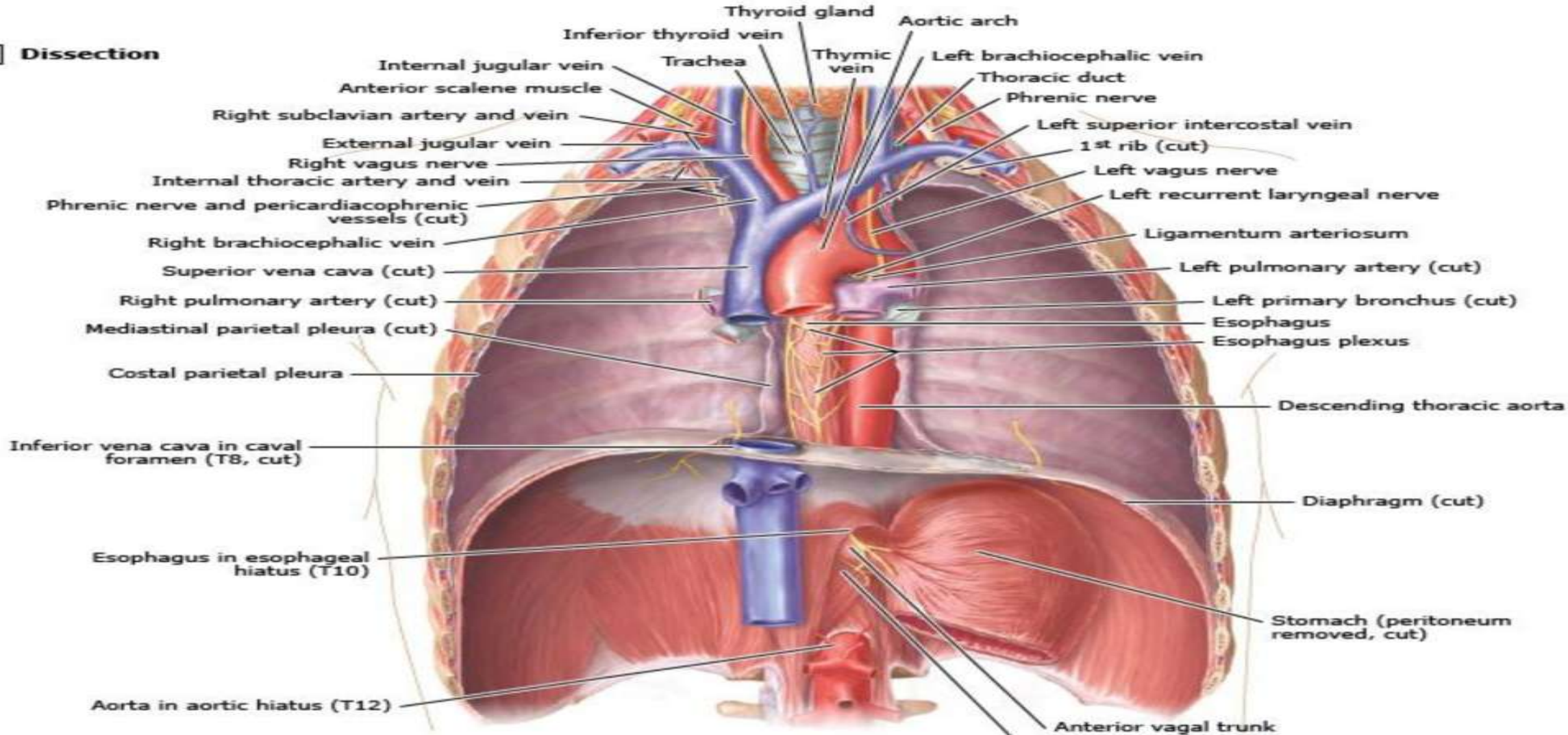
1-trachea
2-oesophagus
3-right braciocephalic ve
4-left brachiocephalic vein
5-right brachiocephalic artery
6-left common carotid artery
7-left subclavian artery


Anterior
compartment
Posterior
compartment
(paraventral
sulcus)
Middle (visceral)
compartment

A **Parts of the mediastinum**



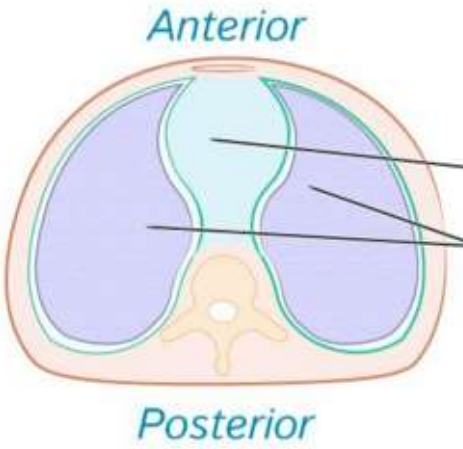
B **Dissection**



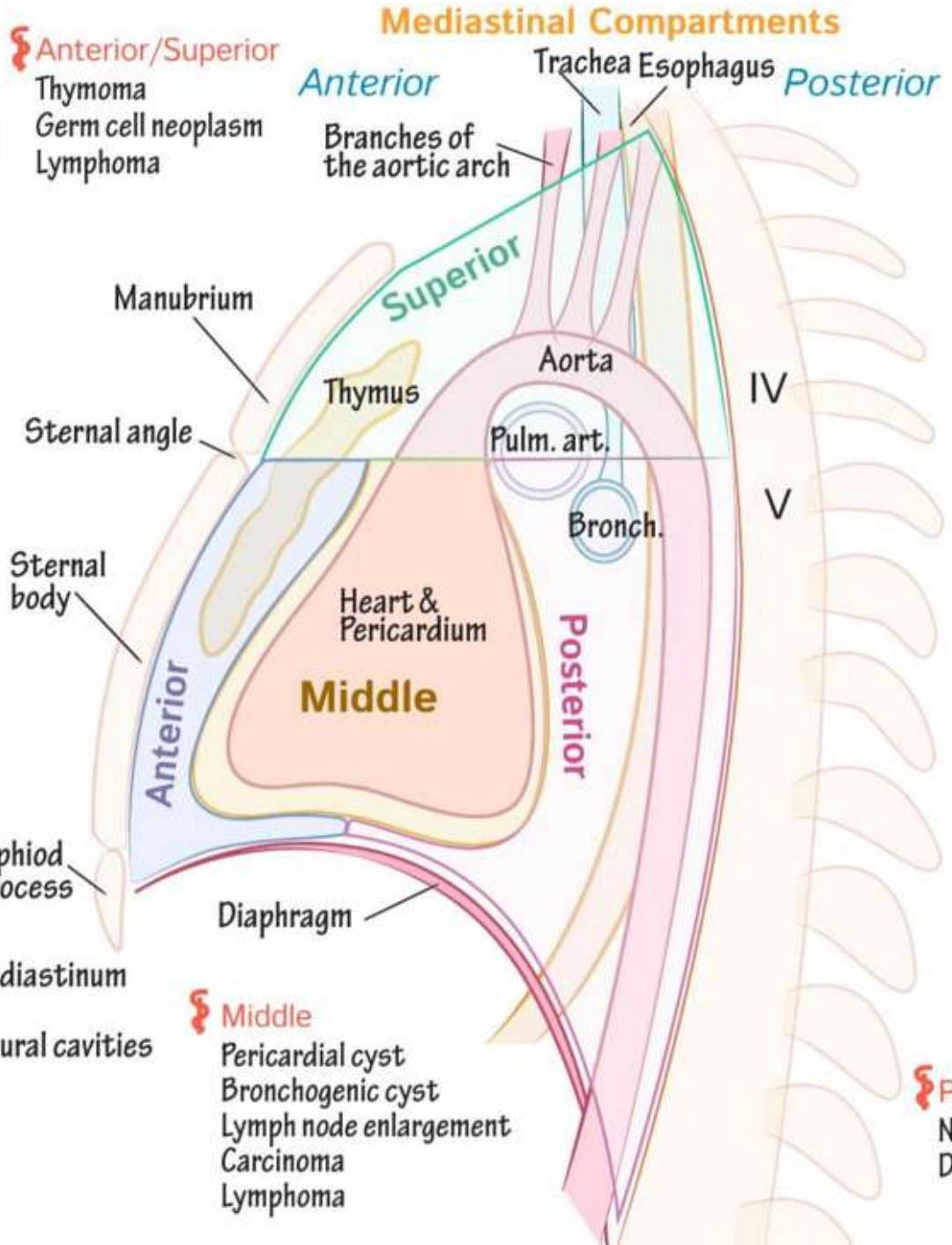
MEDIASTINUM

Definition & Regions

- ✓ *Location* —
  - ✓ Area between R. and L. pleural cavities.
- ✓ *Superior* —
  - ✓ Structures pass from head/neck to thorax.
- ✓ *Anterior* —
  - ✓ Connective/fat tissue that protects deeper structures.
- ✓ *Middle* —
  - ✓ Heart & Great Vessels
- ✓ *Posterior* —
  - ✓ Structures pass through thorax.



Mediastinal Compartments



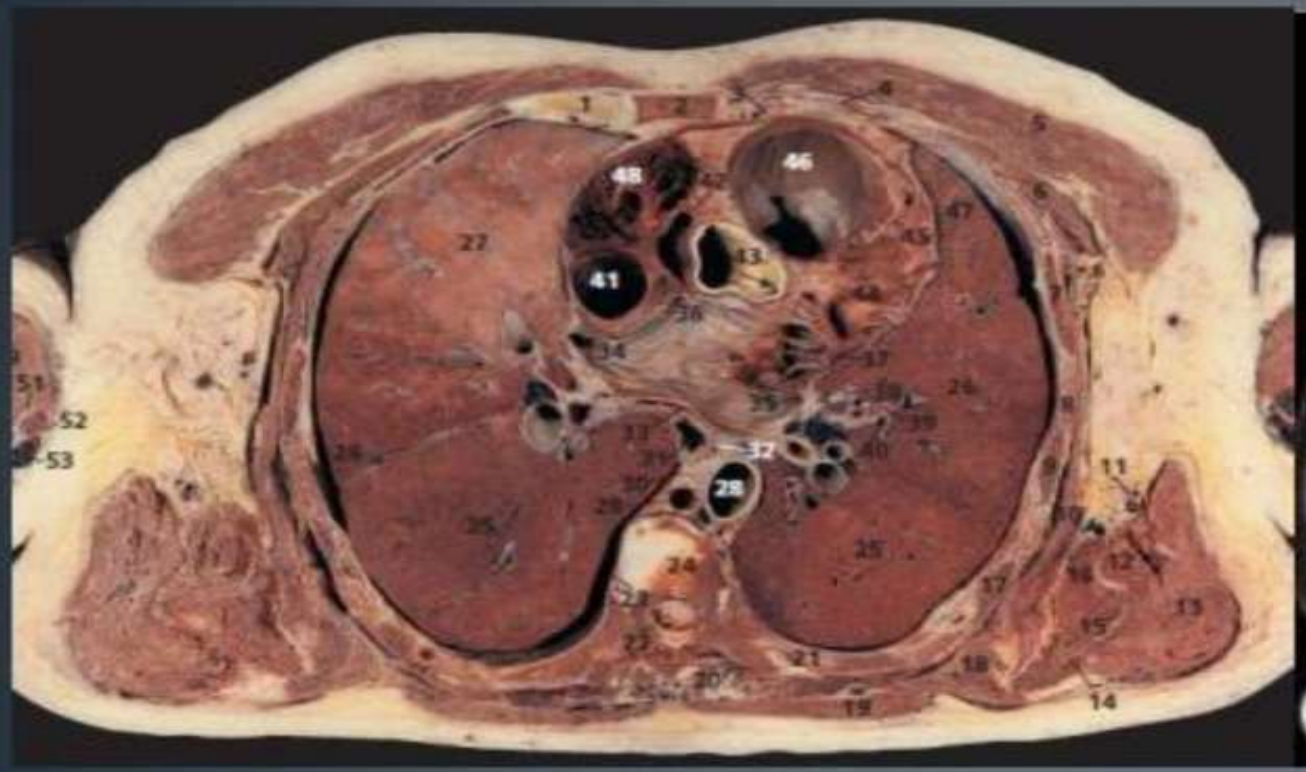
Anterior/Superior
- Thymoma
- Germ cell neoplasm
- Lymphoma

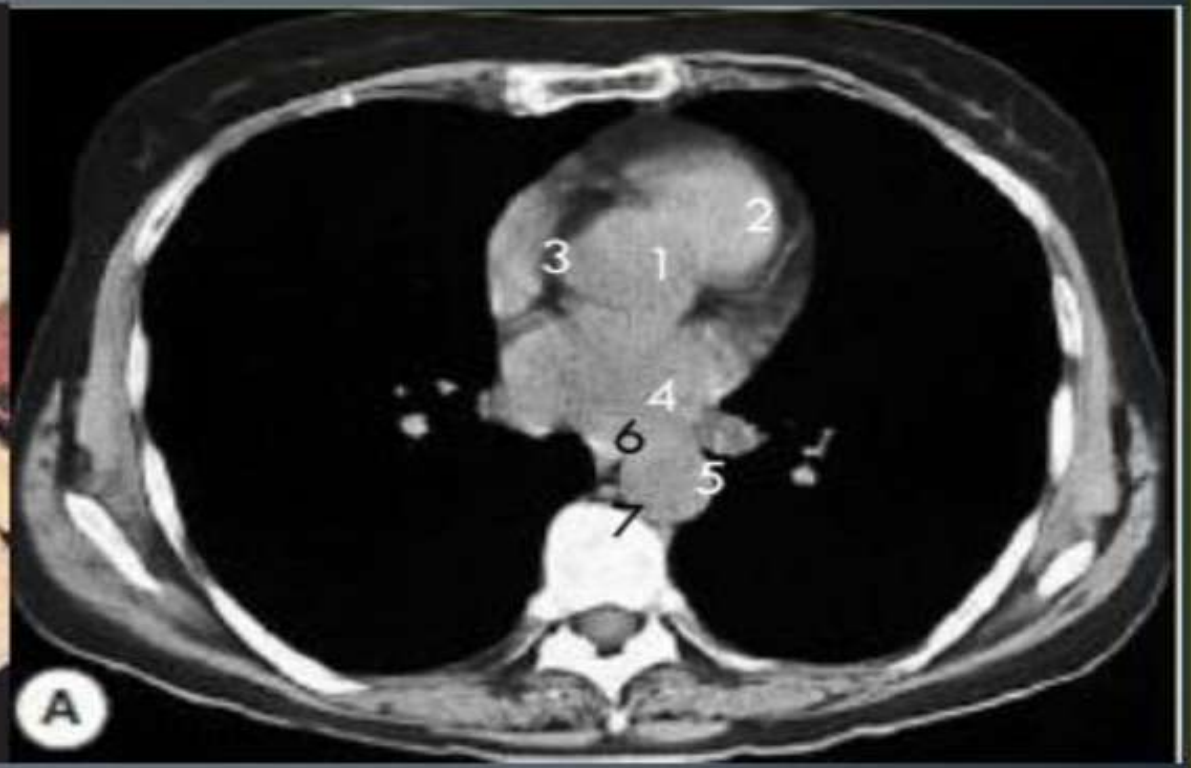
Middle
- Pericardial cyst
- Bronchogenic cyst
- Lymph node enlargement
- Carcinoma
- Lymphoma

Posterior
- Neurogenic tumor
- Diaphgram hernia

Compartment	Contents
Superior	Thymus (esp. children), Trachea, Esophagus
	Aortic Arch and Branches
	Superior Vena Cava, Brachiocephalic vv., Azygos Arch.
	Thoracic duct
	Vagus nn., Recurrent Laryngeal nn., Phrenic nn., Cardiac n.
Anterior	Thymus (esp. children)
	Branches of Internal Thoracic aa.
	Branches of Internal Thoracic vv.
	Parasternal Lymph Nodes
Middle	Heart & Great Vessel Roots
	Ascending Aorta, Pulmonary Trunk, Pericardiacophrenic aa.
	Superior Vena Cava, Pulmonary vv., Pericardiacophrenic vv.
	Vagus nn., Phrenic nn., Sympa. nn.
Posterior	Esophagus
	Thoracic Aorta
	Azygos v., Hemiazygos v., Acces. Hemiazogos vv.
	Thoracic duct
	Vagus nn., Sympa. chains

Cross sectional Anatomy on CT
T6 level

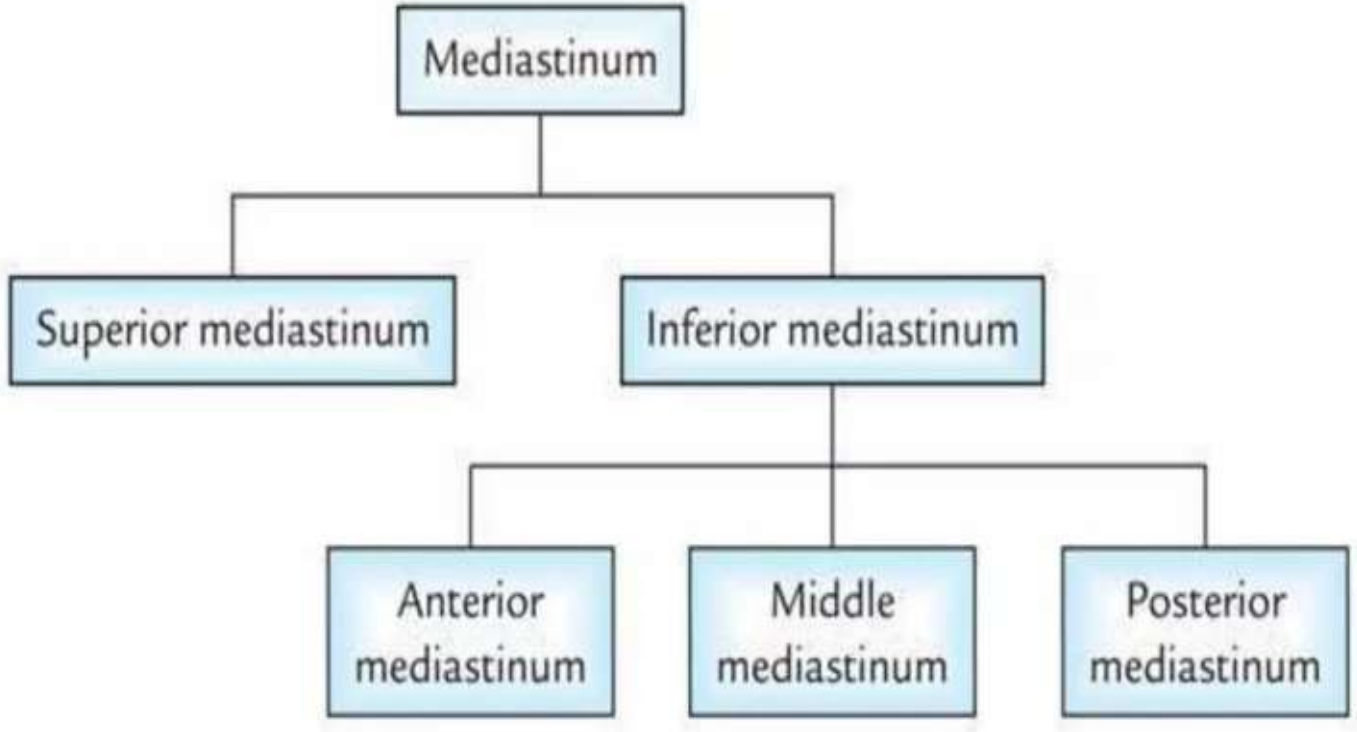


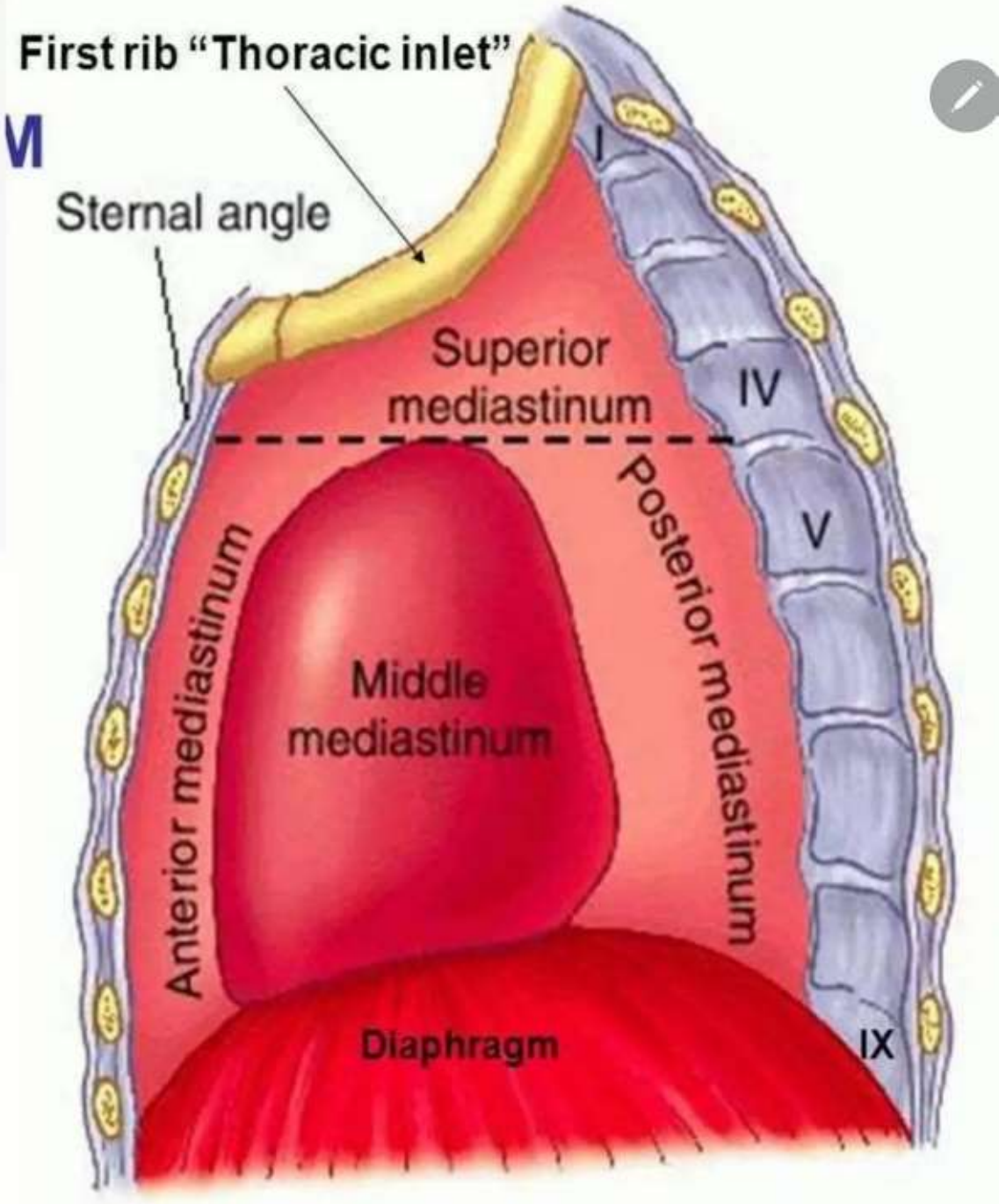


1-aortic root
2-right ventricular outflow tract
3-right atrial appendage
4-left atrium
5-descending aorta
6-oesophagus
7-azygos vein

Mediastinum

- Anatomically, the mediastinum is divided into two parts by an imaginary line that runs from the sternal angle anteriorly and intervertebral disc between T4 and T5 vertebrae posteriorly.





*The 4 subdivision of mediastinum:

1-superior mediastinum

-Boundaries:

(a) anteriorly: manubrium sterni.

(b) posteriorly: the upper 4 thoracic vertebrae.

(C) superiorly: the plane of the thoracic inlet (Page 11)

(d) inferiorly: an imaginary plane passing through the sternal angle (infront) & the lower border of T4 vertebra (behind).

(E) on each side: mediastinal pleura

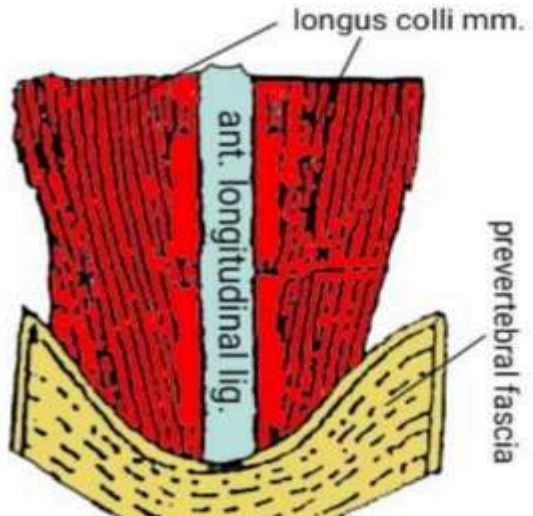
Anatomy of the Contents of Superior Mediastinum

The contents of the sup. mediastinum can be arranged in 5 layers (from behind forwards) as follows:

(A) Prevertebral layer: contains:

(1) the ant. longitudinal ligament.
(2) the 2 longus colli muscles.

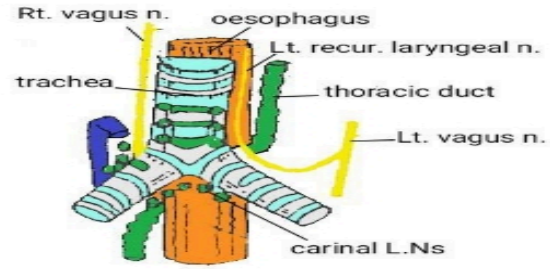
(3) the prevertebral fascia which covers the previous structures & blends with the ant. longitudinal lig. at the lower border of T4.



(B) Tubal layer: contains the oesophagus & trachea (and related structures):

(1) the oesophagus:

descends infront of the vertebral column with 2 structures



(a)Thoracic duct: ascends on the Lt. side of the oesophagus then passes through the thoracic inlet to reach the root of the neck.
(b) Lt. recurrent laryngeal n.: ascends in the Lt. groove between the trachea & the oesophagus.

(2) the trachea: descends infront of the oesophagus & ends at the lower border of T4 by dividing into 2 bronchi. It has the following structures related to it:

(a)Lymph nodes: -Rt. & Lt. paratracheal L.Ns: on each side of the trachea
-Rt. &Lt. tracheobronchial L.Ns: at the 2 angles between trachea & bronchi.
-Carinal L.Ns: on & below the tracheal bifurcation (the carina).

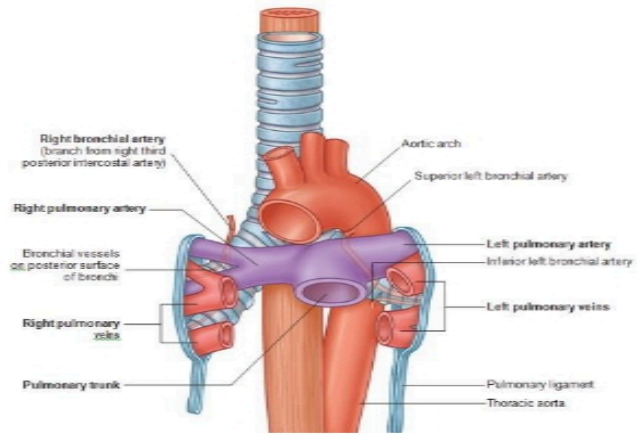
(b) Rt. vagus n.: descends along the Rt. side of the trachea.

(c) the deep cardiac plexus: lying infront of the tracheal bifurcation.

(d) the arch of azygos v.: on the Rt. side of the Rt. tracheobronchial angle.

(C) Arterial layer: includes the aortic arch & its branches together with The termination of the pulmonary trunk:

I- the arch of aorta: ascends upwards & to the Lt. infront of the trachea then turns backwards on the Lt. side of the trachea & oesophagus to reach the Lt. side of the body of T4 where it continues as the descending thoracic aorta.



- The concavity of the arch is occupied by:

(a) the bifurcation of the pulmonary trunk
(b) the Lt. bronchus
(c) ligamentum arteriosum (connecting the Lt. pulmonary a. to the arch of aorta). Ductus Arteriosus

(d) Lt. recurrent laryngeal n.: a branch of the Lt. vagus n. which hooks below the ligamentum arteriosum then ascends deep to the arch of aorta.

(e) superficial cardiac plexus: lies on the ligamentum arteriosum, below the arch of aorta.

-The front of the arch of aorta is related to: Lt. phrenic & Lt. vagus nerves, Lt. sup. intercostal v, cardiac branches of Lt. vagus & Lt. symp. trunk.

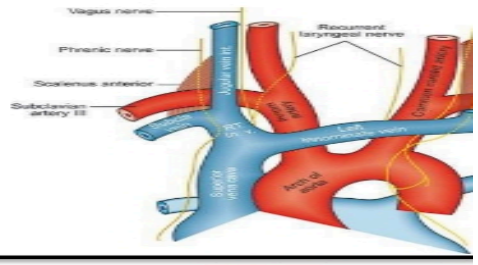
I-Branches of the aortic arch: (1) innominate a.
(2) Lt. common carotid a.
(3) Lt. subclavian a.

(Arranged from Rt. to Lt. & also from before backwards)

III- the termination of the pulmonary trunk: lies within the concavity of the aortic arch.

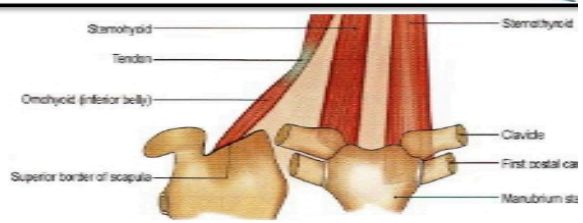
D- Venous layer: contains the 2 innominate veins & the upper 2 of the S.V.C.:

(1) **the Rt. innominate v.:** descends vertically behind the upper part of the Rt. margin of the manubrium.
(2) **The Lt. innominate v.:** runs obliquely behind the upper 1/2 of the manubrium above the aortic arch (crossing its 3 branches).
(3) **the upper 1/2 of the S.V.C.:** descends vertically behind the lower part of the Rt. margin of the manubrium with the Rt. phrenic n. on its Rt. side & the arch of azygos v. joining its back.



E-Retro-sternal layer: contains:

(1) the origins of sternohyoid & sternothyroid muscles
(2) remnants of the thymus gland.



Clinical Importance

***The mediastinal Syndrome:**

- It is a clinical condition caused by compression of the contents of the superior mediastinum by ar space occupying lesion (e.g.: tumor, enlarged L.Ns or retro-sternal goitre).

* **manifestations**:-

- Compression of the veins → venous congestion in the U.L., head & neck.
- Compression of the arteries → ischemia of the U.L., head & neck.
- Compression of the trachea → dyspnea.
- Compression of the oesophagus → dysphagia

II-Anterior Mediastinum

-It is a narrow space in front of the pericardium, overlapped by ant. borders of both lungs.

***Boundaries:** - **Anteriorly:** body of the sternum.
- **Posteriorly:** the pericardium.
-**Superiorly:** the imaginary plane separating the sup. mediastinum from the inf. mediastinum.

- **Inferiorly:** sup. surface of the diaphragm.
- **On each side:** mediastinal pleura.

***Contents:**

(1) remains of the thymus gland & loose areolar tissue.

(2) upper & lower sternopericardial ligaments connecting the pericardium to the back of the sternum.

(3) 2-3 lymph nodes & lymph vessels.

(4) small mediastinal brs. of internal thoracic arteries.

III-Middle Mediastinum

*It is the broadest part of mediastinum & is occupied by the pericardium & its content

***Contents:**

(1) The heart enclosed in the pericardium.
(2) Arteries: (a) ascending aorta (b) pulmonary trunk & the 2 pulmonary arteries:
(3) Veins: (a) lower 1/2 of the S.V.C (b) terminal part of the azygos vein (c) upper part of the I.V.C (d) the Rt. & Lt. pulmonary veins
(4) Nerves: (a) both phrenic nerves (b) the deep cardiac plexus
(5) Tubes: (a) the bifurcation of trachea (b) the Rt. &Lt. principal bronchi
(6) Lymph Ns: tracheobronchial L.NS.

IV-Posterior Mediastinum

***Boundaries:**

- **anteriorly:** (a) pericardium (b) post. part of upper surface of diaphragm
- **posteriorly:** the lower 8 thoracic vertebrae
- **on each side:** mediastinal pleura.

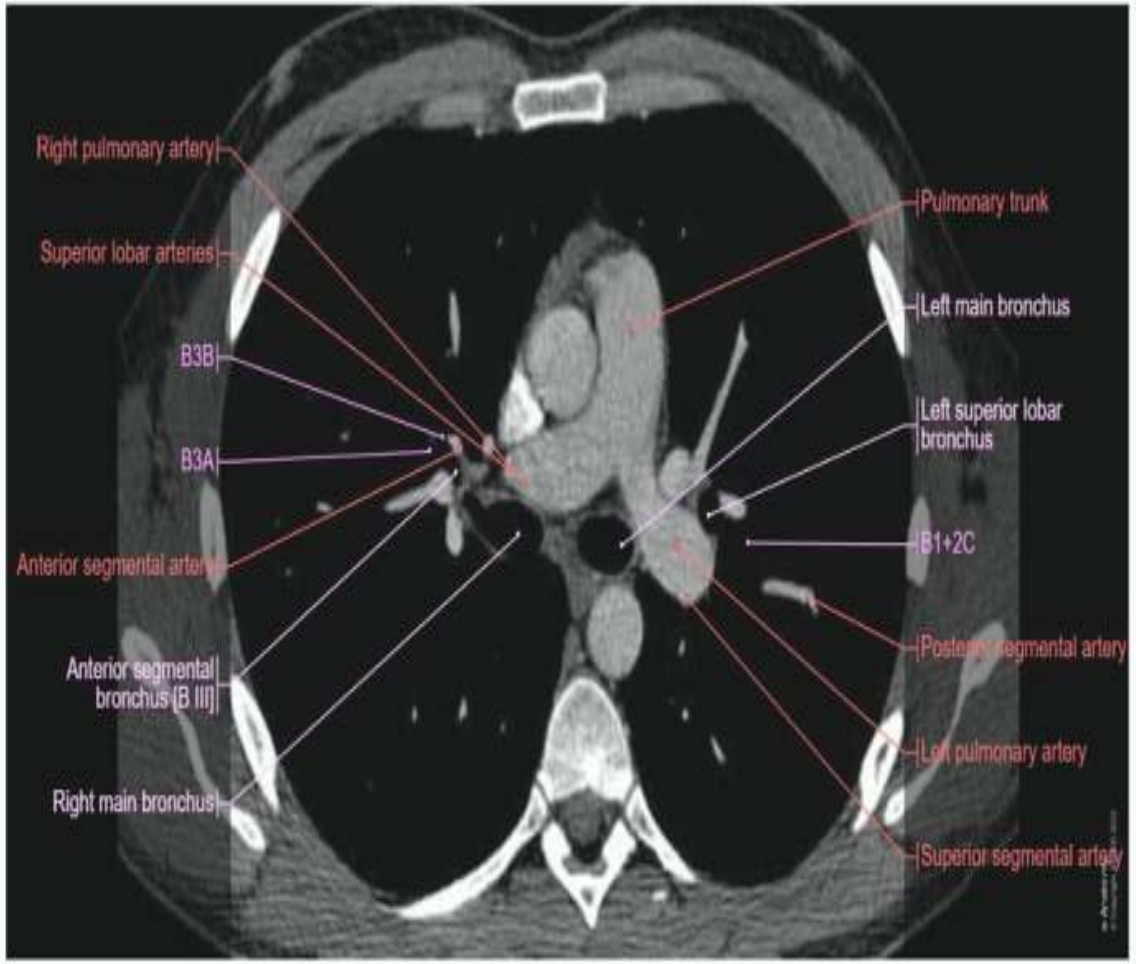
***Contents:**

(1) Tubes: the oesophagus.
(2) Arteries: descending thoracic aorta & its branches.
(3) Veins: azygos & hemiazygos veins.
(4) Nerves: (a) vagi (forming oesophageal plexuses around the oesophagus).
(b) both sympathetic chains & their splanchnic nerves.

thoracic duct.

Mediastinal Compartments & Contents

An Educational Module for Medical Students on Thoracic Anatomy and Clinical Correlations



MEDIASTINUM

Definition

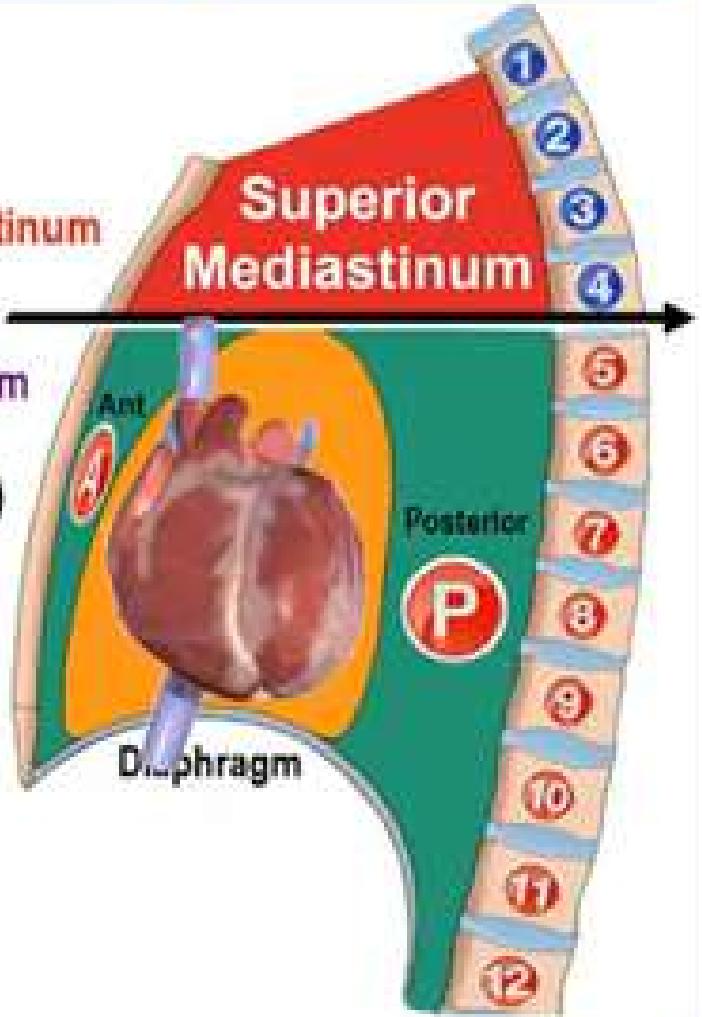
Divisions:

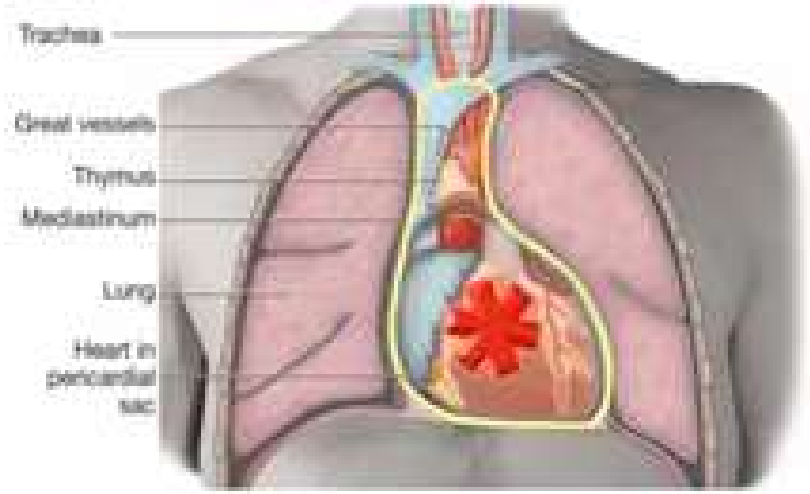
1. **Superior**
2. **Inferior:**
   - **a. Anterior**
   - **b. Middle**
   - **c. Posterior**

Superior Mediastinum

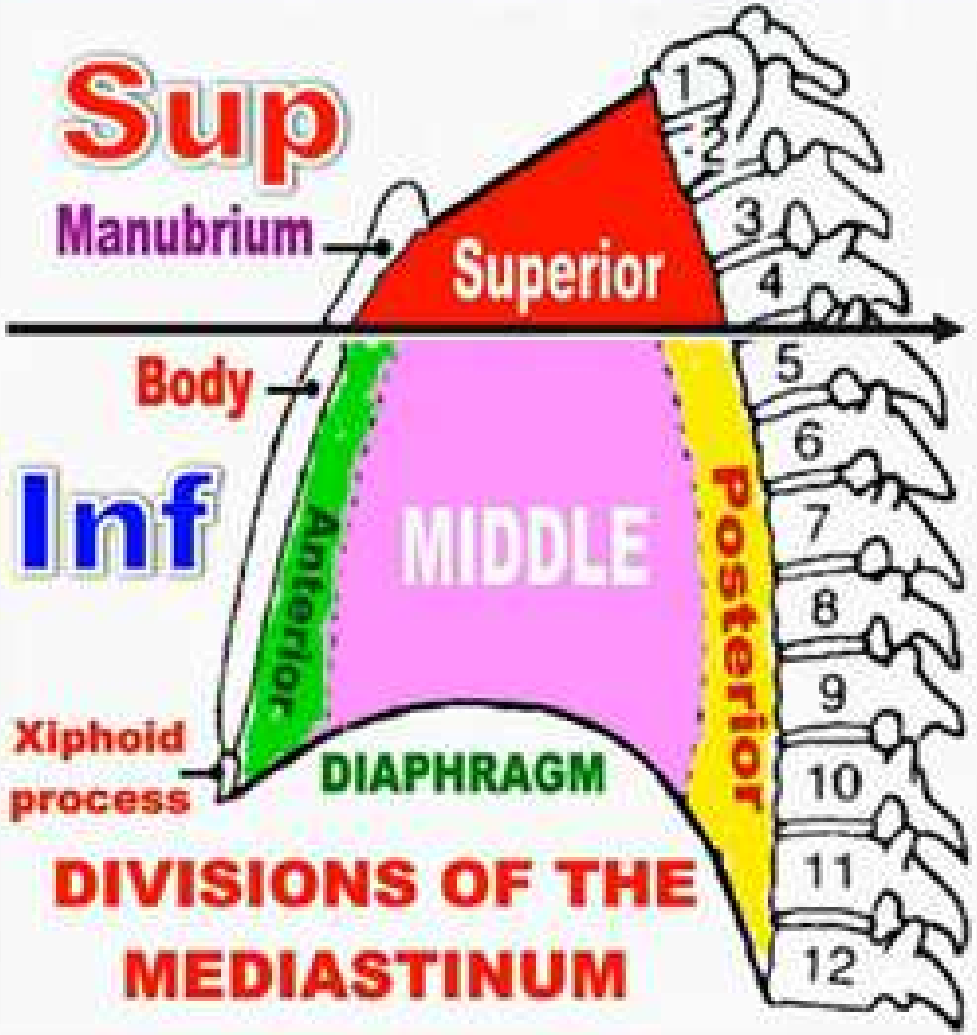
Angle of Louis

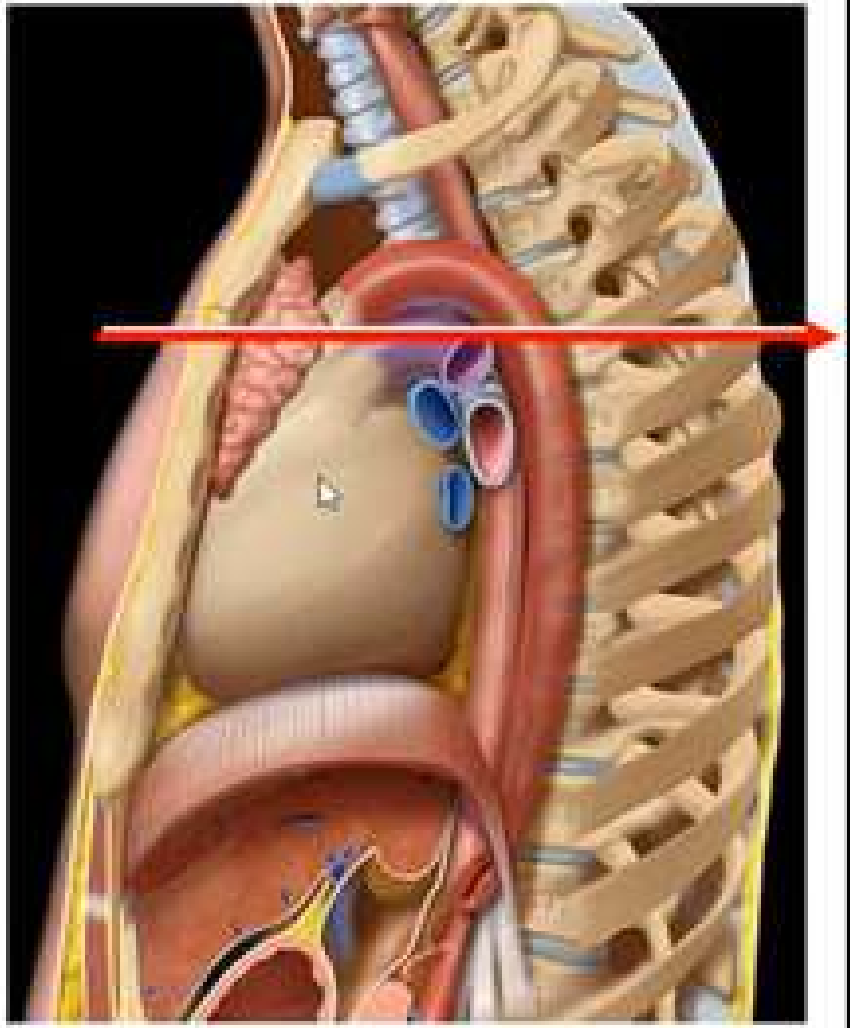
Inferior Mediastinum
- a: Anterior
- b: Middle (heart)
- c: Posterior



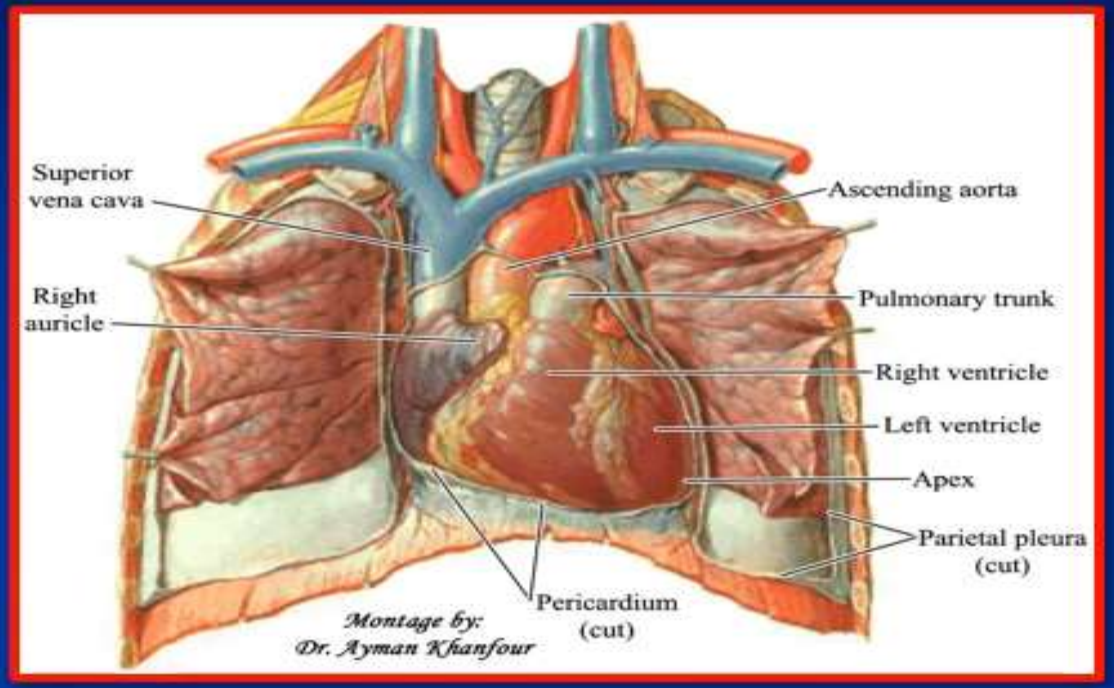


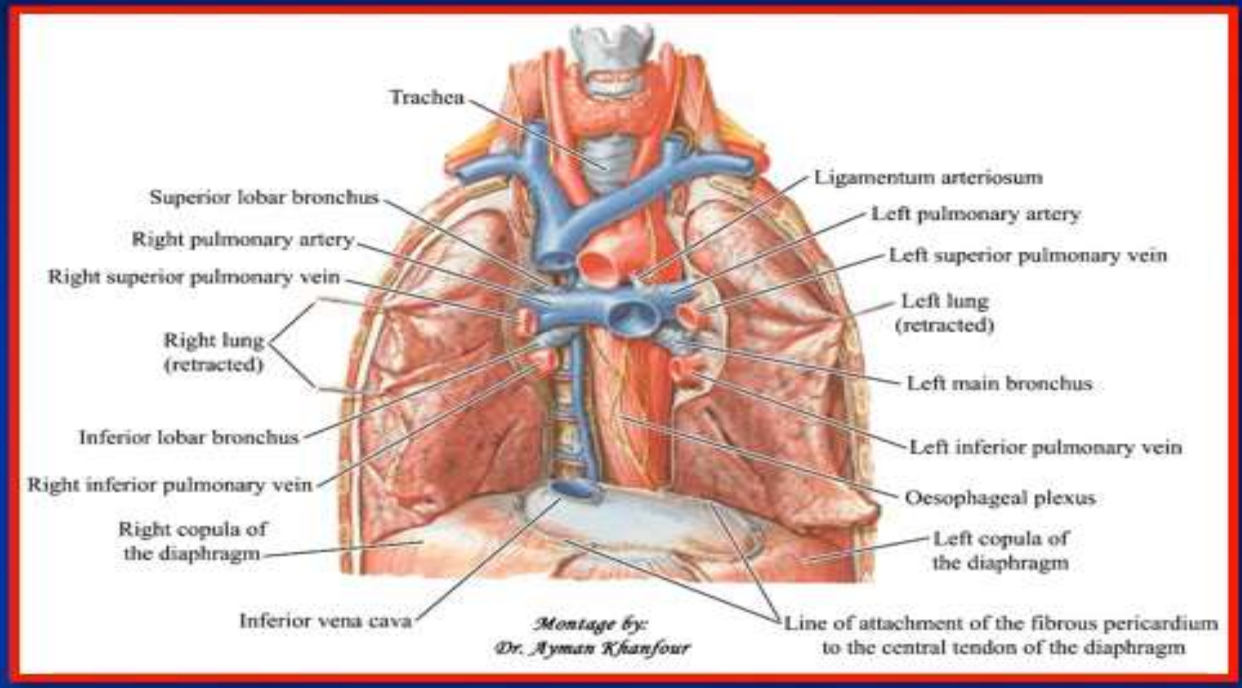
DIVISIONS OF THE MEDIASTINUM





Definition





- It is the area between the right and left lungs.

Subdivisions

- It is subdivided by an imaginary line from the *STERNAL ANGLE* to the level of *T. 4* into:

  I. Superior mediastinum
  (*above* the imaginary line).

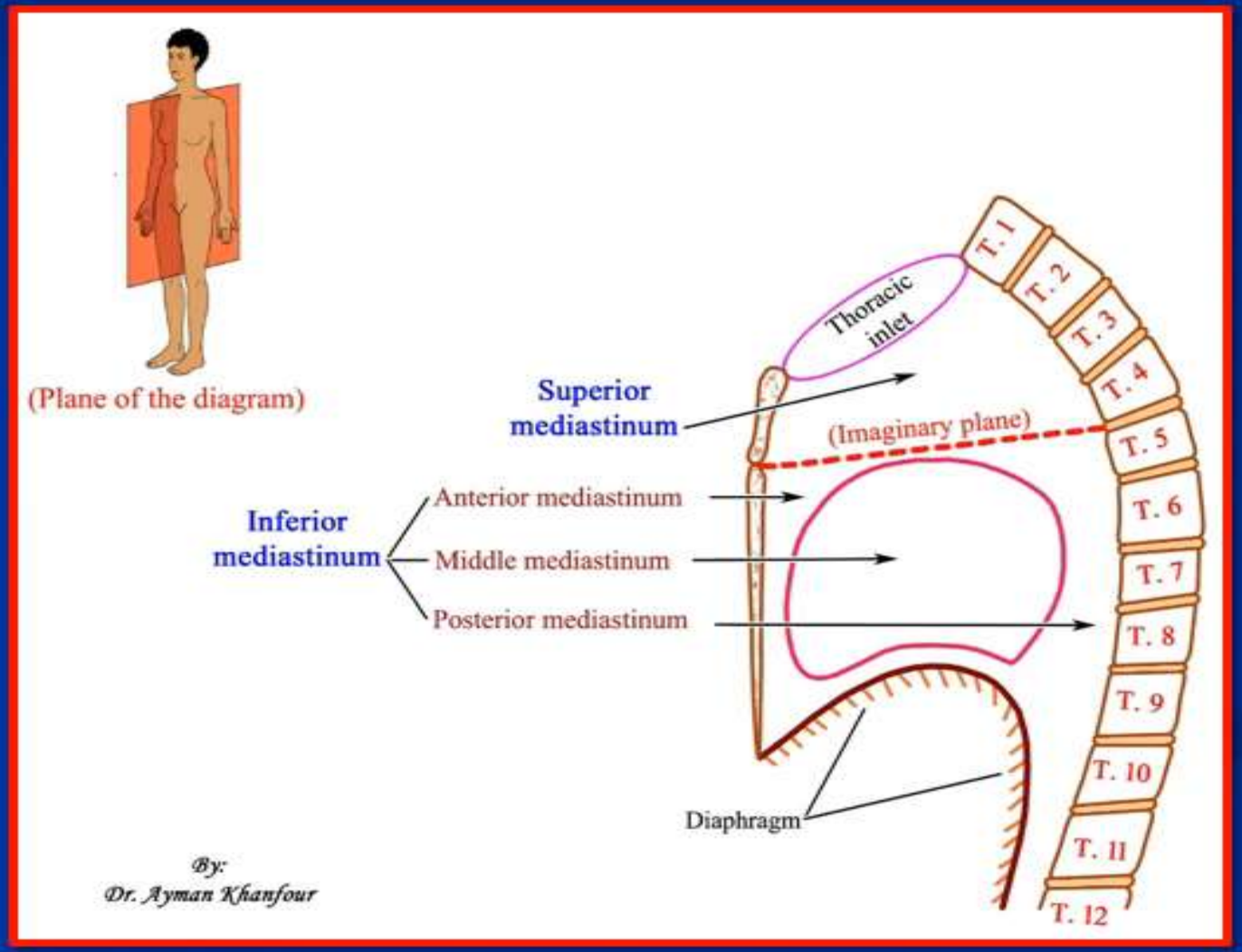
  II. Inferior mediastinum
  (*below* the imaginary line)

- The **inferior mediastinum** is subdivided into:

  1. Anterior mediastinum
  - ***Anterior*** to the pericardium and the heart.

  2. Middle mediastinum
  - ***Includes*** the pericardium and the heart.

  3. Posterior mediastinum
  - ***Posterior*** to the pericardium and the heart.



Superior Mediastinum

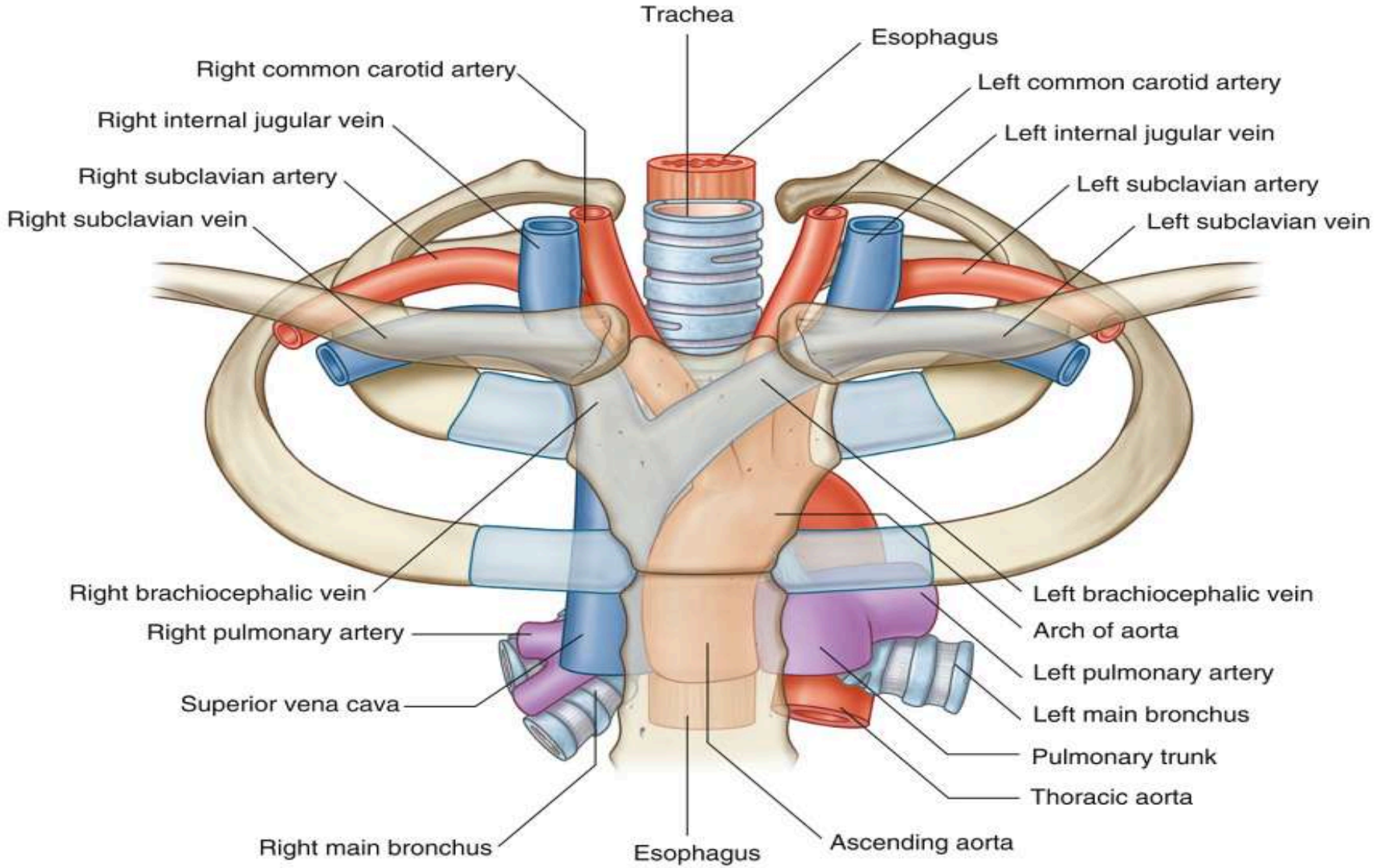


Fig. 3.89 Structures in the Superior Mediastinum.

Contents of Superior Mediastinum

I. Tubes:

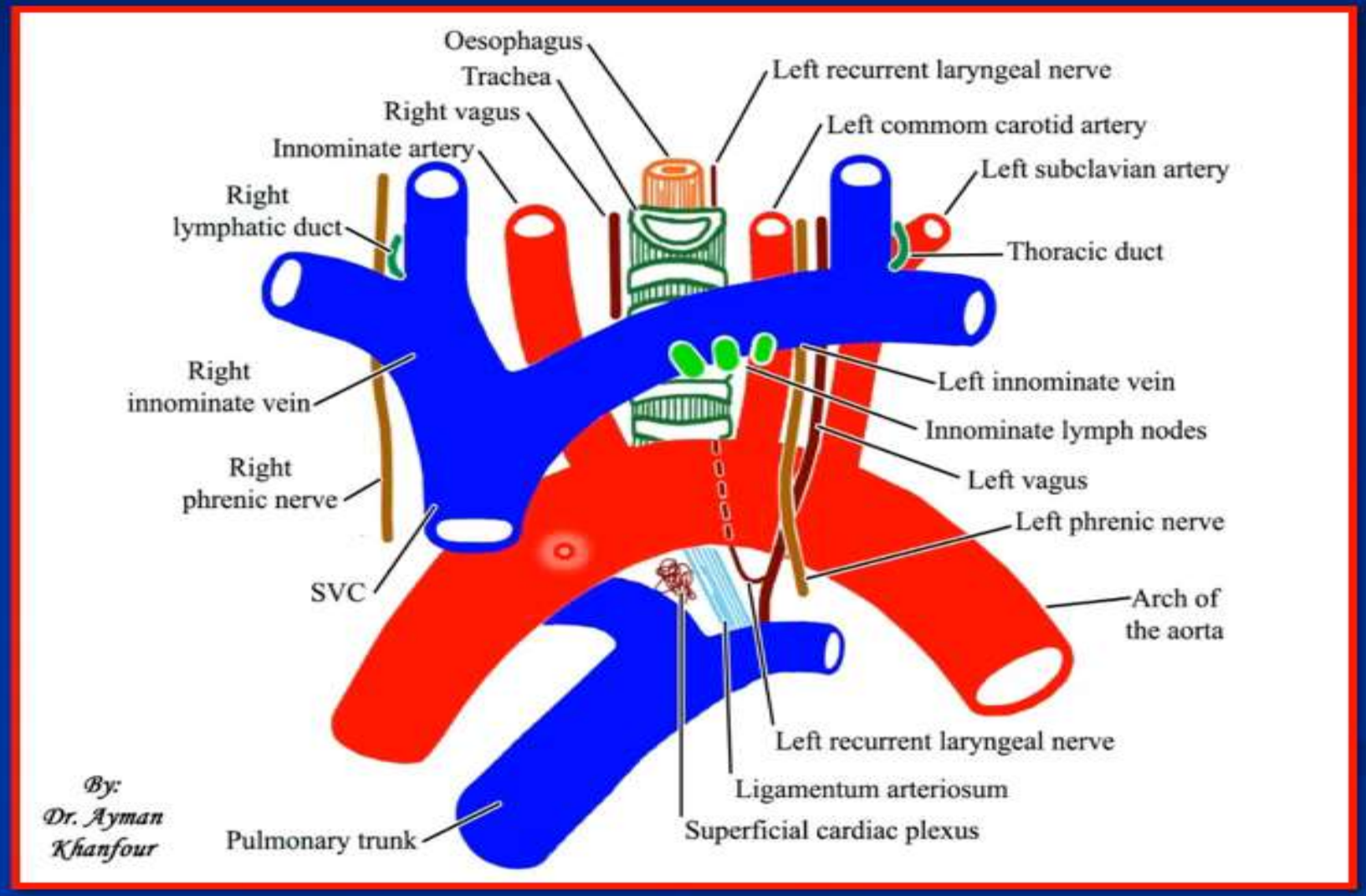
- Oesophagus.
- Trachea.

II. Veins:

- Innominate veins (right and left).
- Superior vena cava (upper ½).

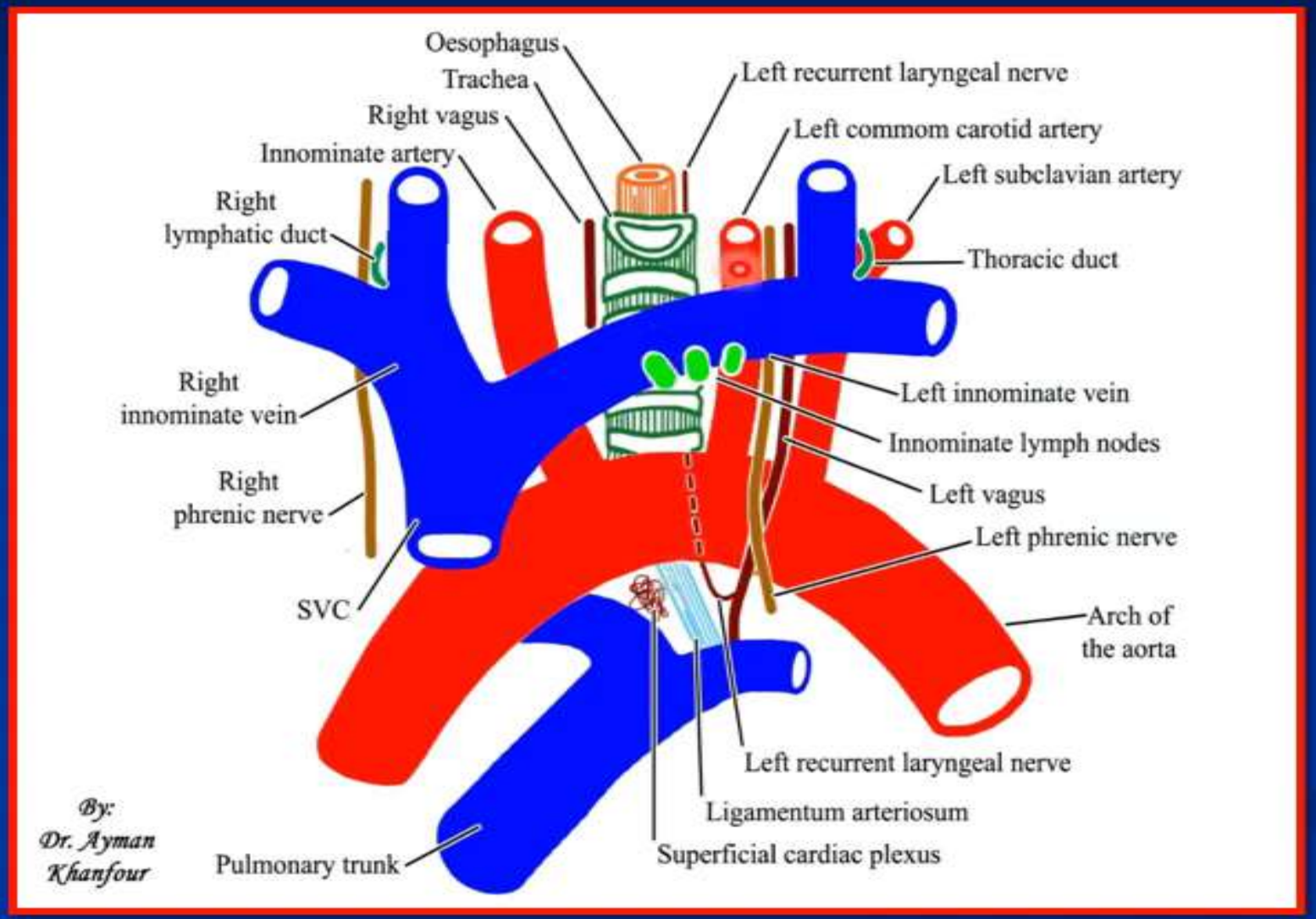
III. Arteries:

- Arch of the aorta and its big branches.



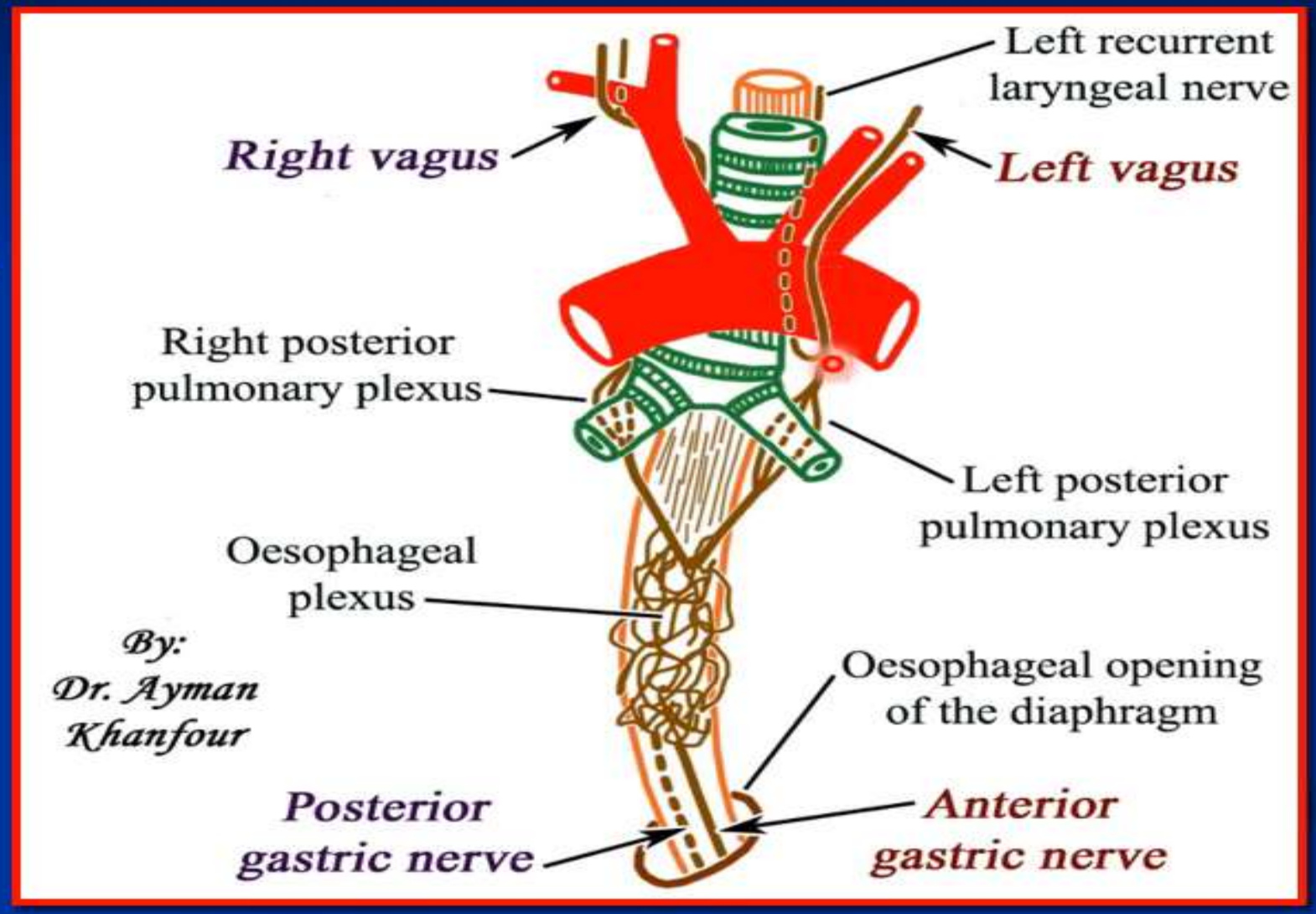
IV. Nerves:

- Phrenic nerves.
- Vagi n.
- Superficial part of cardiac plexus.
- Lt. recurrent laryngeal n.
- Sympathetic trunk



IV. Nerves:

- Phrenic nerves.
- Vagi n.
- Superficial part of cardiac plexus.
- Lt. recurrent laryngeal n.
- Sympathetic trunk

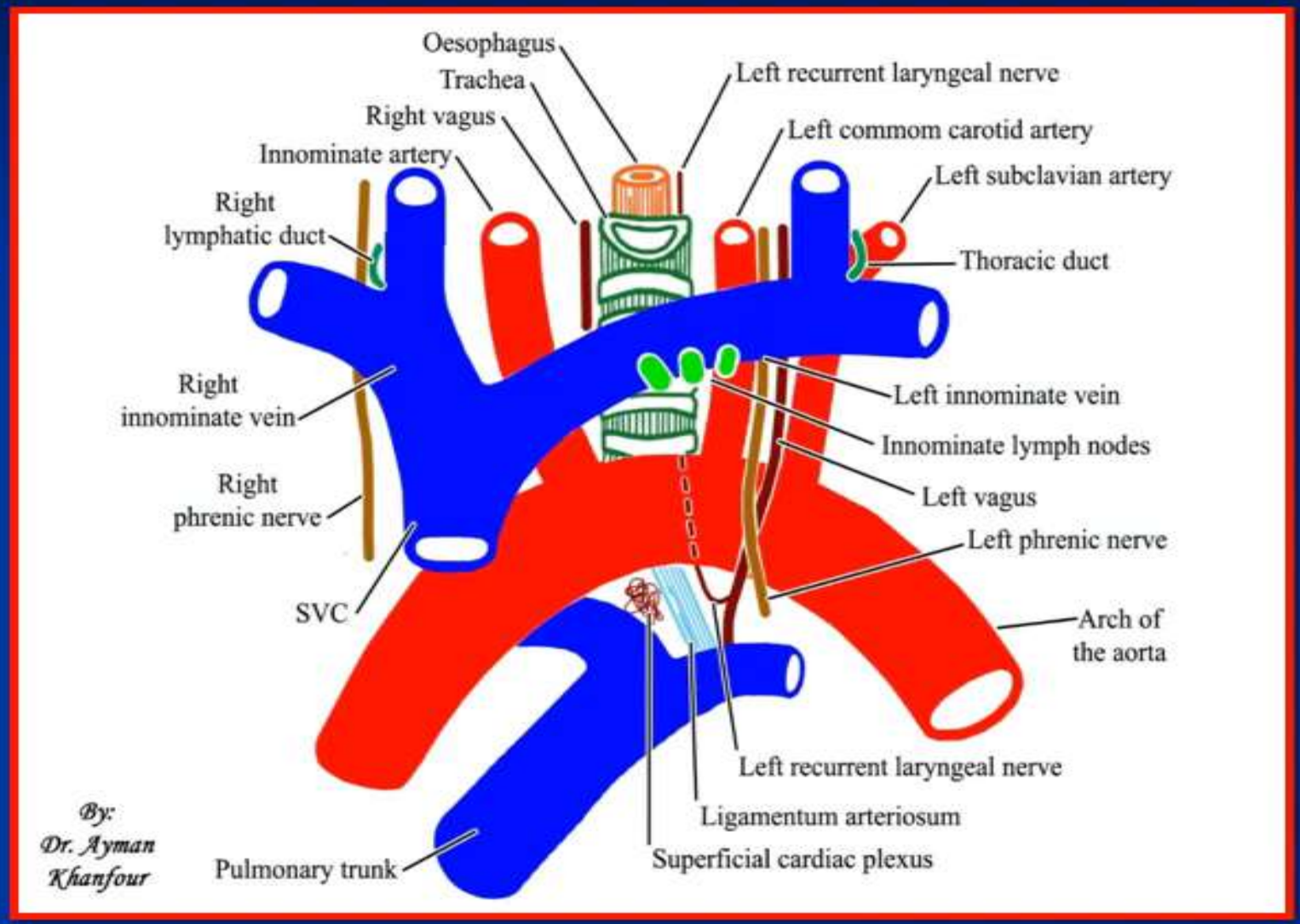


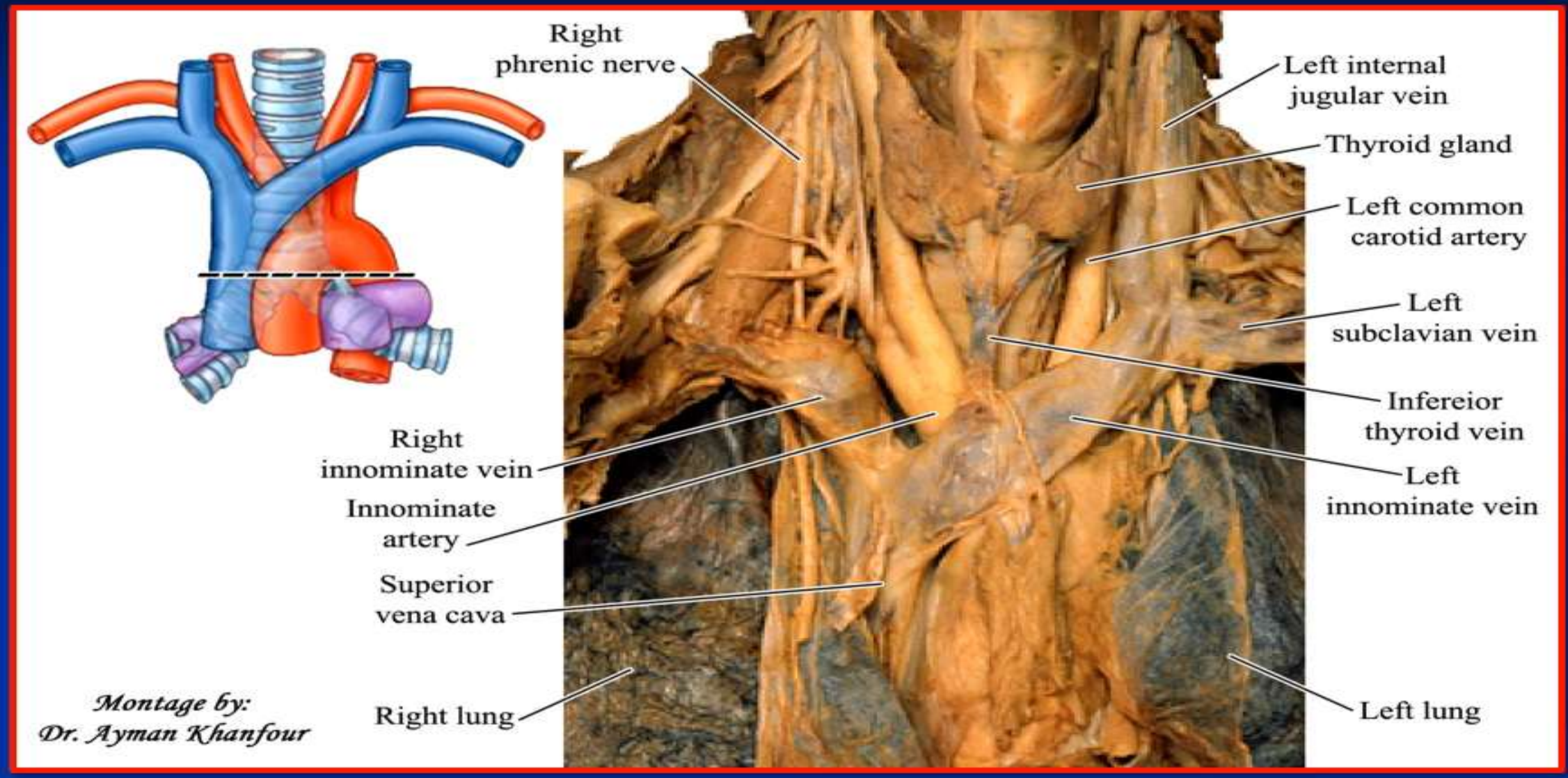
IV. Nerves:

- Phrenic nerves.
- Vagi n.
- Superficial part of cardiac plexus.
- Lt. recurrent laryngeal n.
- Sympathetic trunk

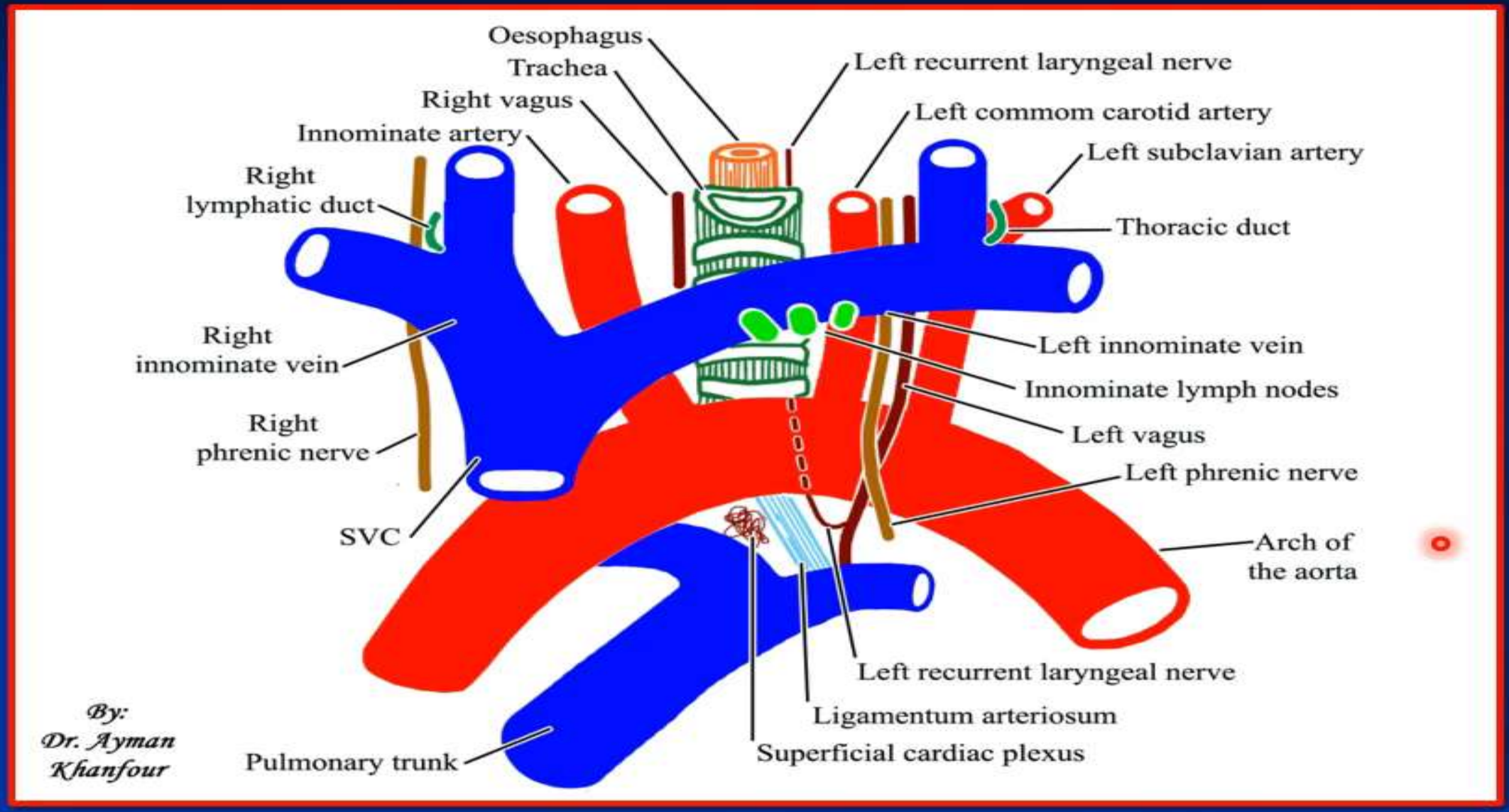
V. Lymphatics:

- Innominate, paratracheal + superior mediastinal lymph nodes.
- Thoracic duct.





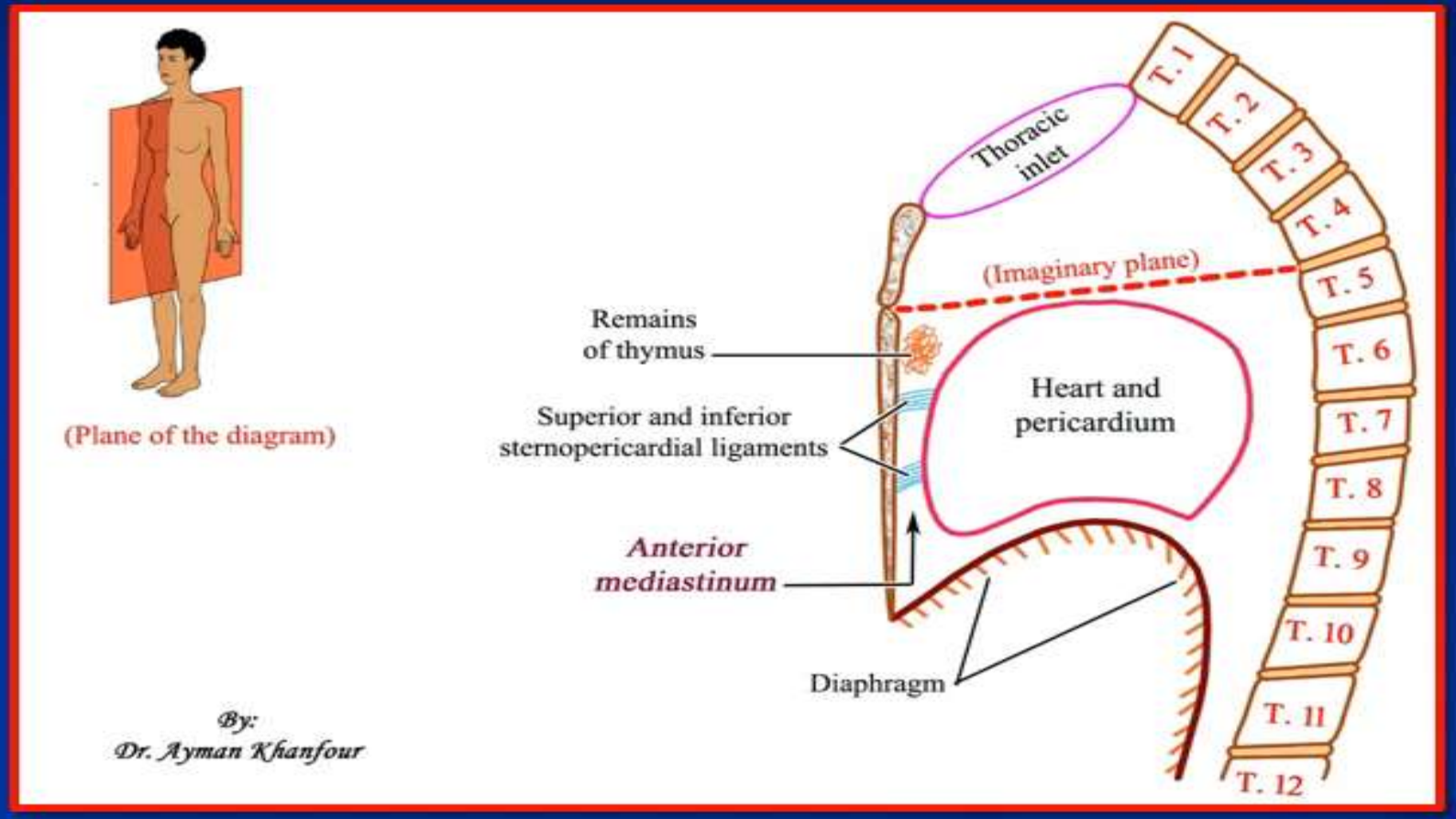
(Contents of the superior mediastinum)

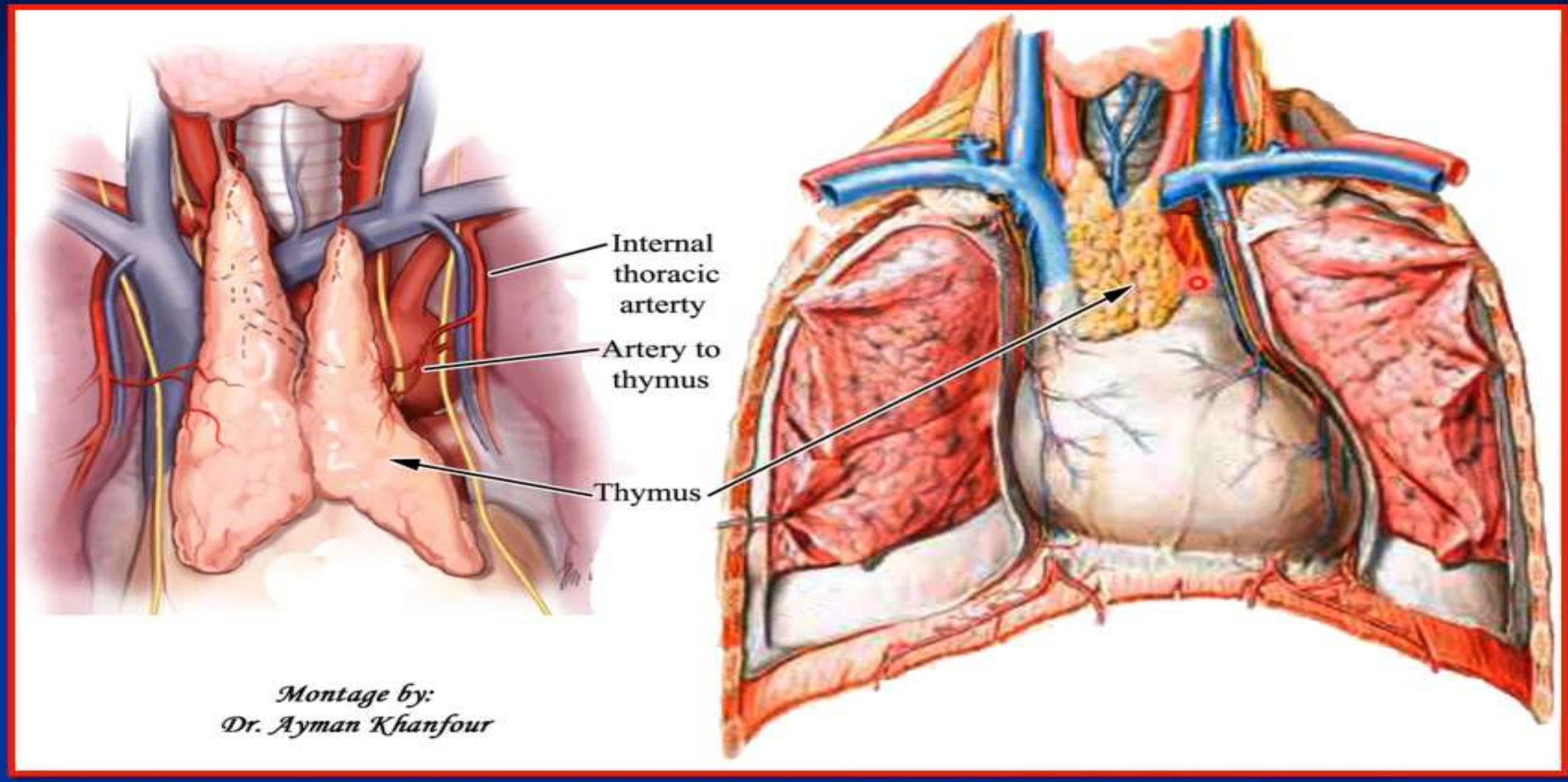


(Contents of the superior mediastinum)

Contents of Anterior Mediastinum

1. Thymus (or its remains).
2. Sterno-pericardial ligaments.
3. Mediastinal branches of internal thoracic artery.
4. Lymph nodes.





(Contents of the anterior mediastinum)

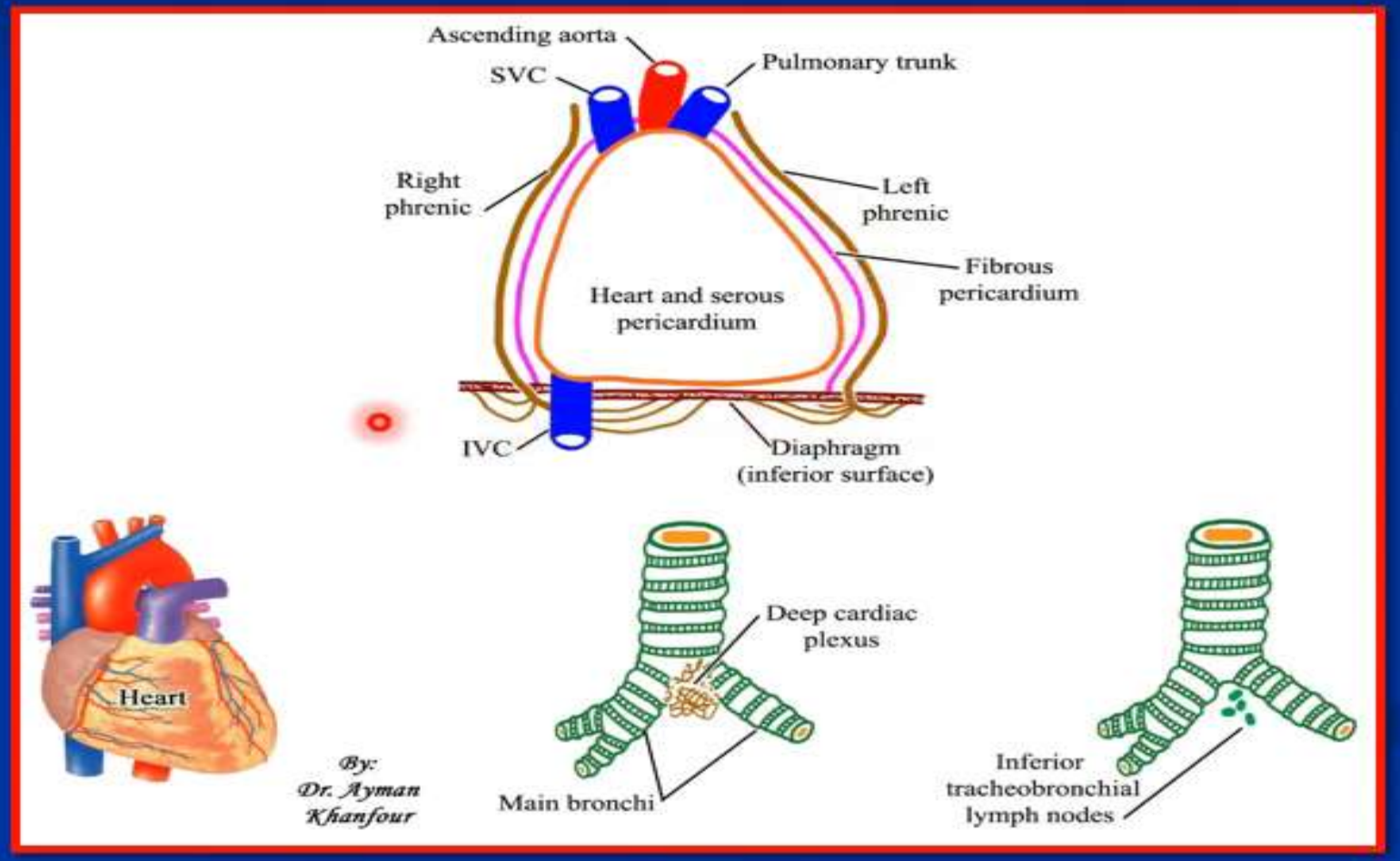
Middle Mediastinum

Contents of Middle Mediastinum

1. Heart and pericardium.

2. Great vessels of the heart
   a. Ascending aorta.
   b. Pulmonary trunk.
   c. Pulmonary veins (terminal parts).
   d. SVC (lower part).
   e. IVC (upper part).

3. Phrenic nerves

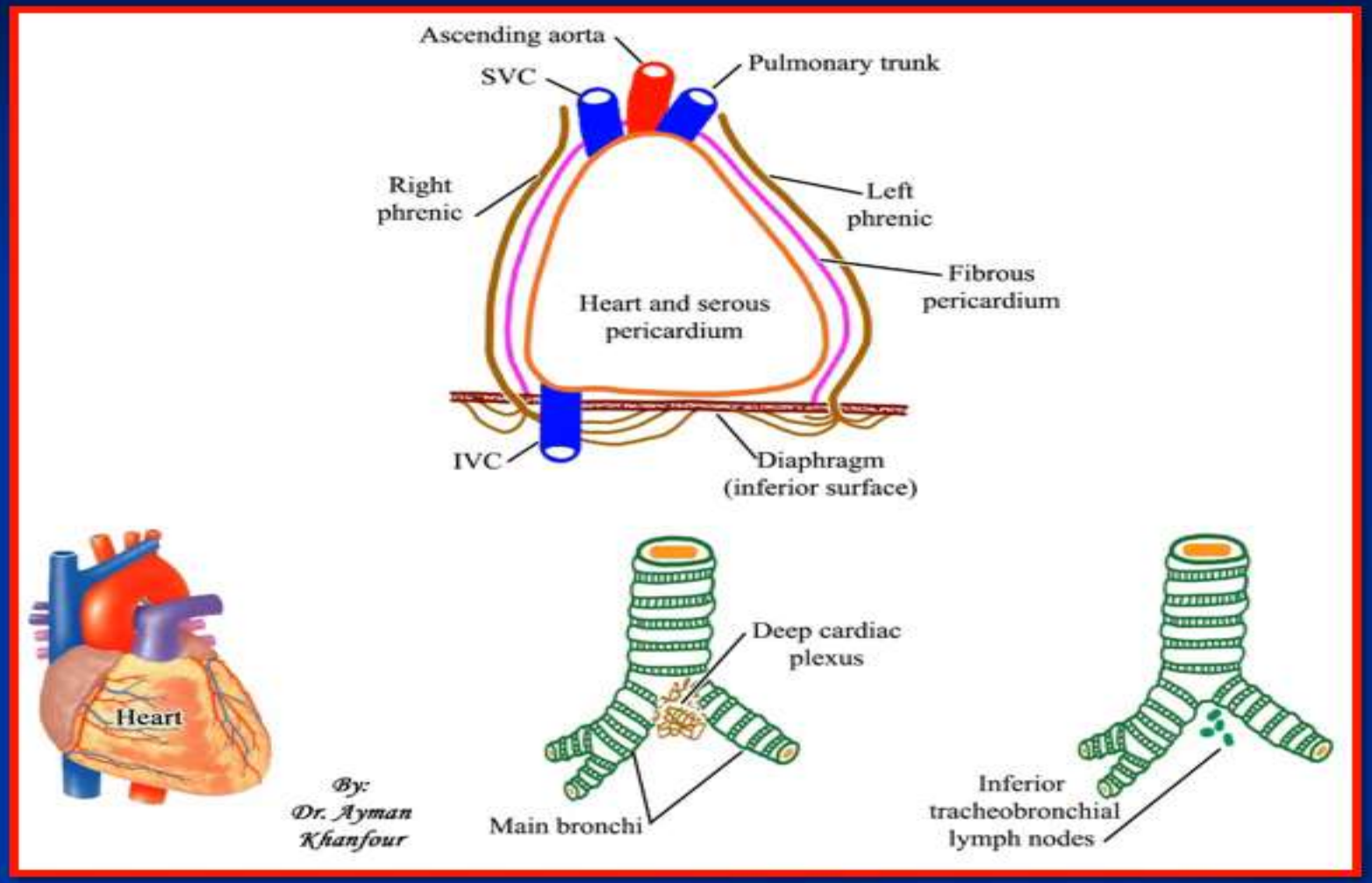


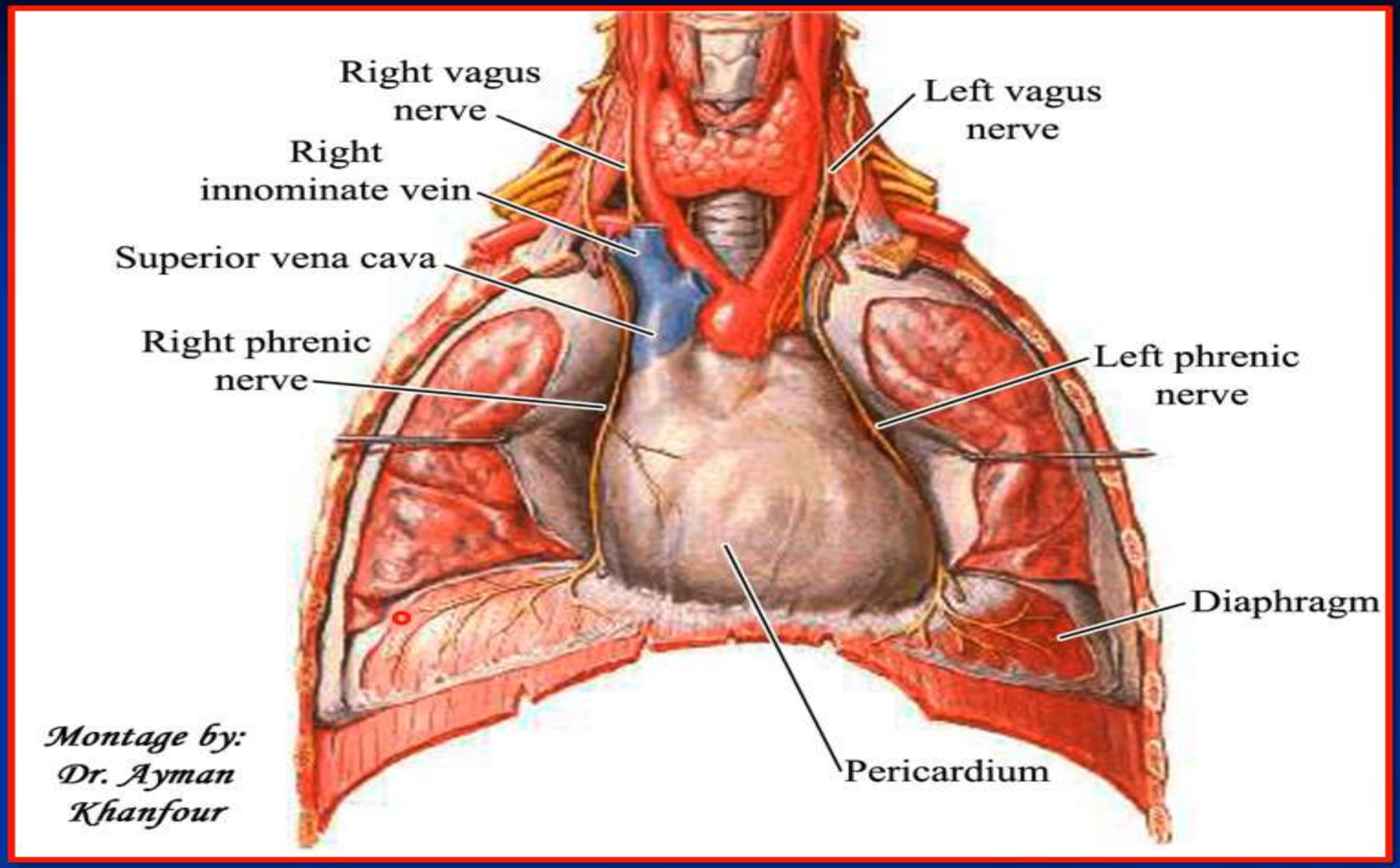
5. Bifurcation of trachea.

6. Inferior trachea
-bronchial LN.

7. Deep part of the
cardiac plexus.

8. Pericardiaco-phrenic
vessels.





(Contents of the middle mediastinum)

Posterior Mediastinum

POSTERIOR MEDIASTINUM

BOUNDARIES

1. Superior: line of the sternal angle
2. Inferior: diaphragm
3. Anterior: heart and pericardium
4. Posterior: lower 8 thoracic vertebrae

Superior Mediastinum

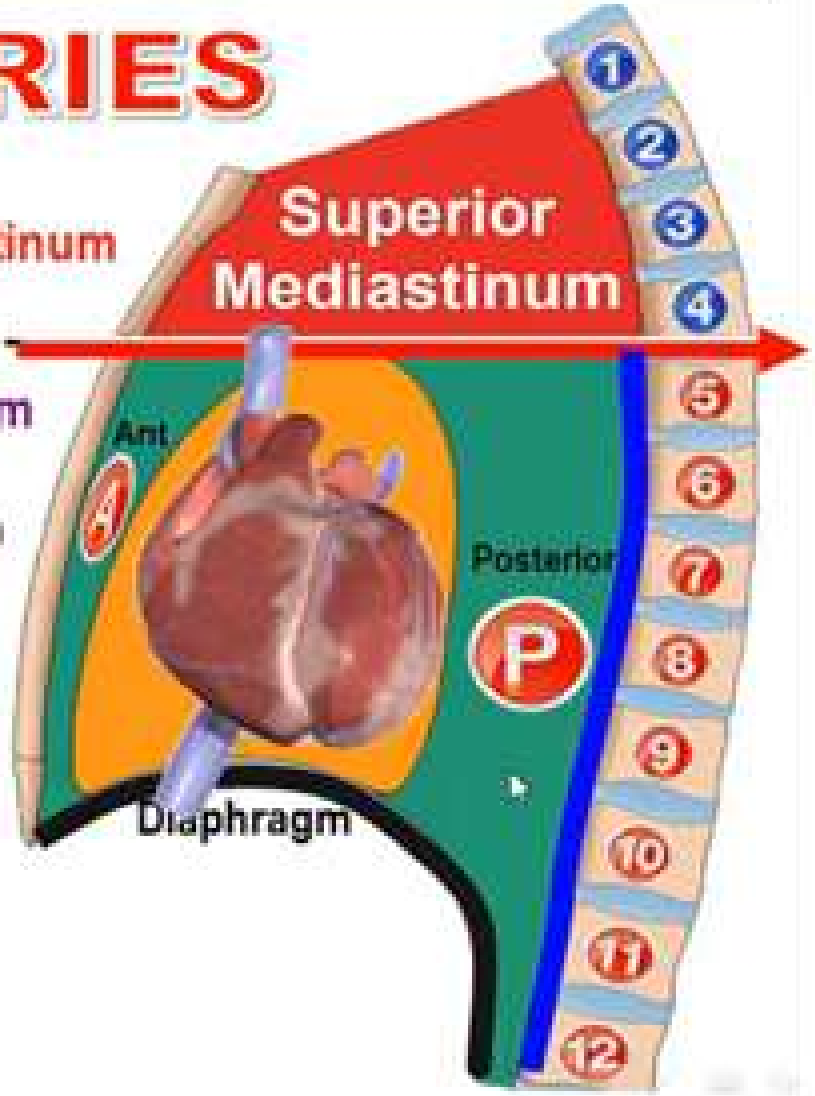
Angle of Louis

Inferior Mediastinum

a: Anterior

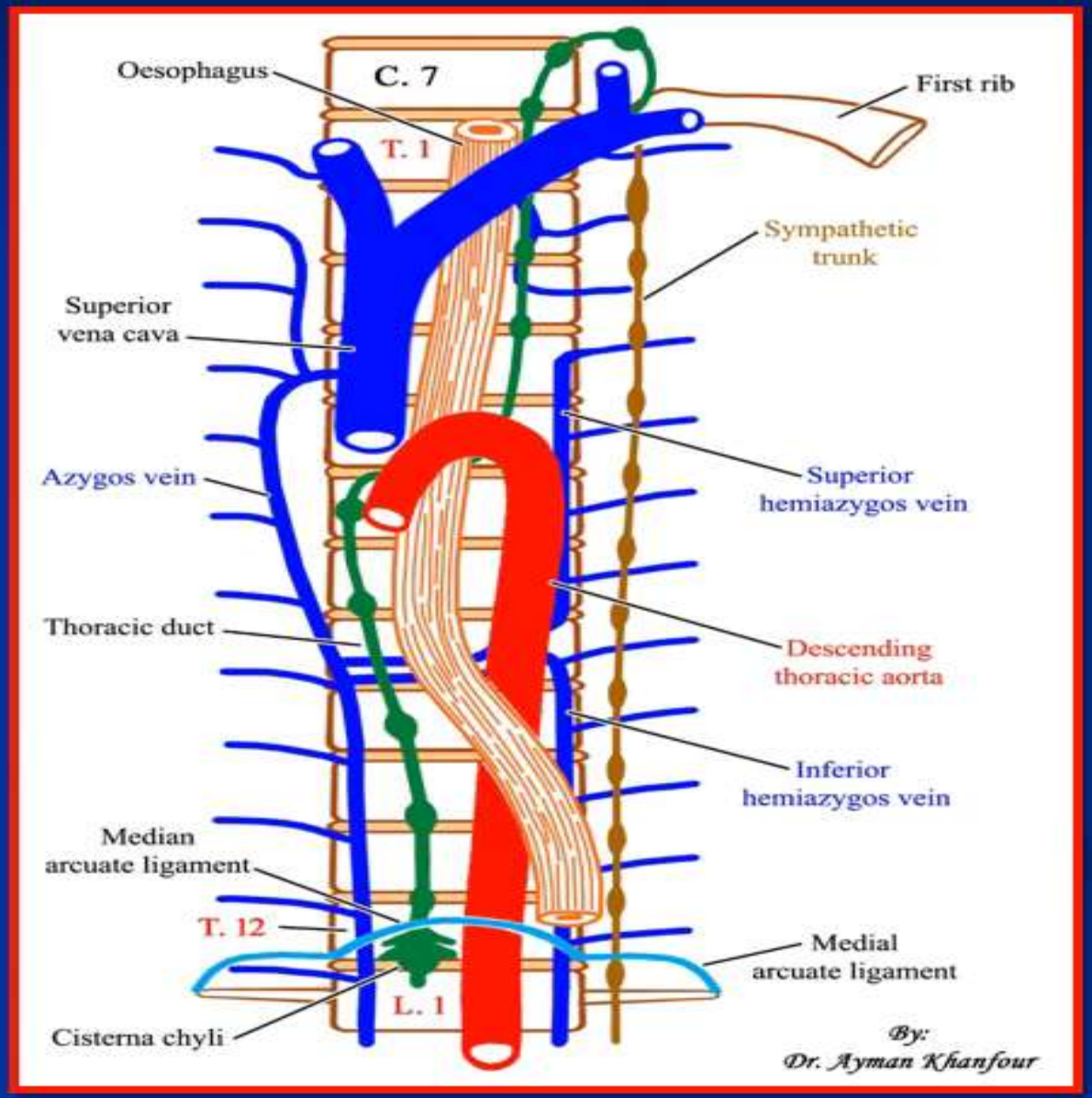
b: Middle (heart)

c: Posterior



Contents of Posterior Mediastinum

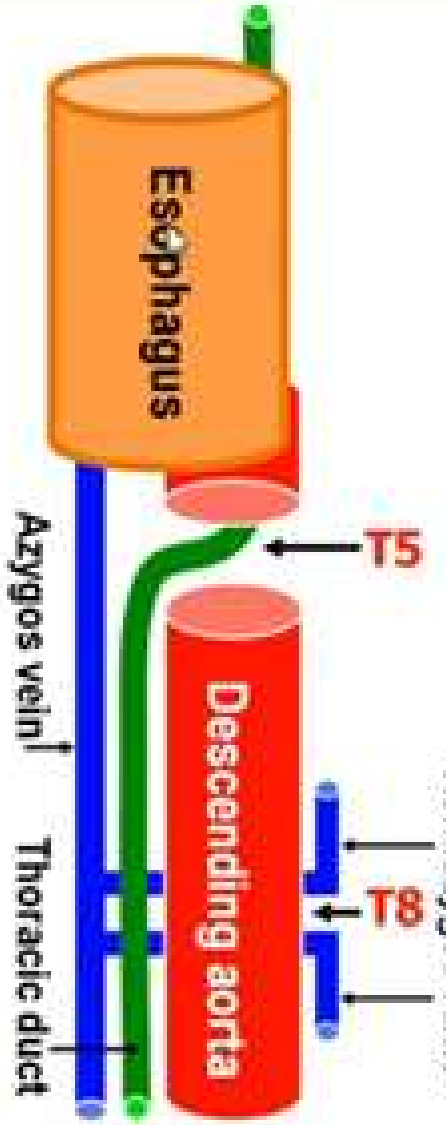
1. Descending thoracic aorta.
2. Oesophagus.
3. Azygos venous system.
4. Vagi nerves.
5. Sympathetic trunk.
6. Thoracic duct.
7. Posterior mediastinal lymph nodes.

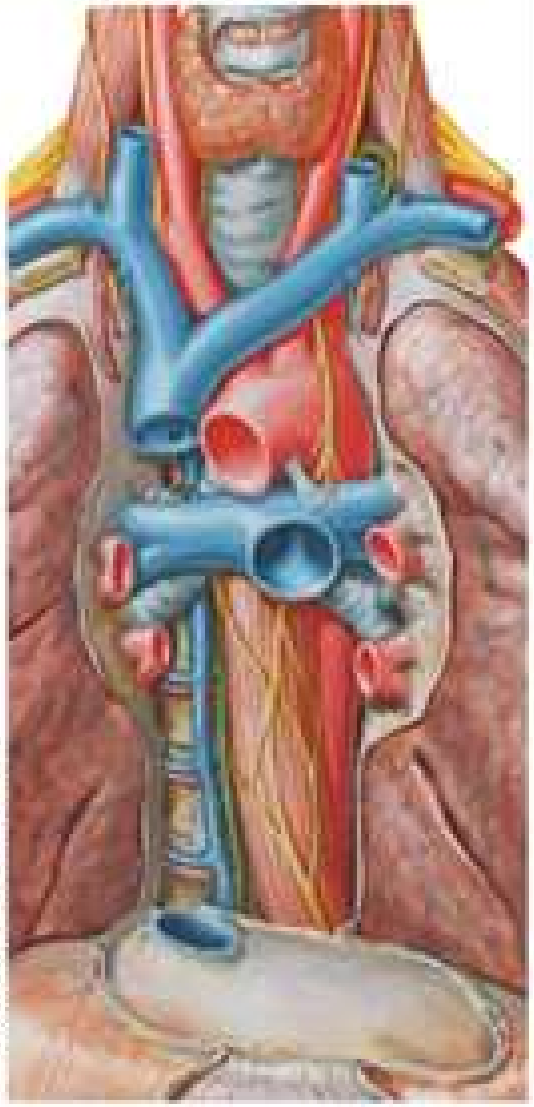


POSTERIOR MEDIASTINUM

CONTENTS

1. Esophagus
2. Esophageal plexus: 2 vagi
3. Descending Thor. aorta
4. Azygos vein
5. 2 hemiazygos veins
6. Thoracic duct
7. Sympathetic trunks
8. Mediastinal lymph nodes





@Smart.doctor1
@DrBahaaDmour

Normal Chest X-Ray Anatomy

Key landmarks on a frontal PA radiograph

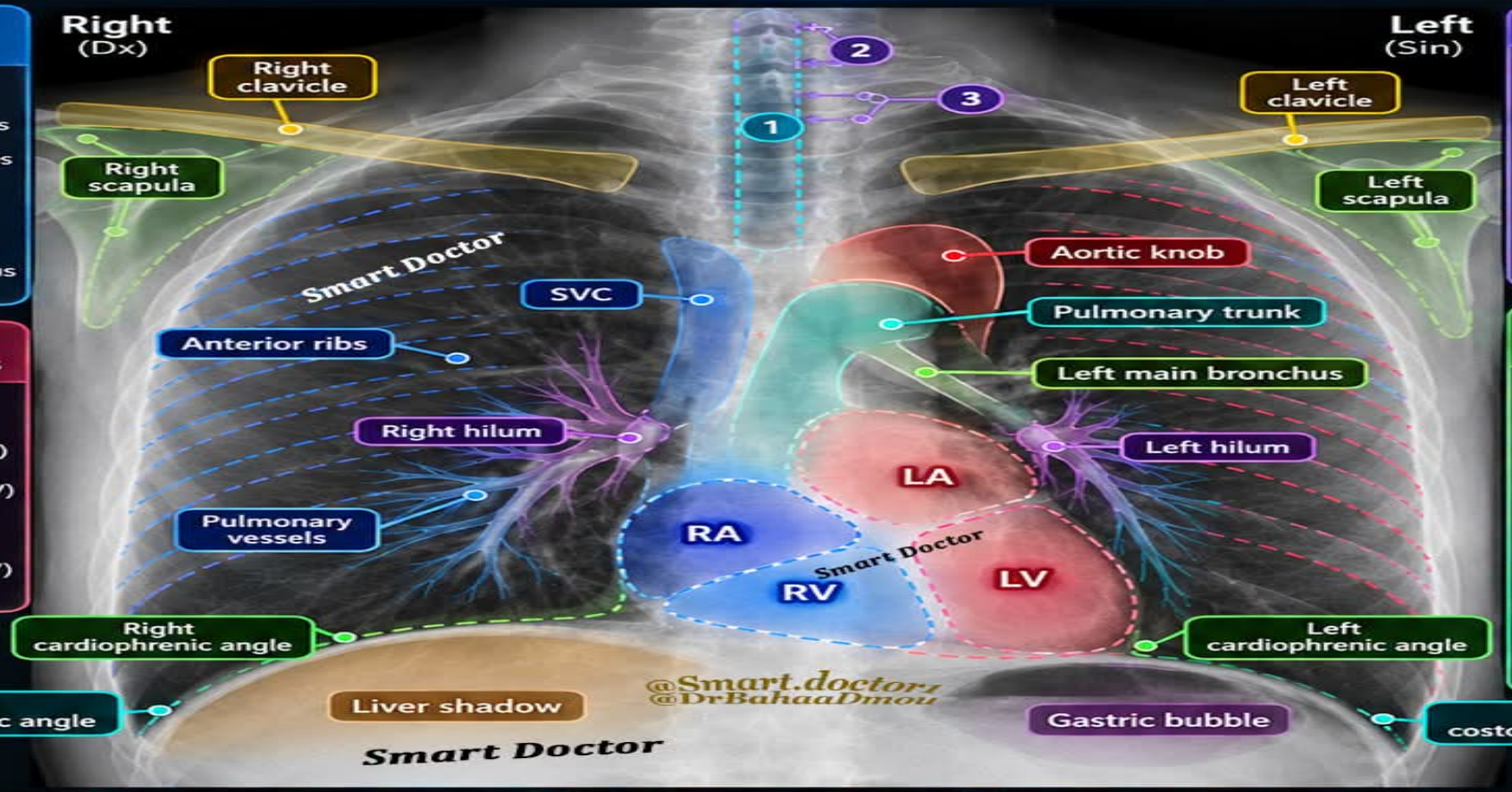
Smart Doctor

Airway & Mediastinum

1. Trachea
2. Spinous processes
3. Transverse processes
4. SVC
5. Pulmonary trunk
6. Left main bronchus

Heart Chambers & Major Contours

7. Aortic knob
8. Right atrium (RA)
9. Right ventricle (RV)
10. Left atrium (LA)
11. Left ventricle (LV)



Bony Landmarks

12. Right clavicle
13. Left clavicle
14. Right scapula
15. Left scapula
16. Anterior ribs
17. Posterior ribs

Diaphragm & Abdominal Structures

18. Right cardiophrenic angle
19. Left cardiophrenic angle
20. Right costophrenic angle
21. Left costophrenic angle
22. Liver shadow
23. Gastric bubble

Trachea & Spine



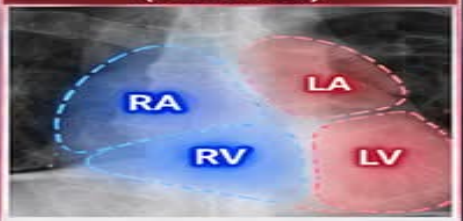
Trachea midline with spinous processes posterior to it. Transverse processes project laterally.

Hila & Pulmonary Vessels



Hila contain pulmonary arteries, veins and bronchi.

Heart Chambers (on PA view)



Normal heart size and contours on PA view.

Ribs: Anterior vs Posterior



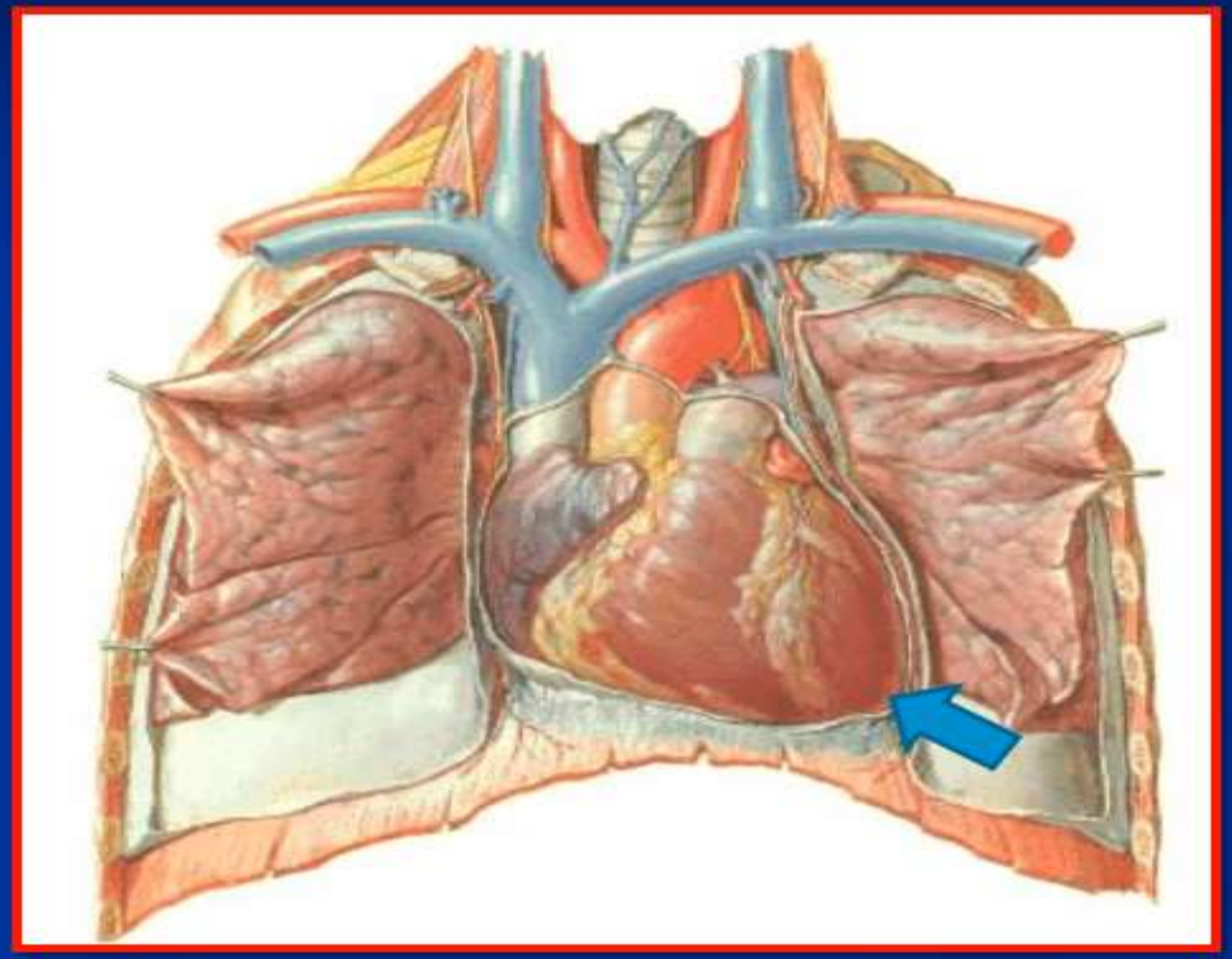
Anterior ribs slope downward. Posterior ribs are more horizontal.

Diaphragm & Angles



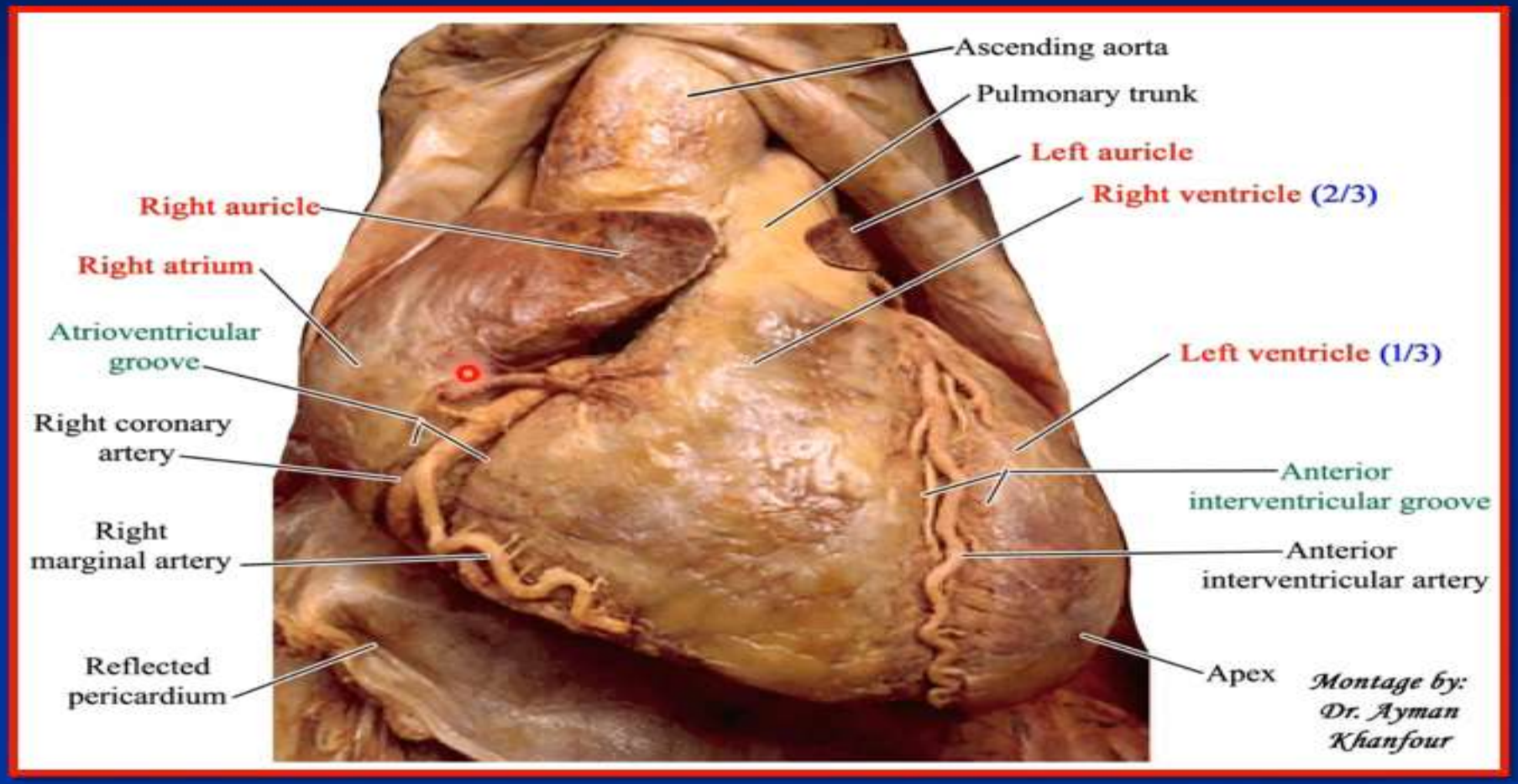
Costophrenic angles should be sharp. Right hemidiaphragm usually higher than left. Gastric bubble under left.

Anatomical Site

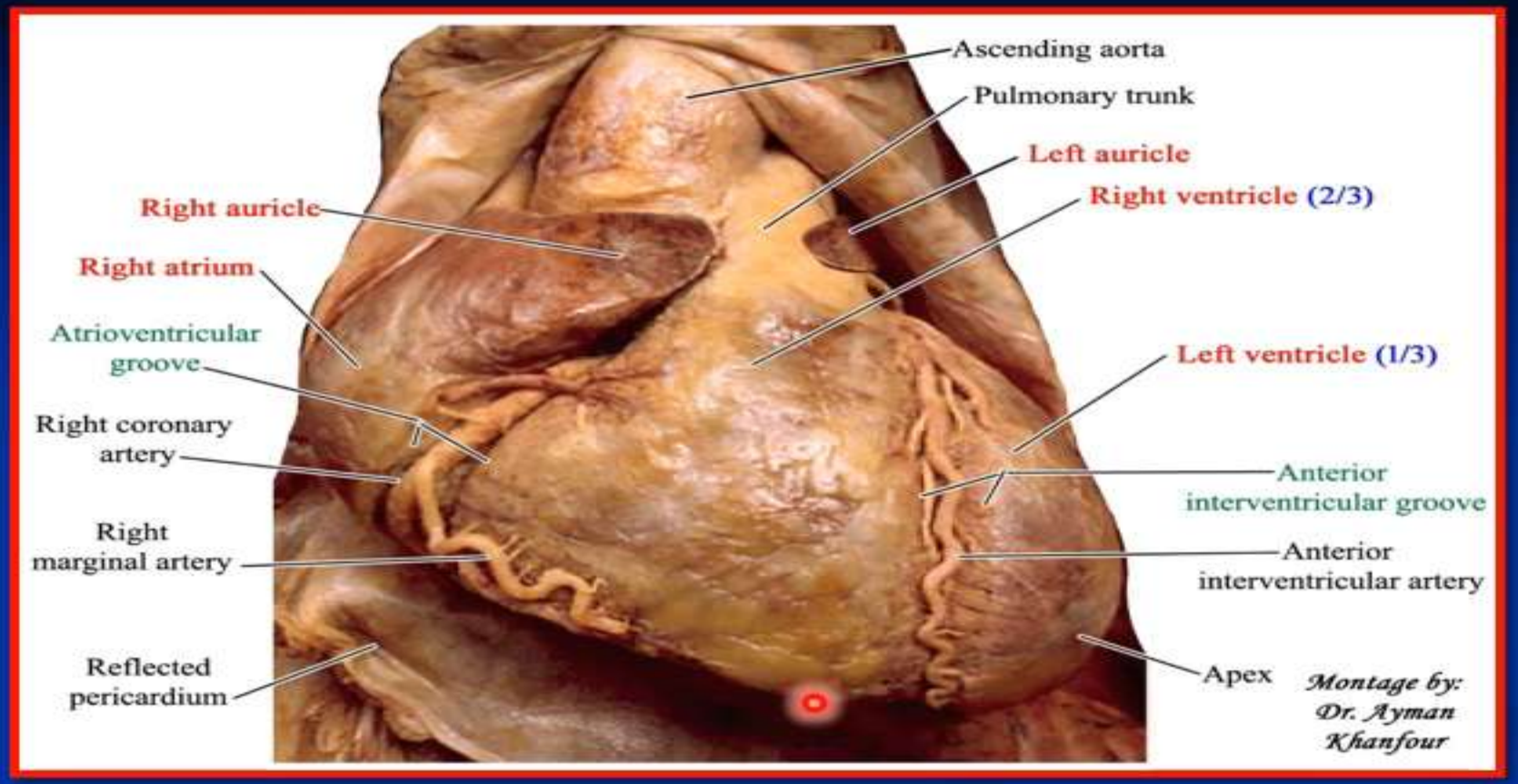


- It lies between both lungs (middle mediastinum) with slight deviation of its apex toward the left side.

Parts



- The heart is formed of FOUR chambers: two atria and two ventricles.
- Both atria and ventricles are separated from each other on the outer surface of the heart by a *circular groove called atrioventricular groove (coronary sulcus).*



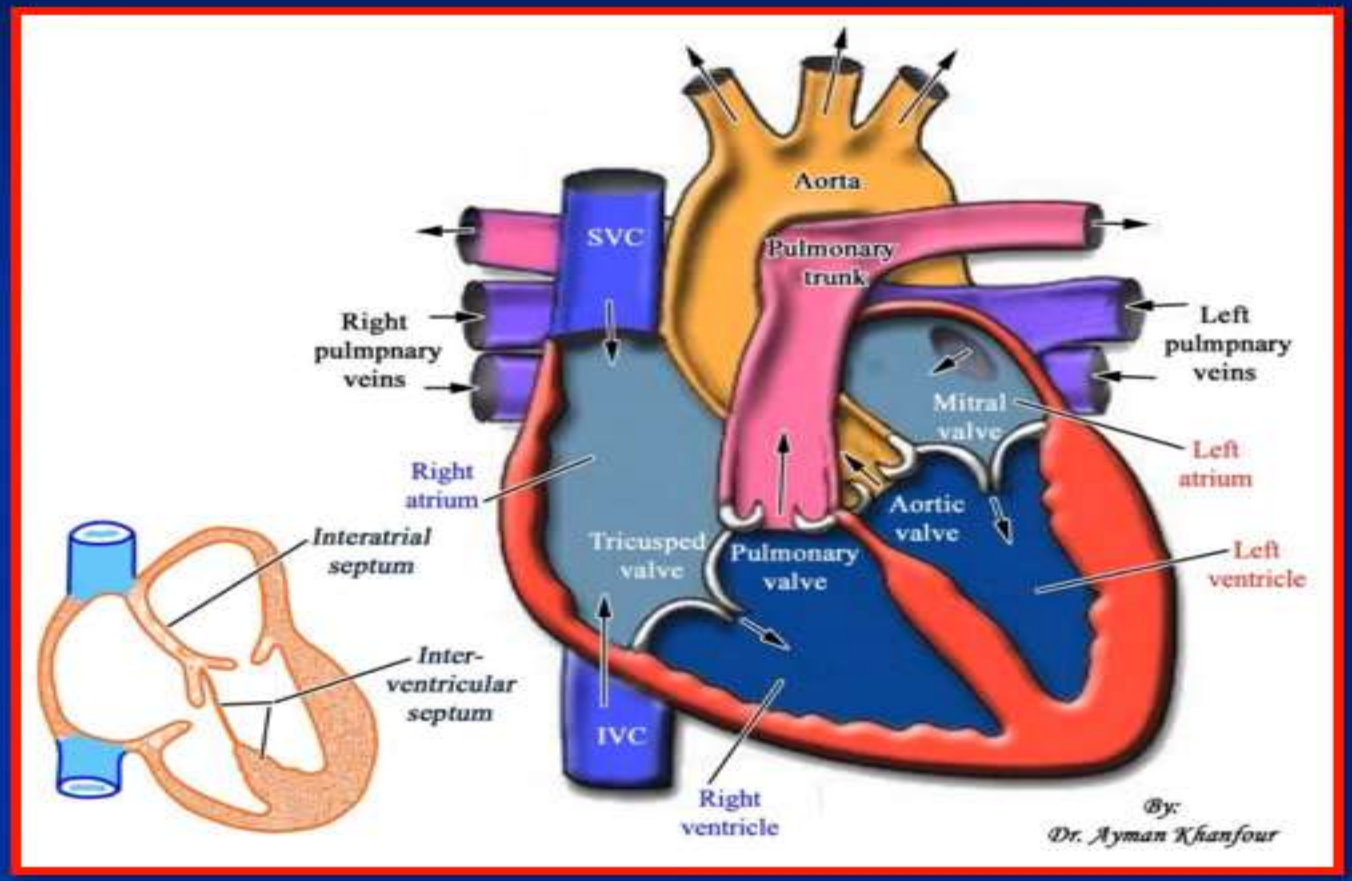
I. Atria

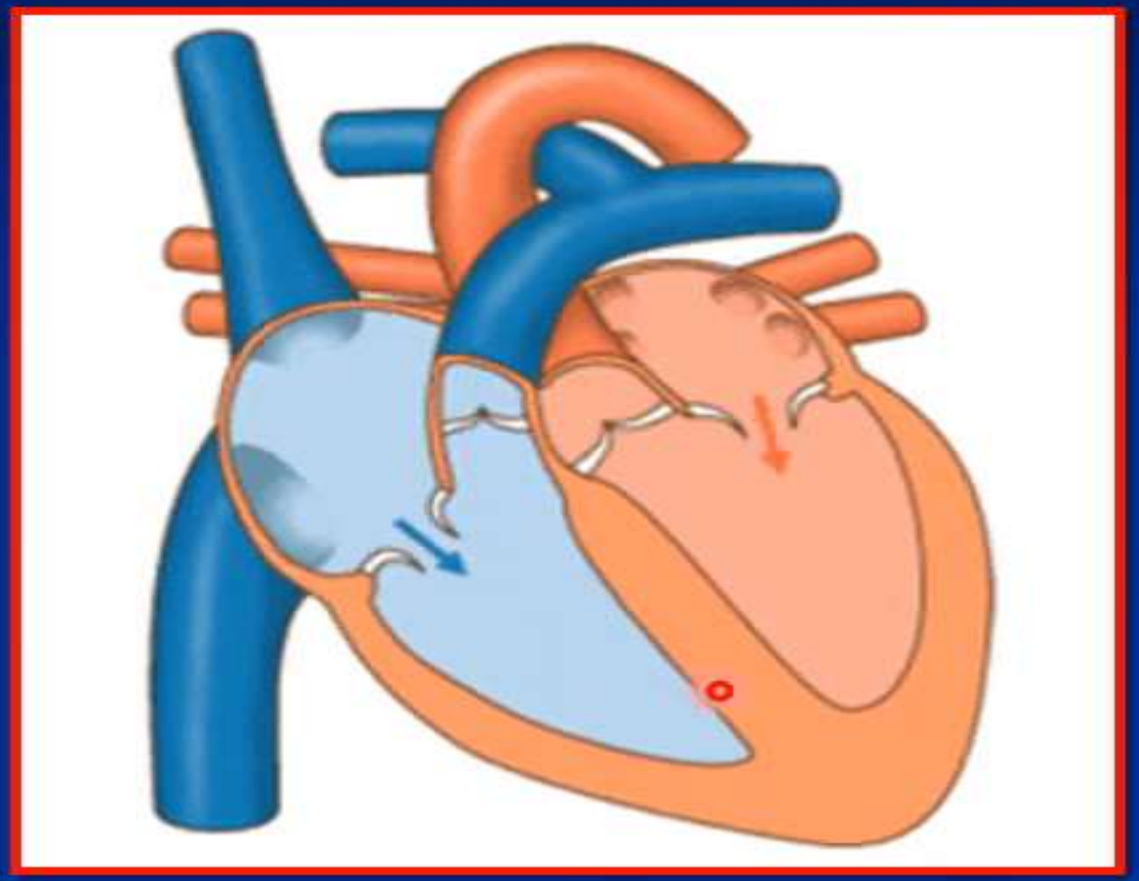
- **Each atrium has a triangular outgrowth called auricle.**
- **The two atria are separated from each other from inside by interatrial septum.**

II. Ventricles

- **The two ventricles are separated from each other from inside by the interventricular septum and from outside by the *anterior and posterior interventricular grooves*.**

❖ How blood circulates inside the heart?



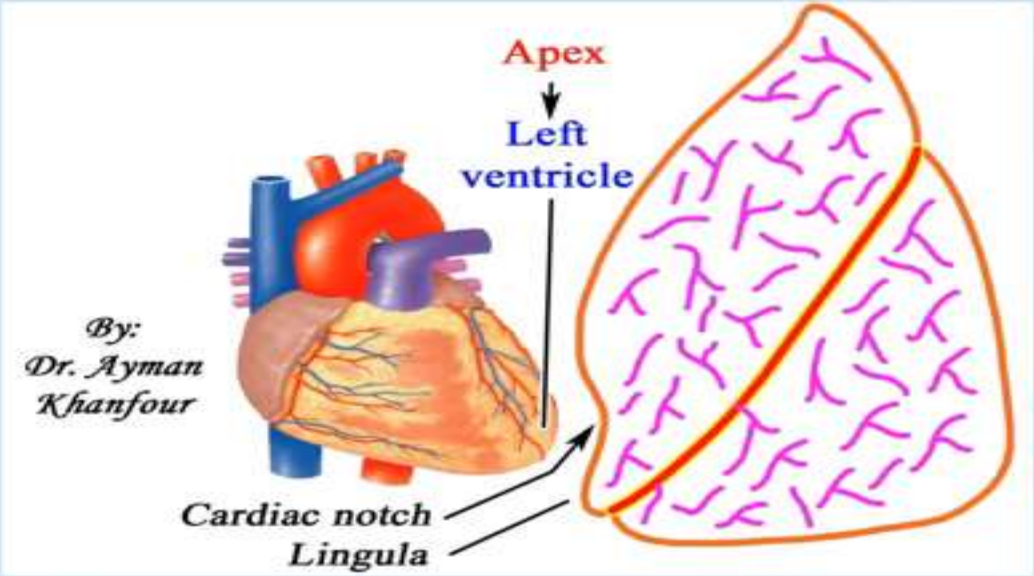
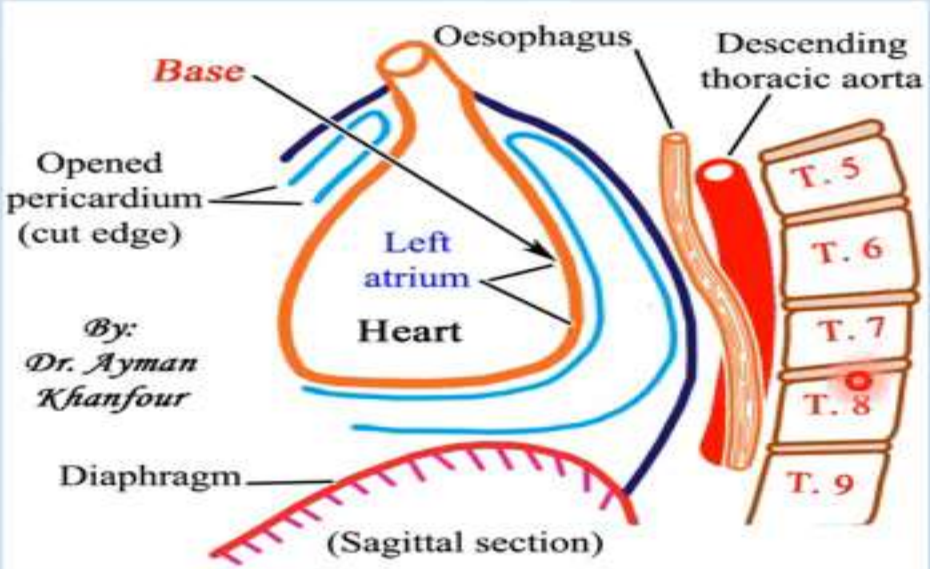


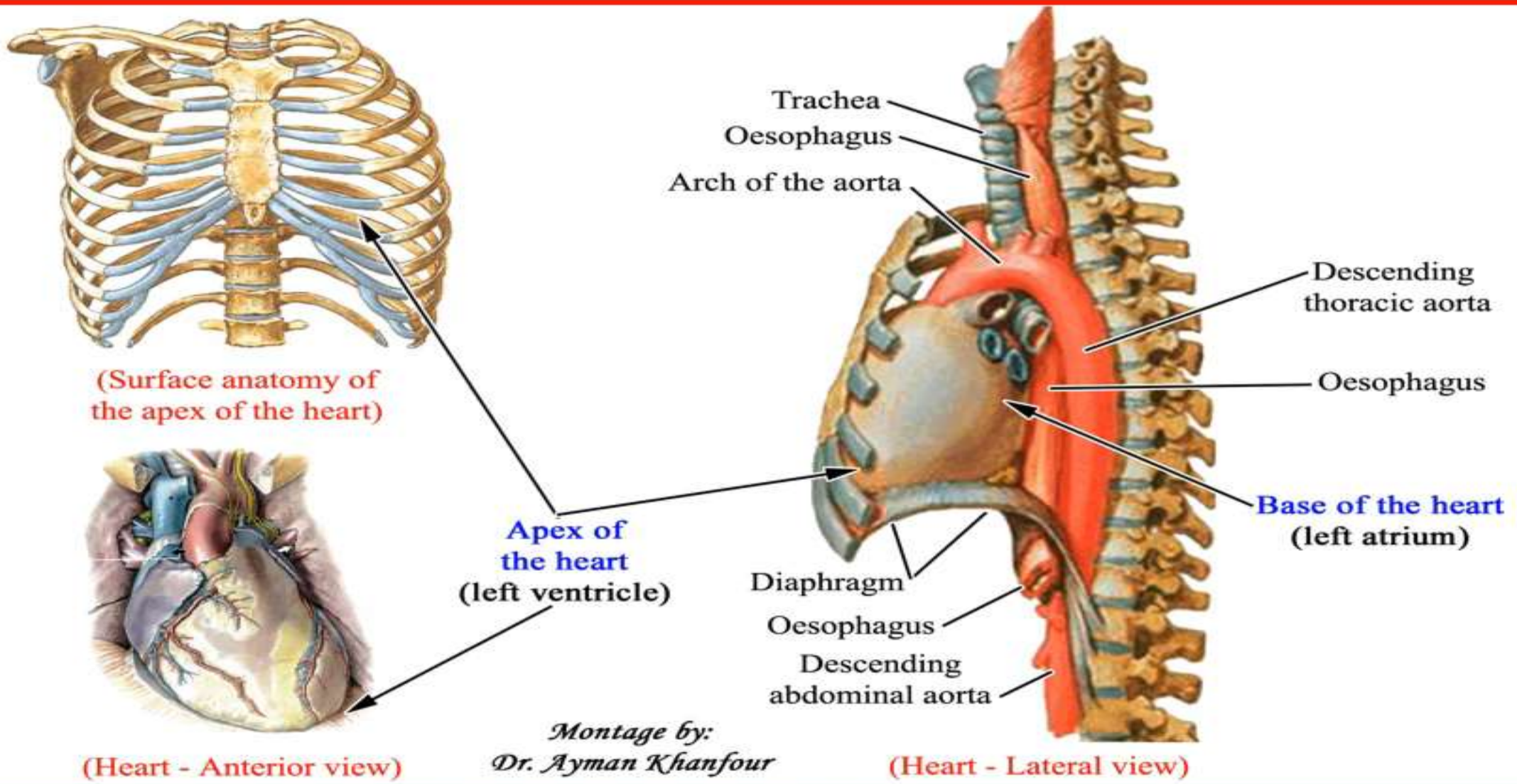
(Champers, valves and great vessels of the heart)



Apex and Base

Apex and Base

	Apex	Base
		
Chamber forming	**- Left ventricle.**	**- Left atrium mainly (and small part of the right atrium).**
Site (surface anatomy)	**- Left fifth intercostal space 3 ½ inches (9 cm) from the midline).**	**- Middle four thoracic vertebrae.**



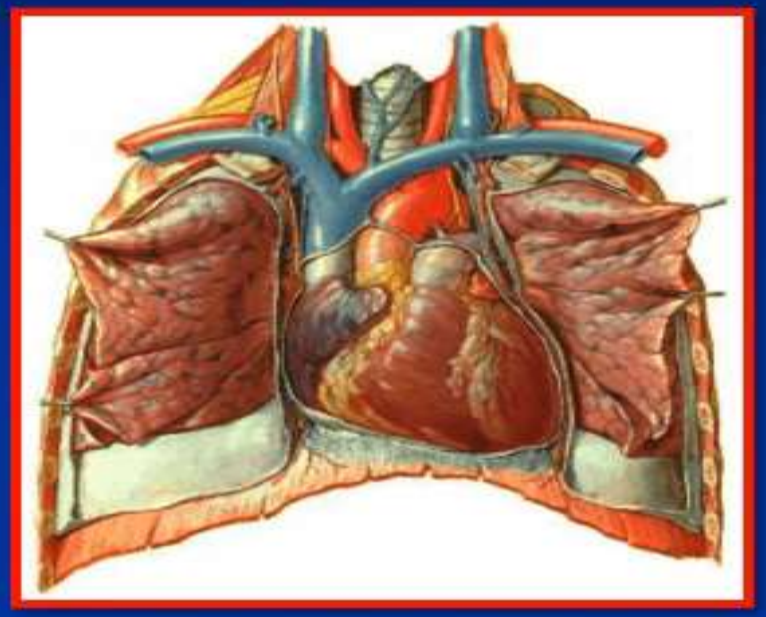
(Apex and base of the heart of the heart)

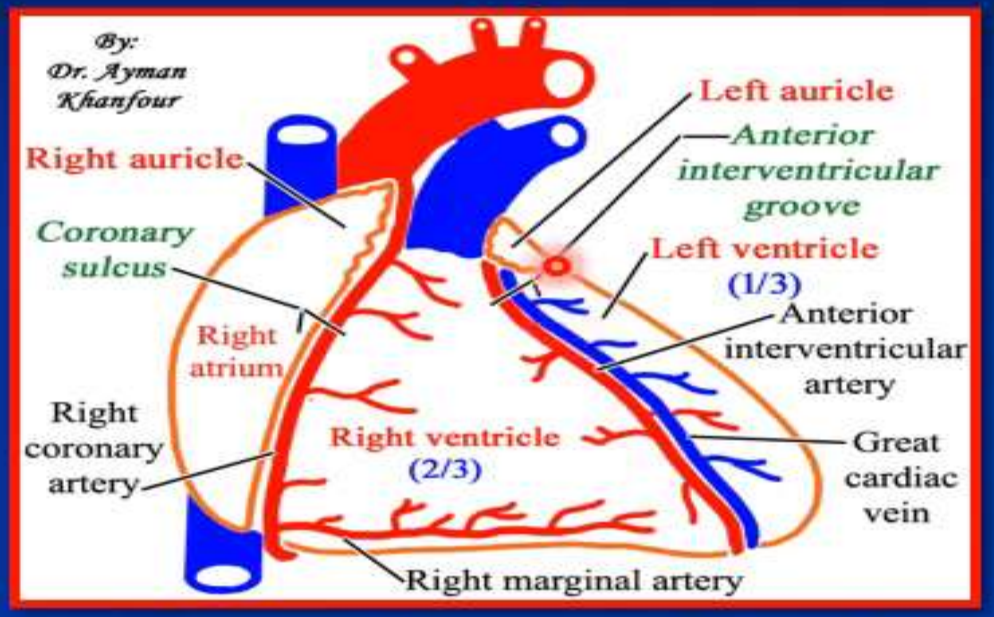


Surfaces

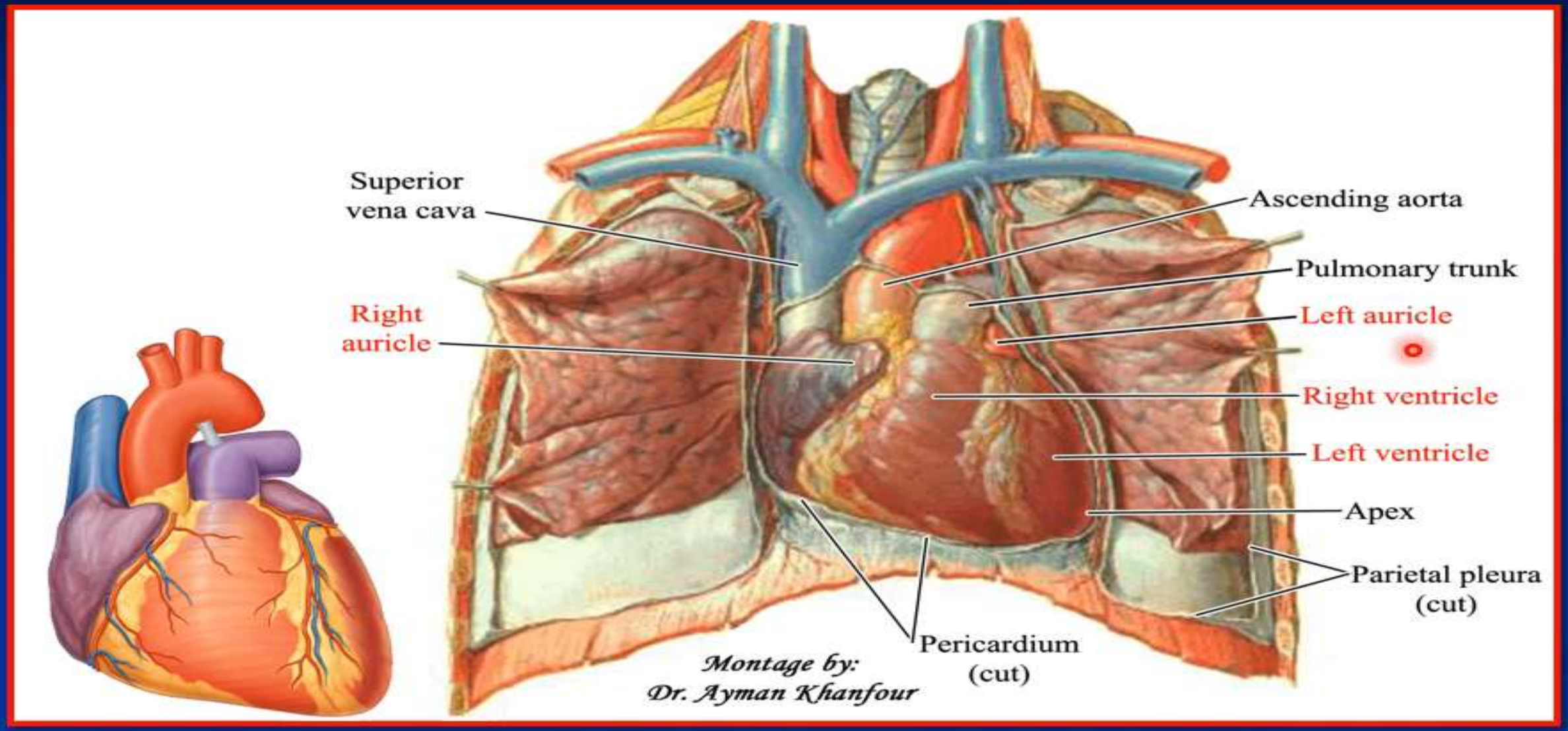
Surfaces

I. Sterno-costal Surface



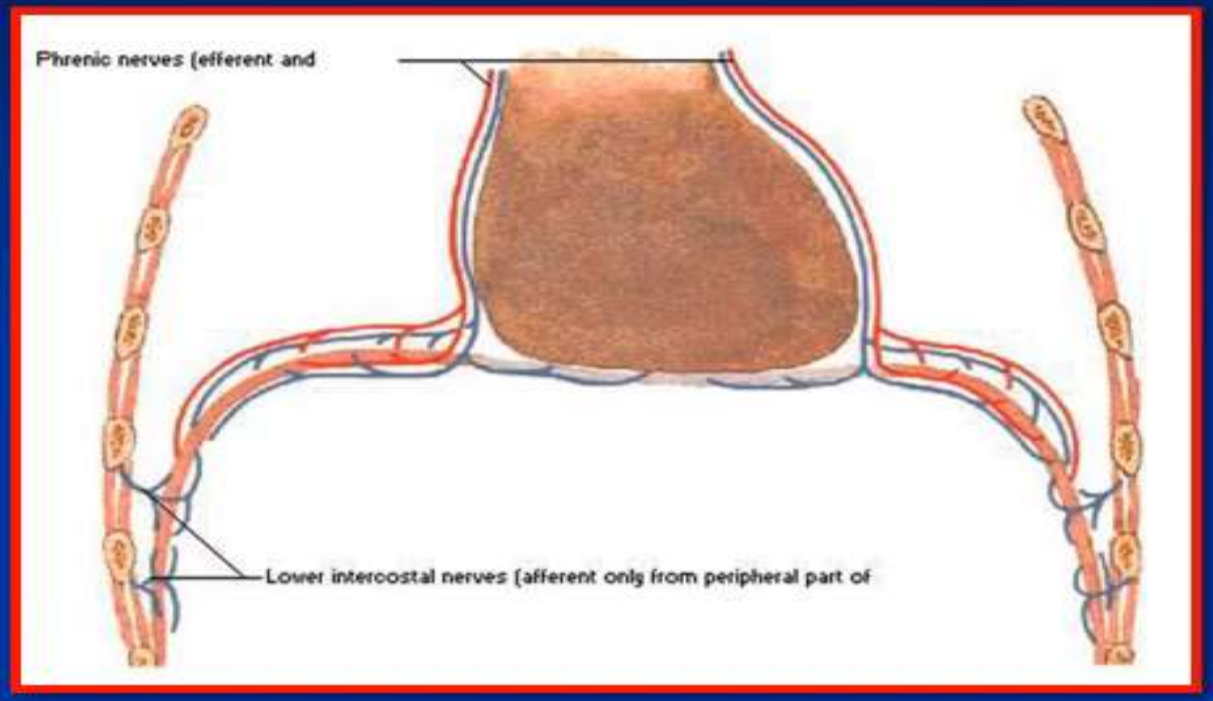


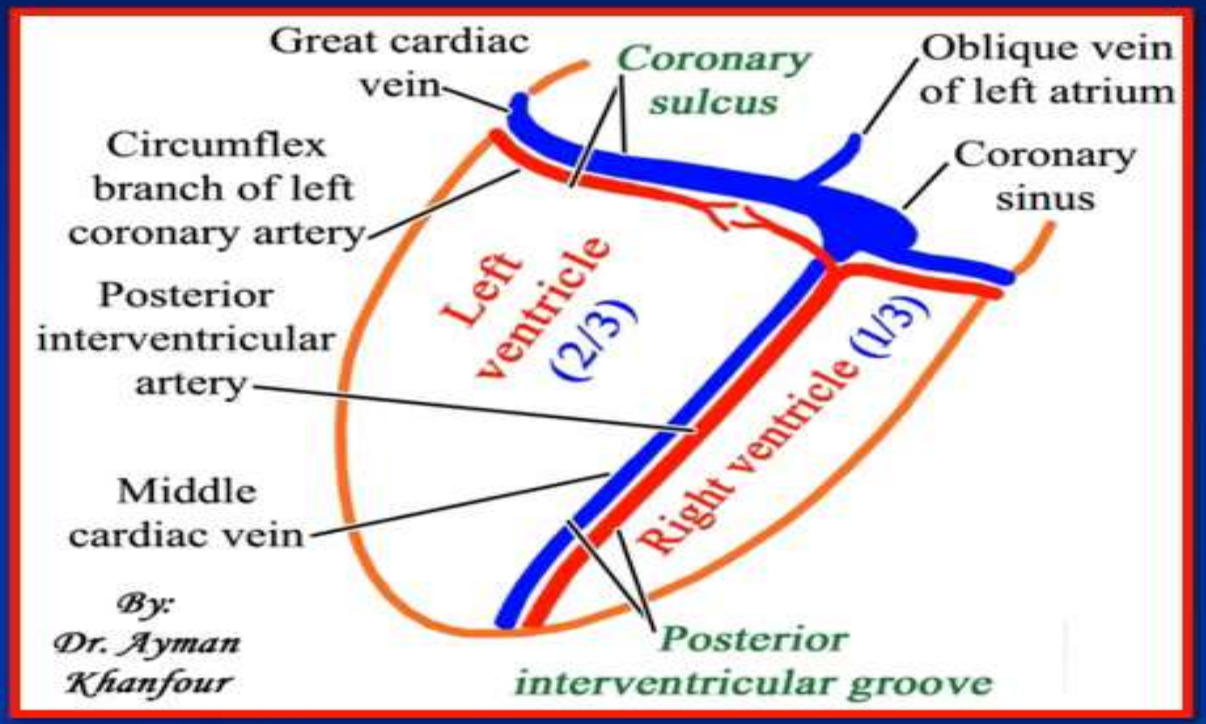
- **The sternocostal surface is formed of: (four chambers)**
  - **A. Ventricular part, consists of:**
    1. **Right ventricle (2/3).**
    2. **Left ventricle (1/3).**
  - **B. Atrial part, consists of:**
    1. **Right atrium (mainly).**
    2. **Left auricle (forms small part).**



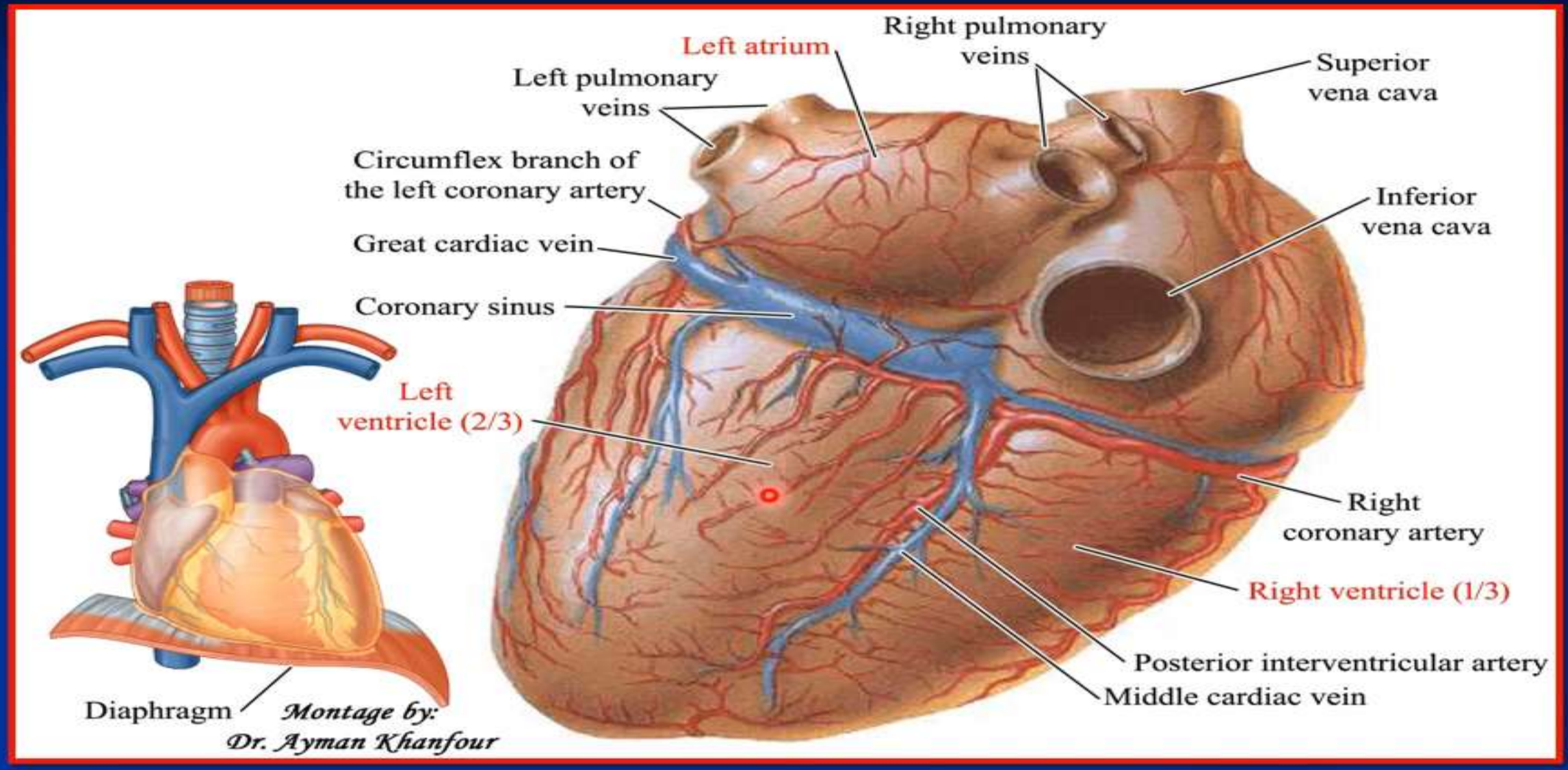
(Sternocostal surface of the heart)

II. Diaphragmatic Surface





- **The diaphragmatic surface is formed of:**
  1. **Left ventricle (2/3).**
  2. **Right ventricle (1/3).**

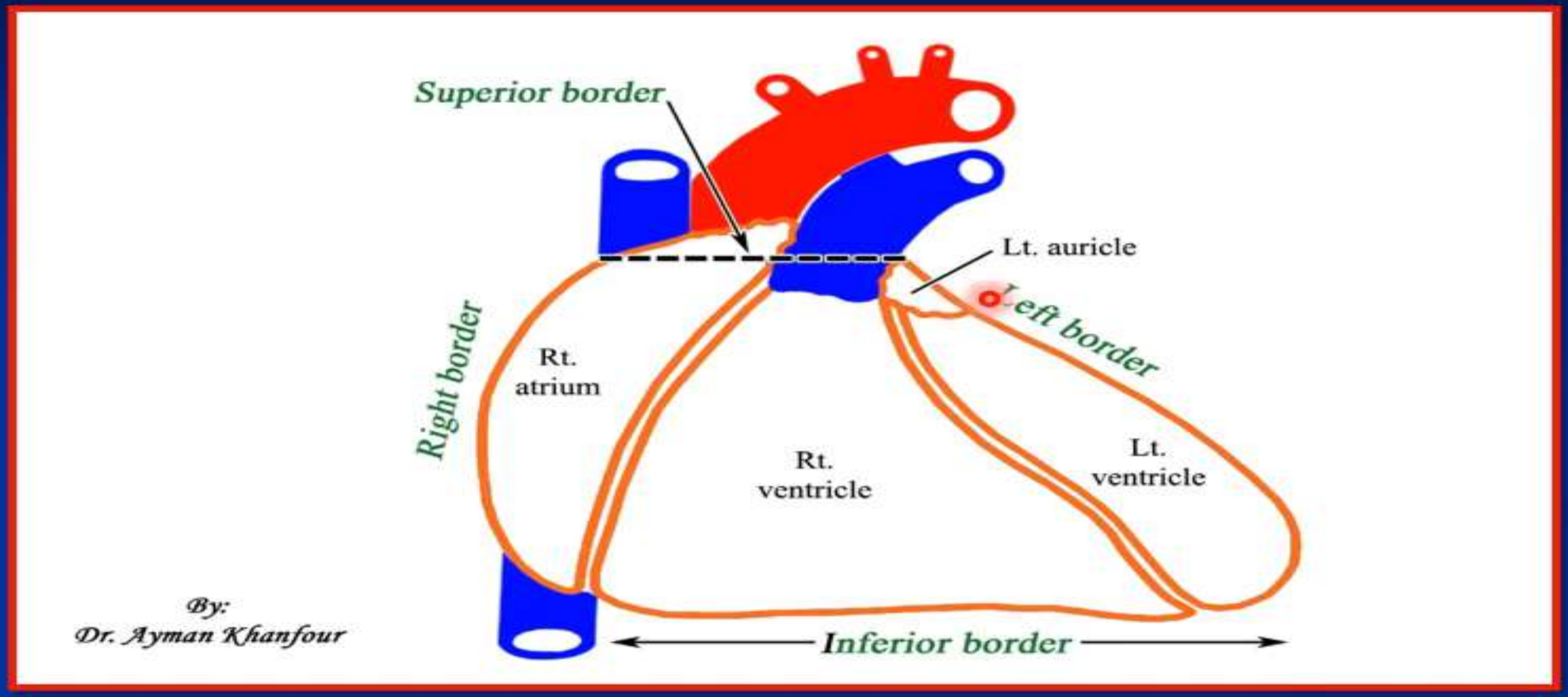


(Diaphragmatic surface of the heart)

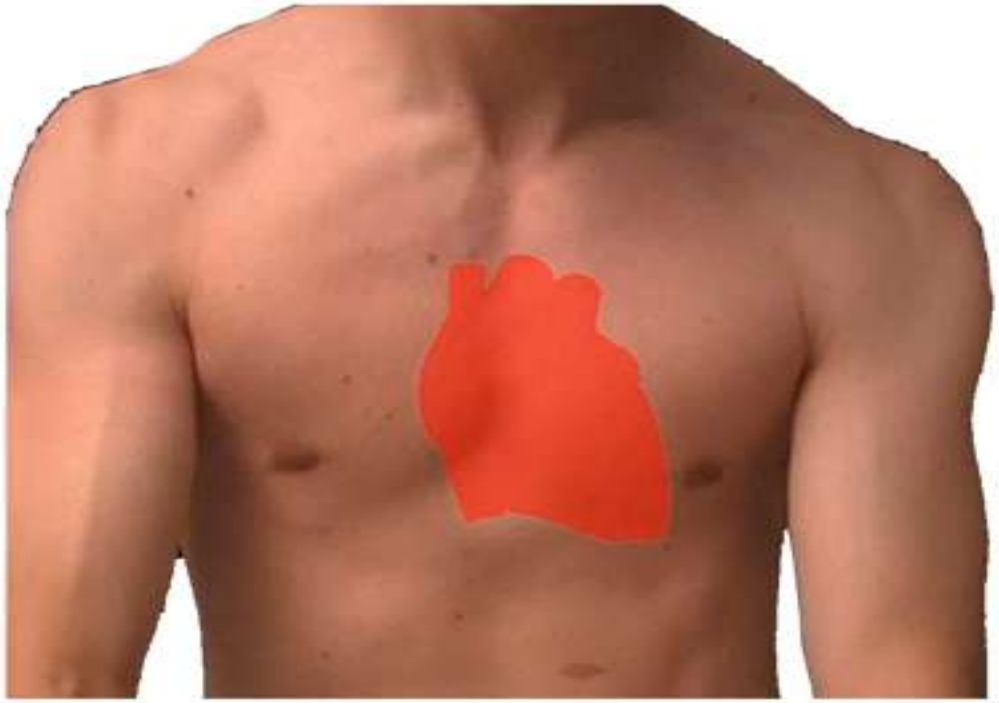


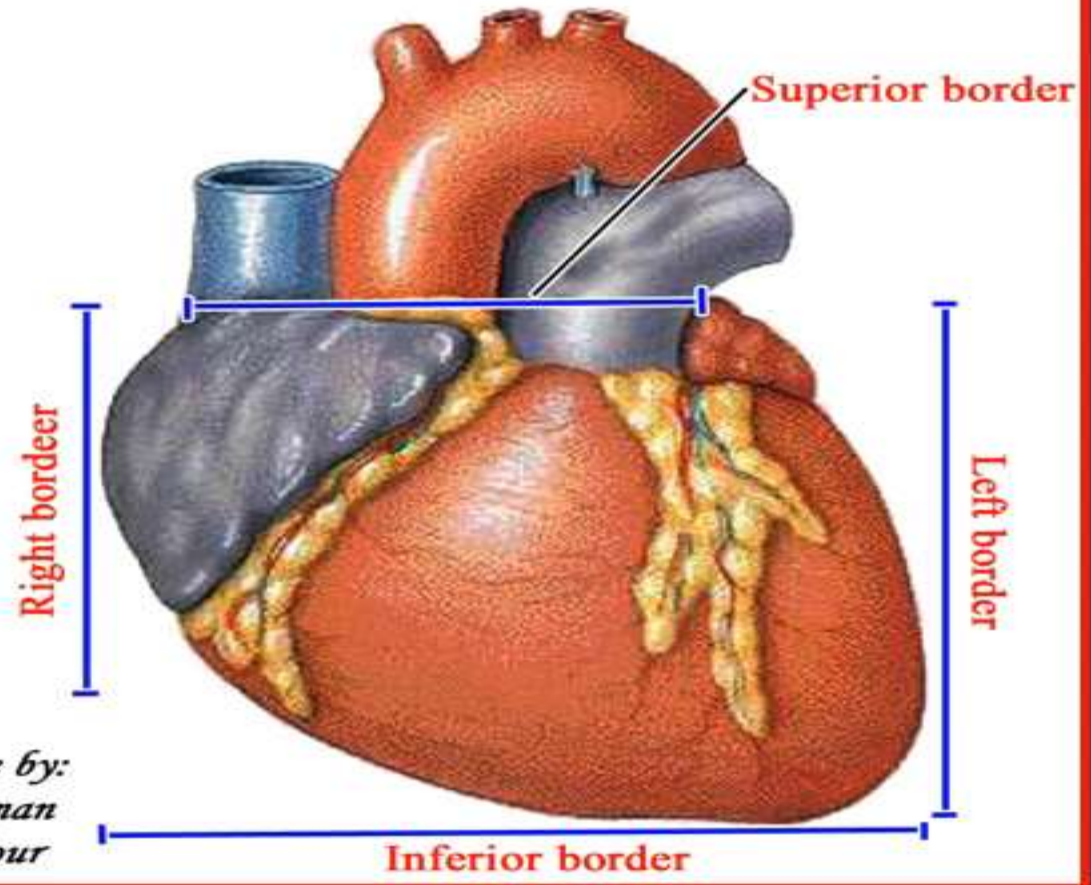
Borders

Borders



1. Superior border: Formed by both atria (mainly the Lt.).
2. Inferior border: Formed by both ventricles (mainly the Rt.).
3. Right border: Formed by the right atrium.
4. Left border: Formed mainly by the left ventricle and its upper part is formed by the left auricle.



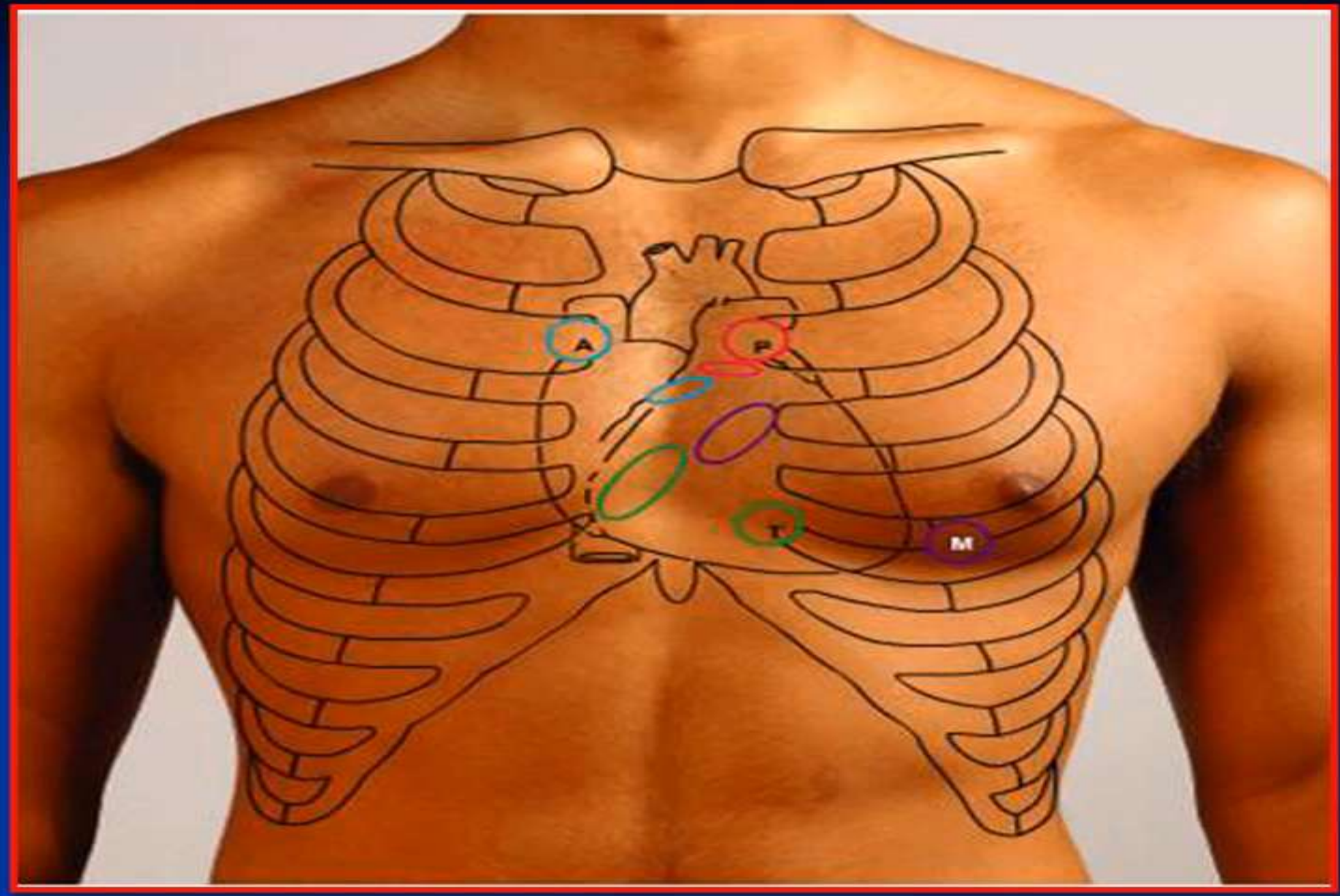


Montage by:
Dr. Ayman
Khanfour

(Diaphragmatic surface of the heart)

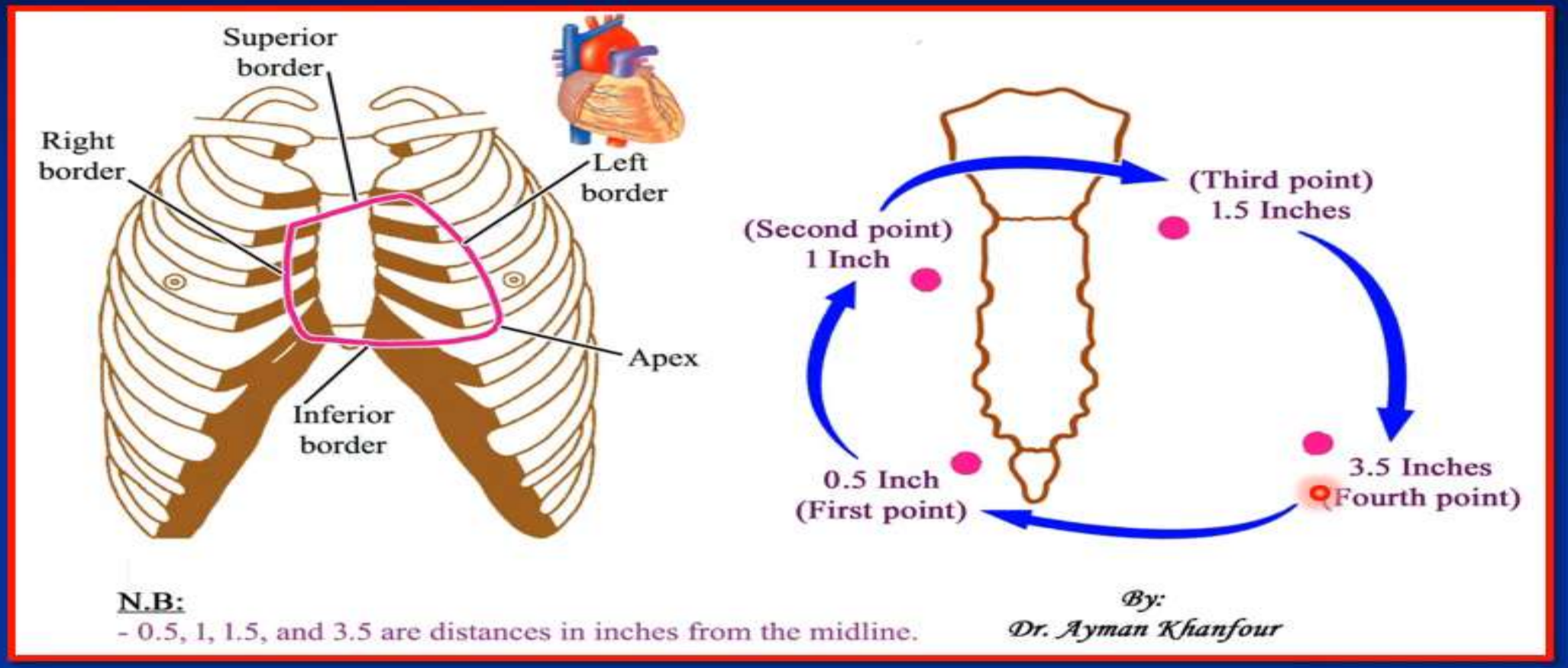


Surface Anatomy of the Heart



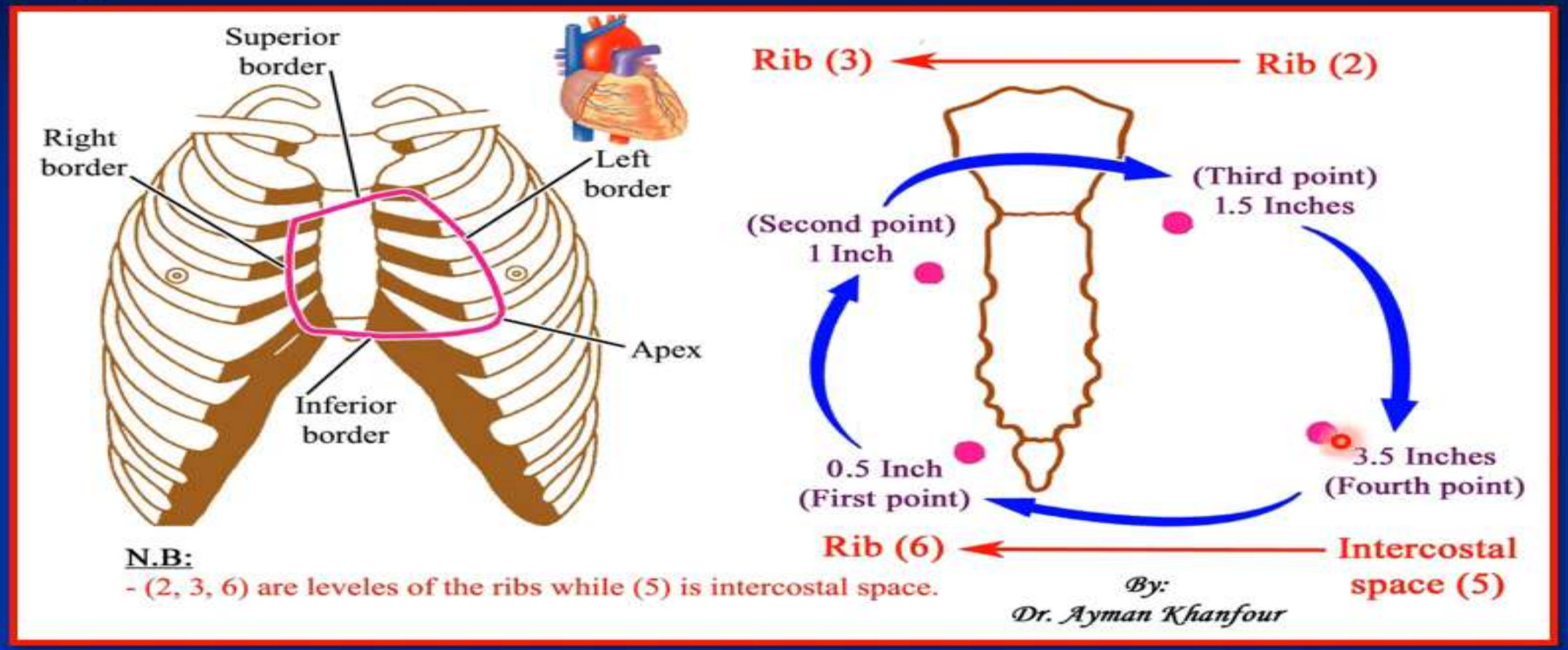
(Surface anatomy of the heart)

➢ Step one: (Measures on the horizontal plane)



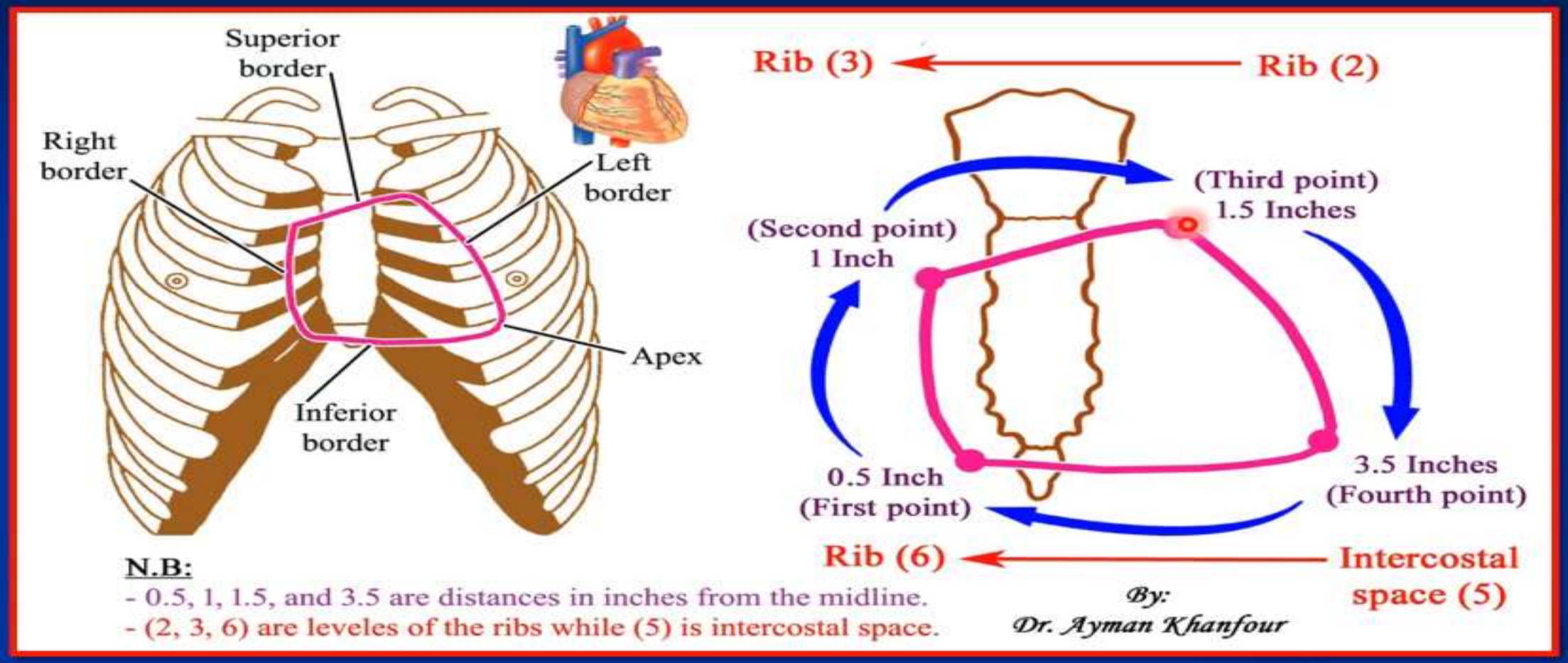
* **First point: ½ inch** *from the midline.*
* **Second point: 1 inch** *from the midline.*
* **Third point: 1 ½ inches** *from the midline.*
* **Fourth point: 3 ½ inches** *from the midline.*

➢ Step two: (Measures on the vertical plane)



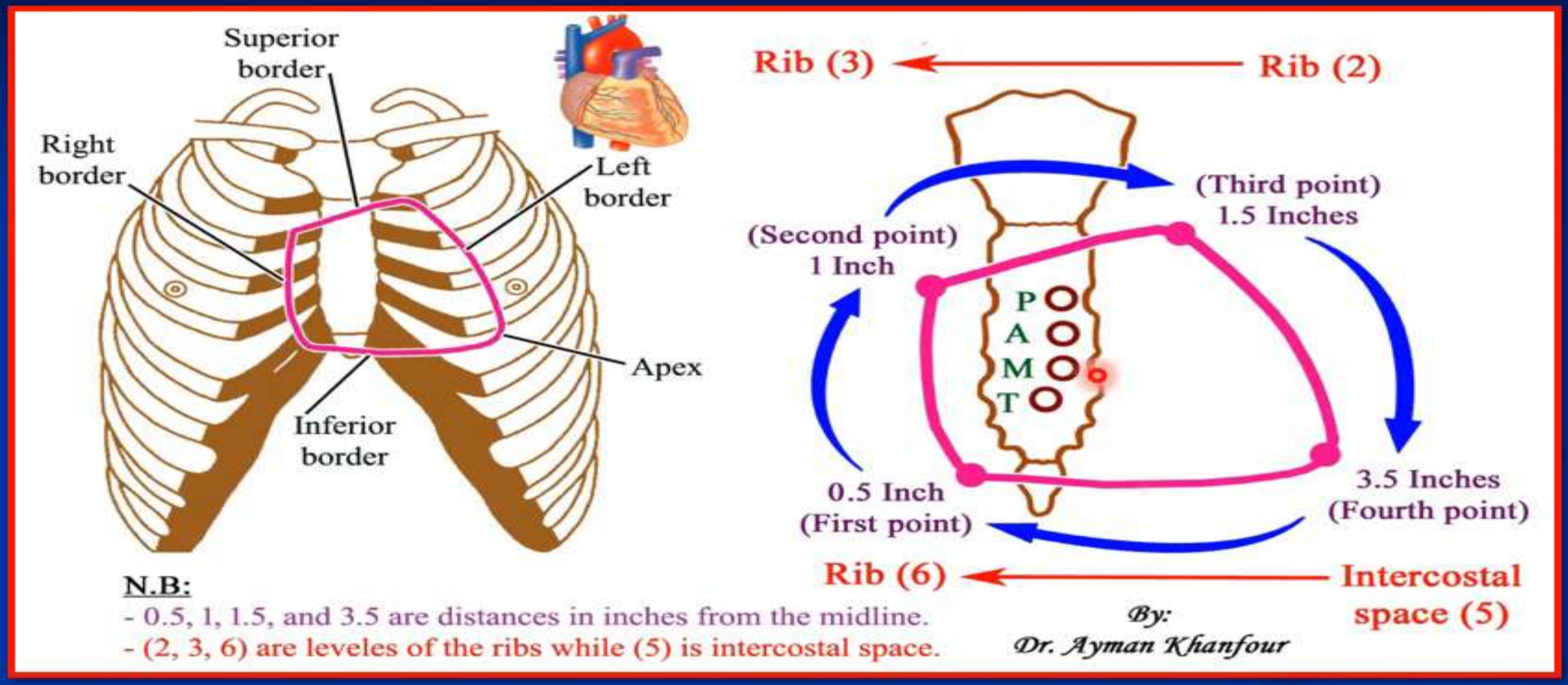
* **First point:** At the **"right sixth"** costal cartilage.
* **Second point:** **At** the **"right third"** costal cartilage.
* **Third point:** **At** the **"left second"** costal cartilage.
* **Fourth point:** **At** the **"left fifth"** intercostal space.

➢ Step three: (Merge the two sets of numbers)



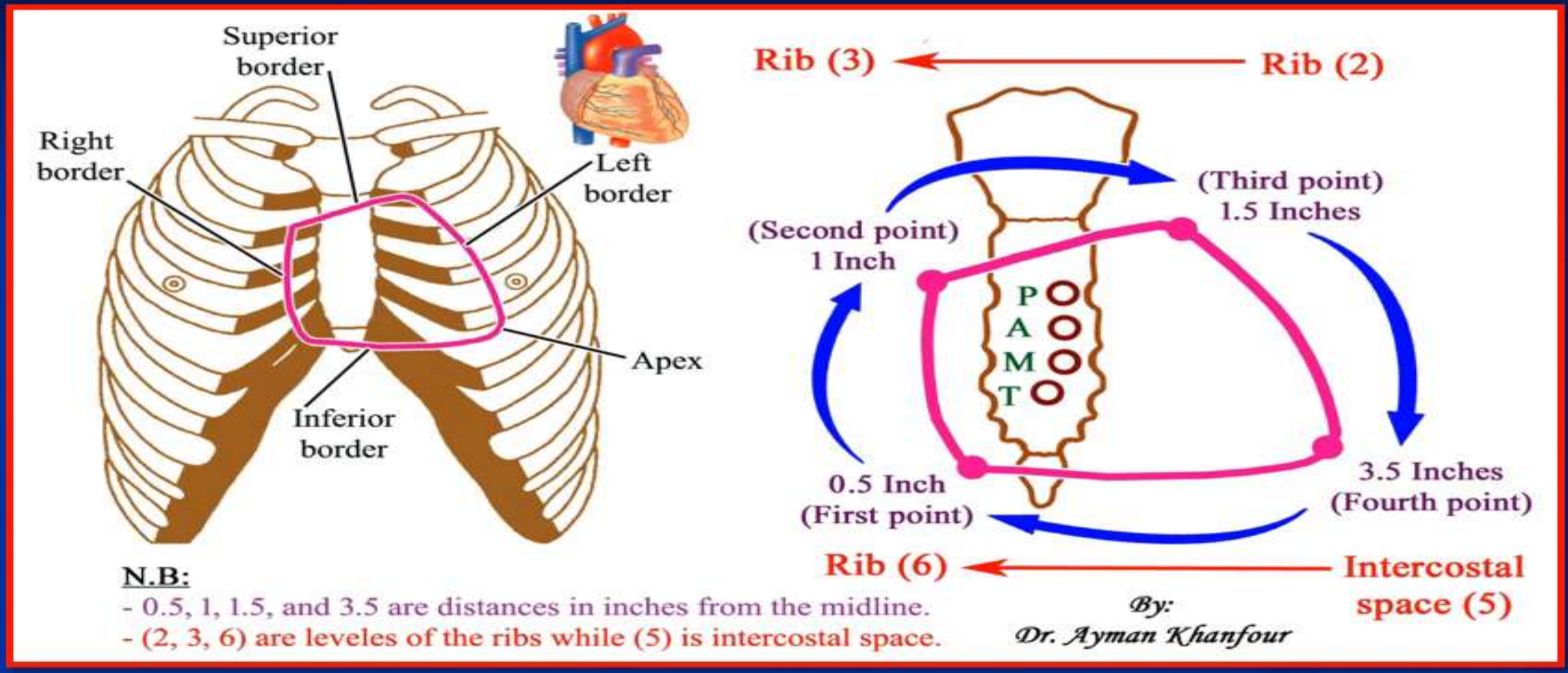
* **First point: ½ inch** *from the midline* at the **"right sixth"** costal cartilage.
* **Second point: 1 inch** *from the midline* at the **"right third"** costal cartilage.
* **Third point: 1 ½ inches** *from the midline* at **"left second"** costal cartilage.
* **Fourth point: 3 ½ inches** *from the midline* at the **"left fifth"** intercostal space.

➢ Step four: (Valves of the heart)

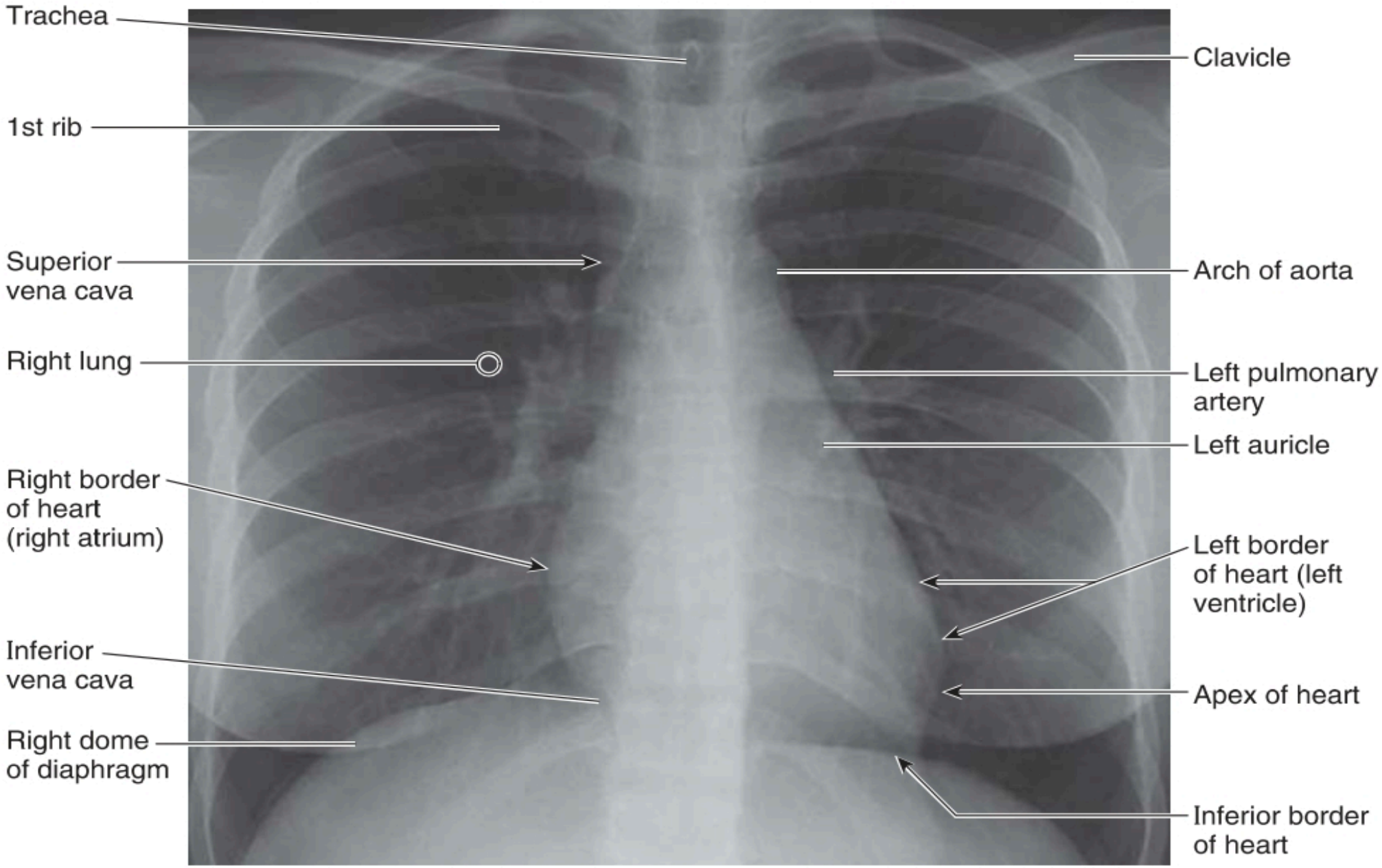


* 3 = Third rib.
* 3 ½ = Third intercostal space.
* 4 = Fourth rib.
* 4 ½ = Fourth intercostal space.

➢ Step four: (Valves of the heart)



1. **P**ulmonary valve: It lies behind the left border of the sternum opposite the **3rd** *costal cartilage.*
2. **A**ortic valve: It lies behind the left border of the sternum opposite the **3rd** *intercostal space.*
3. **M**itral valve: It lies behind the left border of the sternum opposite the **4th** *costal cartilage.*
4. **T**ricusped valve: It lies at the midline opposite the **4th** *intercostal space.*
5. The ***coronary sulcus*** corresponds to a line between the *first* and the *third points.*



(A) Anteroposterior view

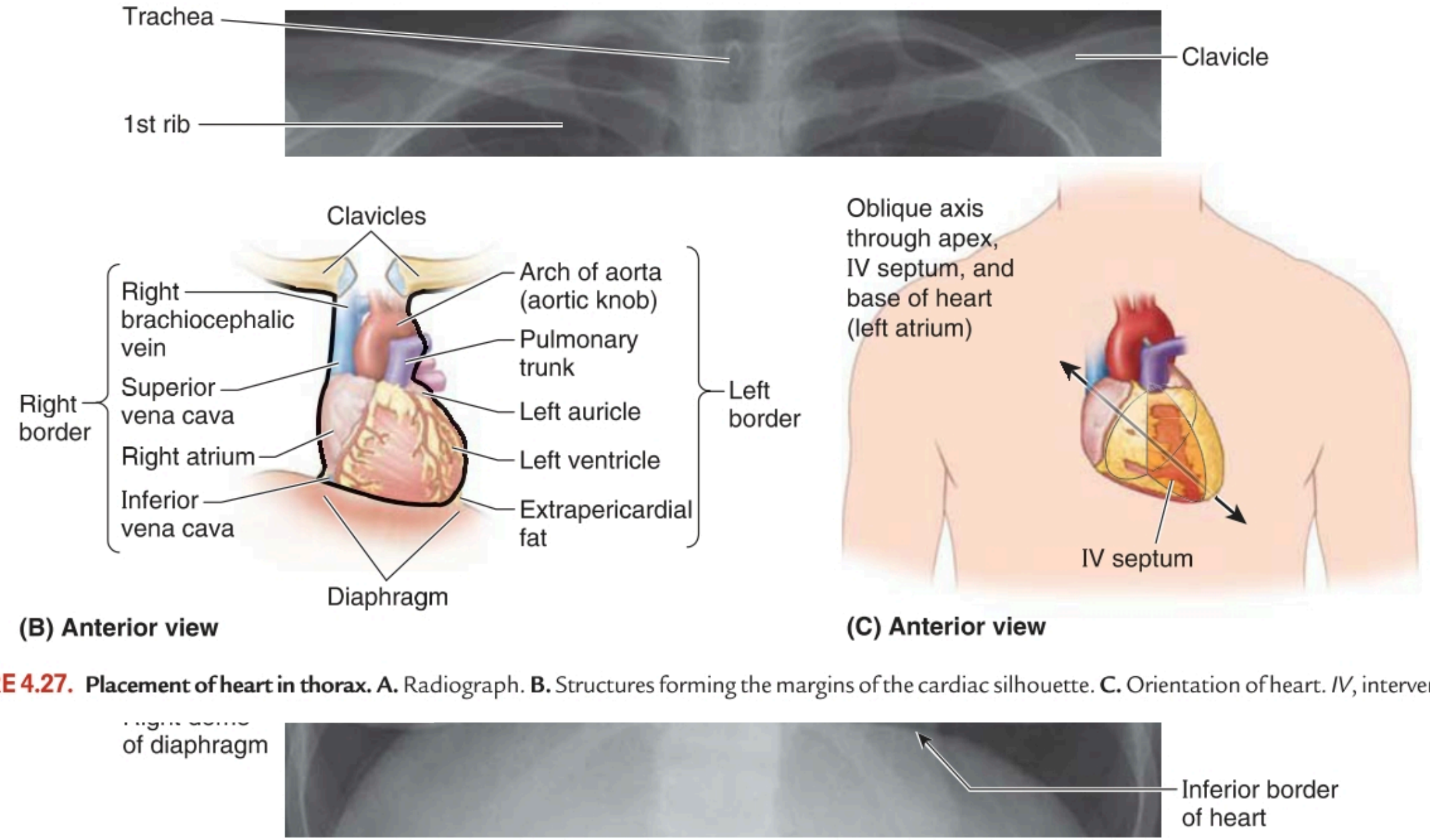
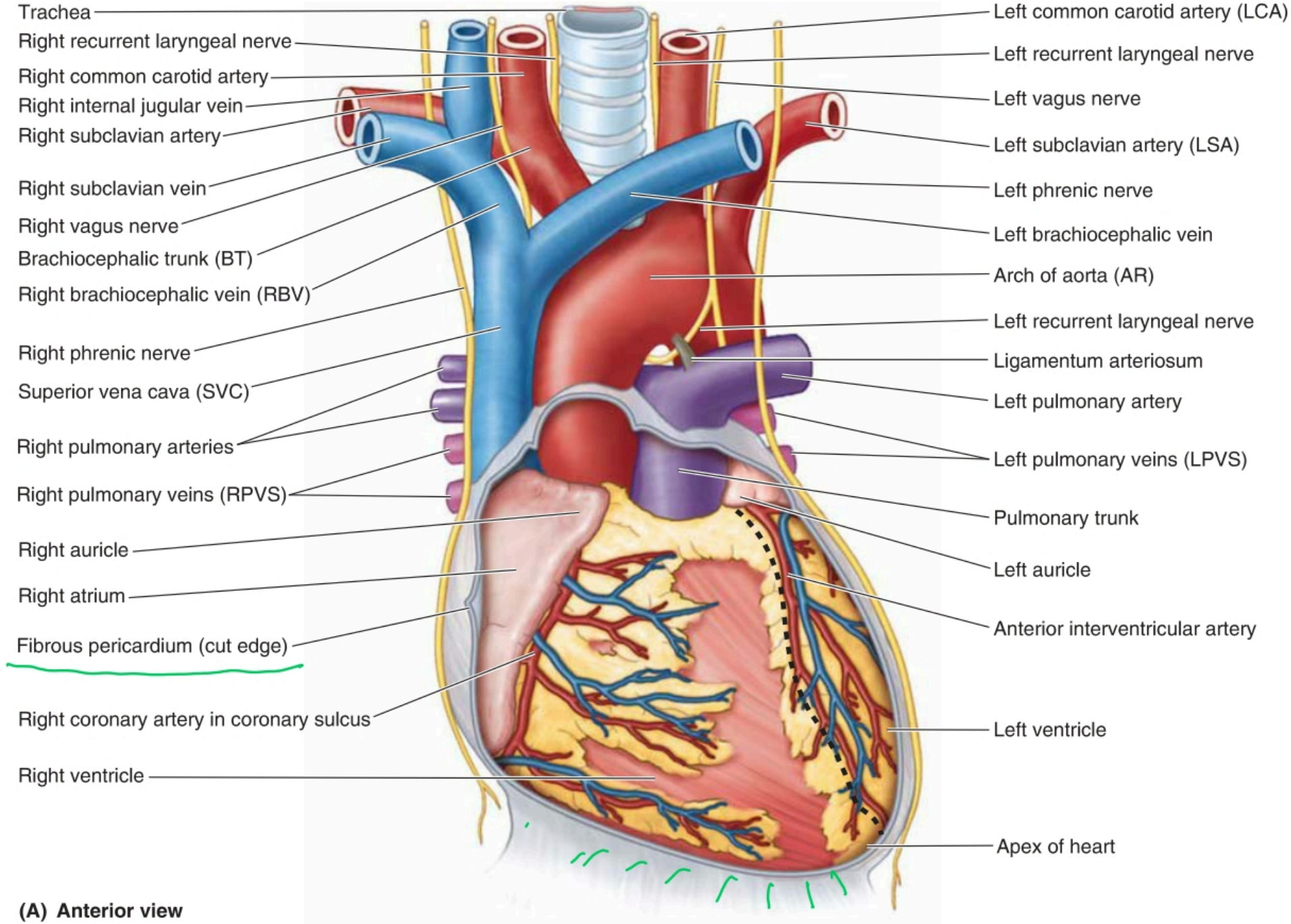
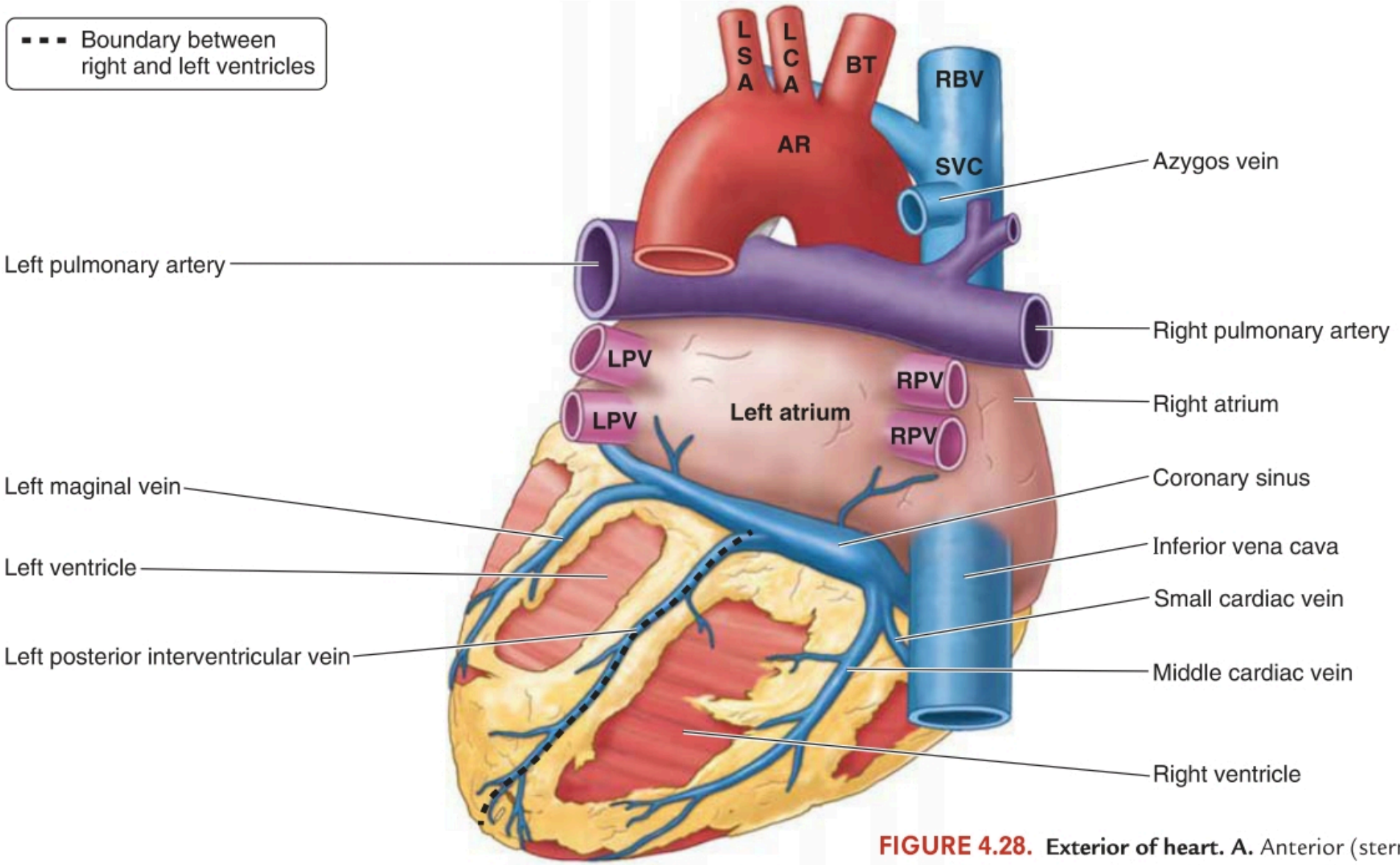


FIGURE 4.27. Placement of heart in thorax. A. Radiograph. **B.** Structures forming the margins of the cardiac silhouette. **C.** Orientation of heart. *IV*, interventricular.

(A) Anteroposterior view



(A) Anterior view



(B) Postero-inferior view

FIGURE 4.28. Exterior of heart. A. Anterior (sternocostal) and left pulmonary and right pulmonary surfaces. **B.** Diaphragmatic (inferior) surface and base (posterior aspect).

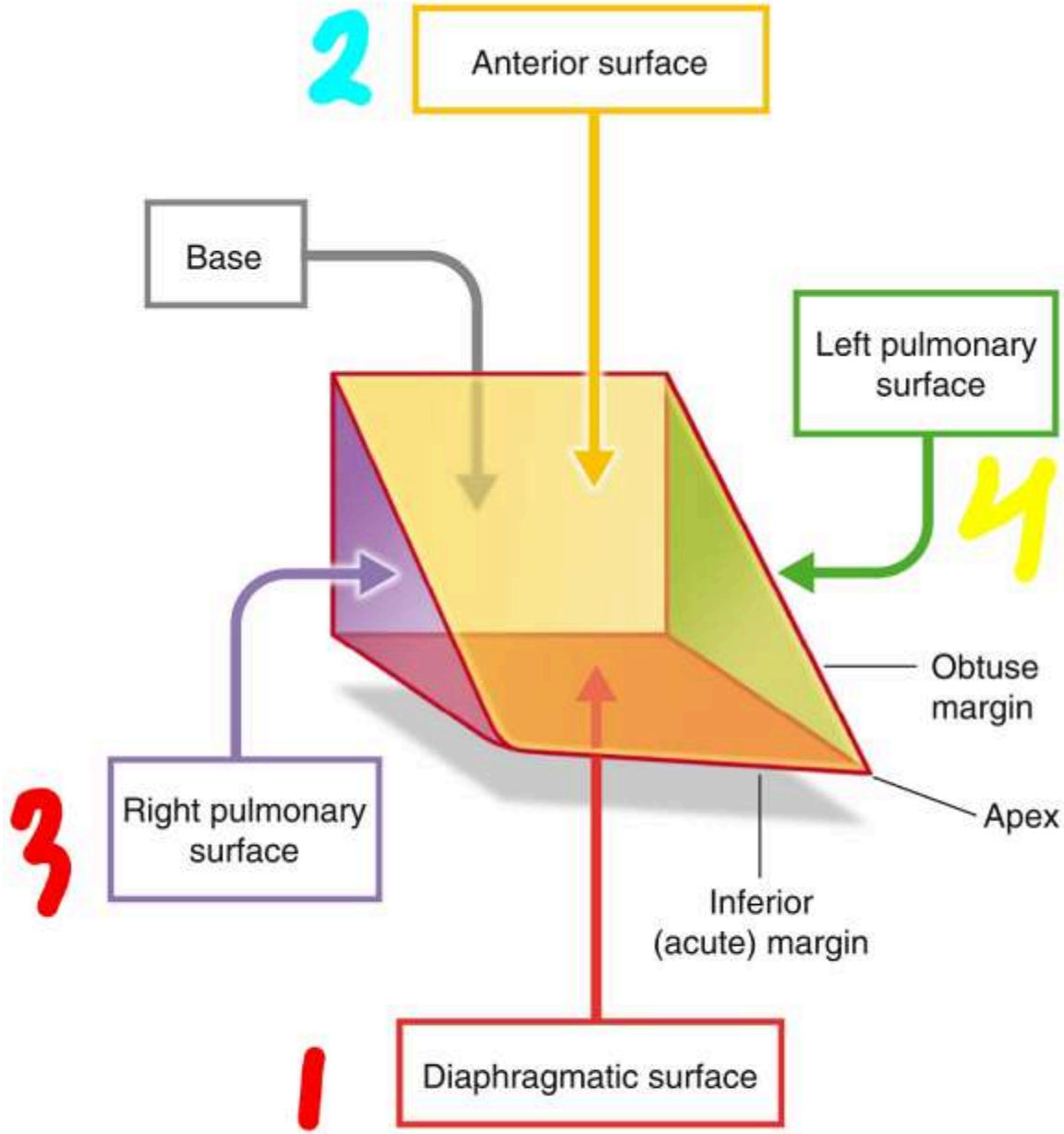


Fig. 3.65 Schematic Illustration of the Heart Showing Orientation, Surfaces, and Margins.

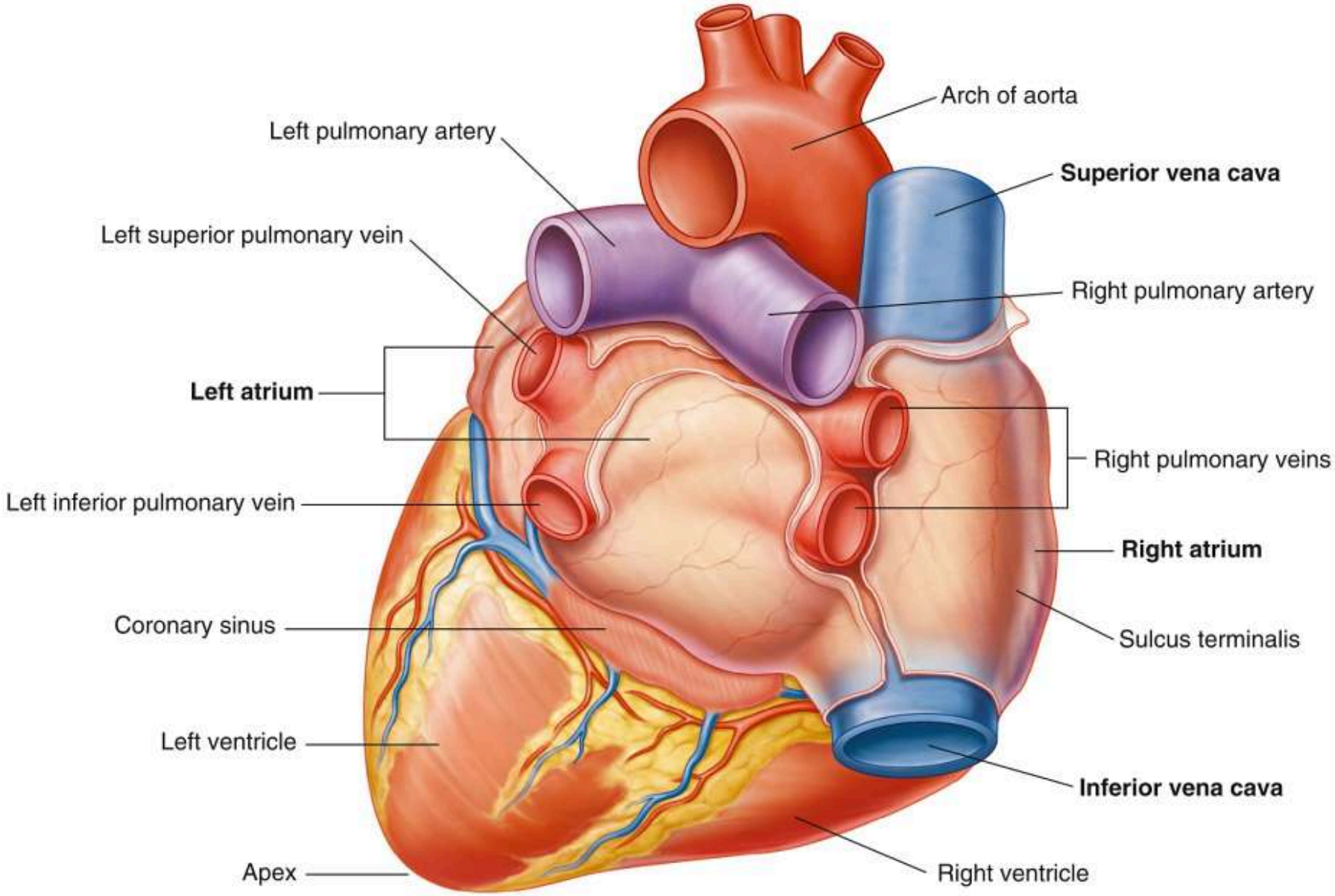


Fig. 3.66 Base of the Heart.

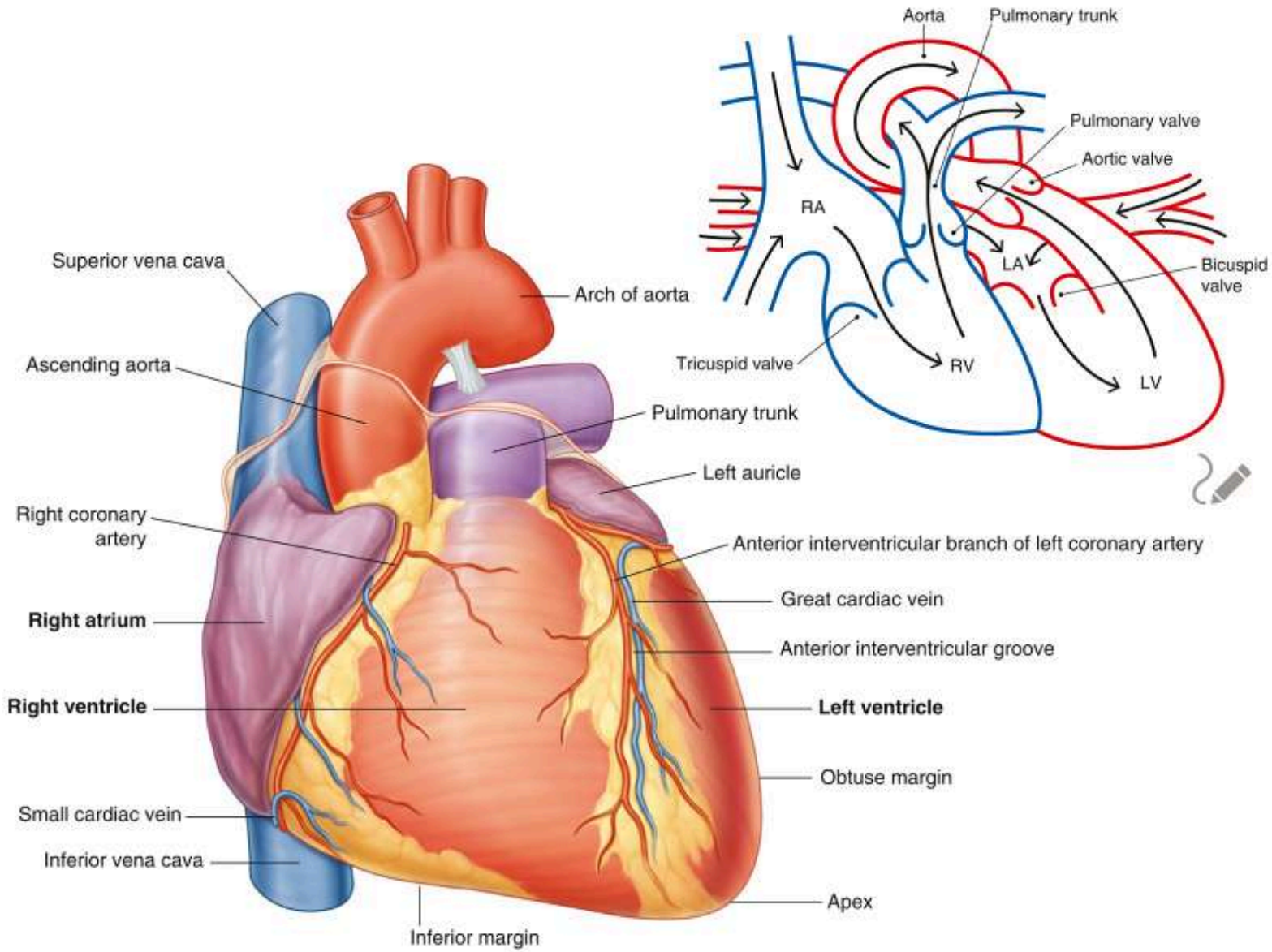


Fig. 3.67 Anterior Surface of the Heart. *LA*, left atrium; *LV*, left ventricle; *RA*, right atrium; *RV*, right ventricle.

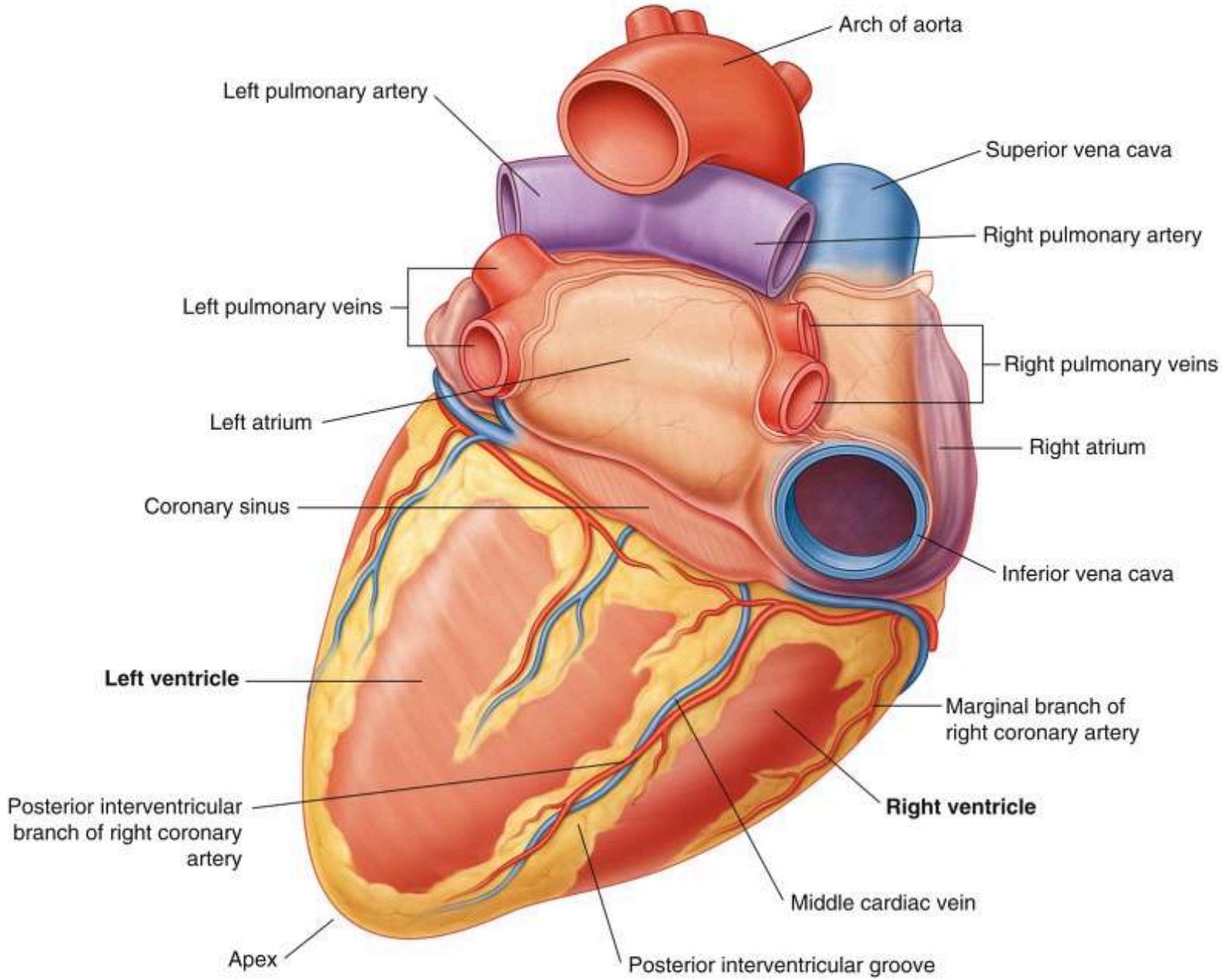


Fig. 3.68 Diaphragmatic Surface of the Heart.

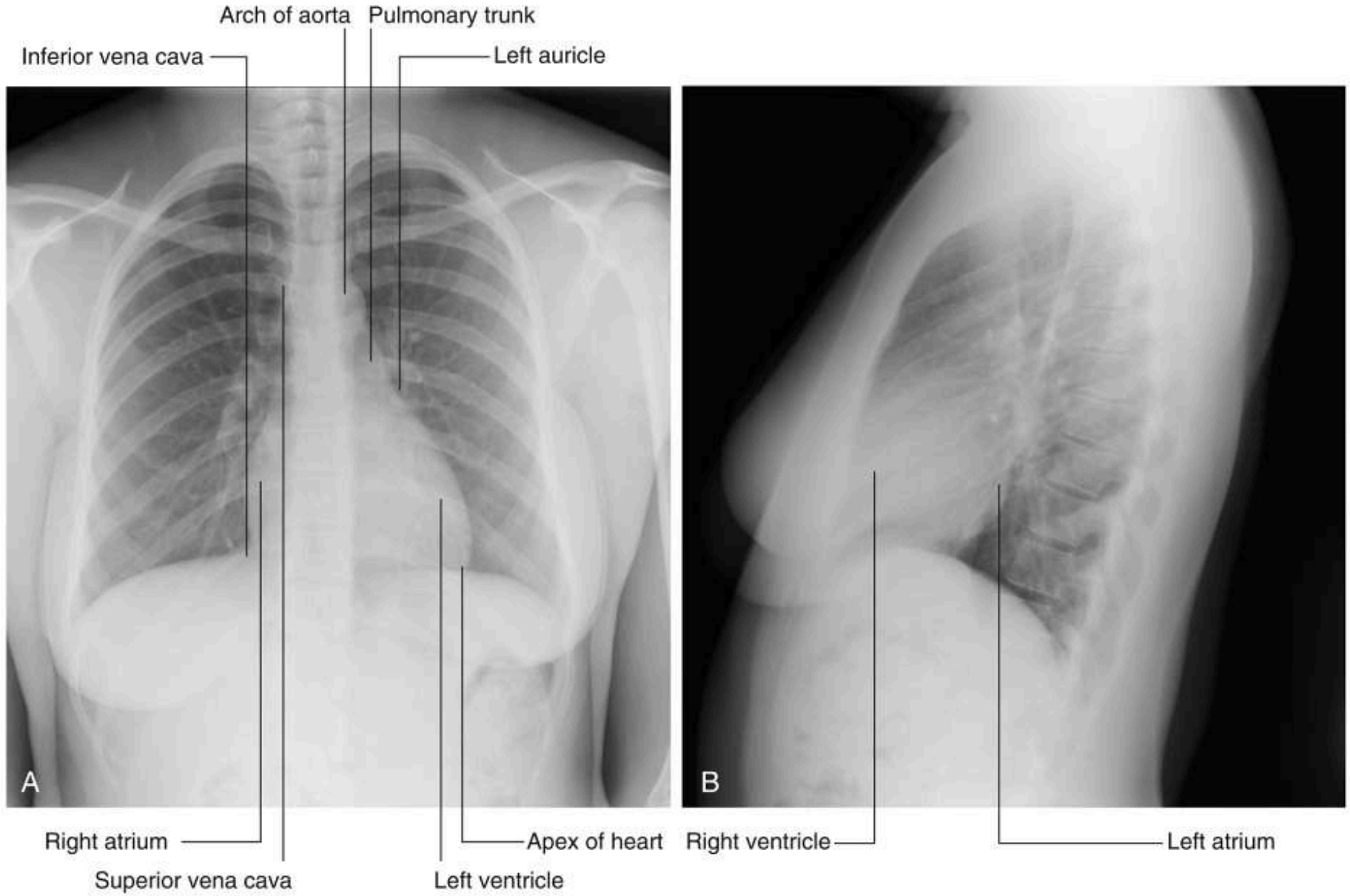


Fig. 3.69 Chest Radiographs. (**A**) Standard posteroanterior view of the chest. (**B**) Standard lateral view of the heart.

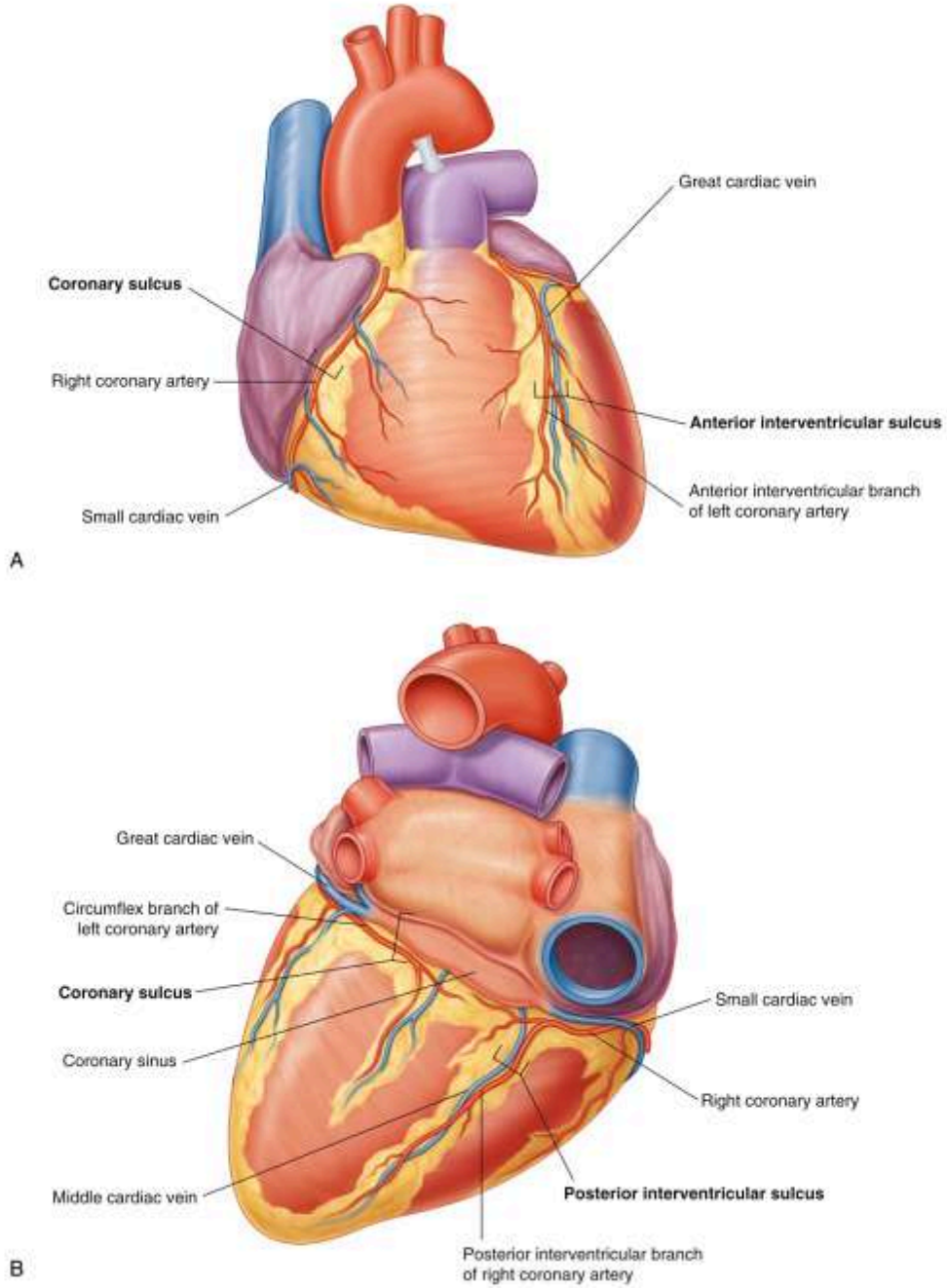
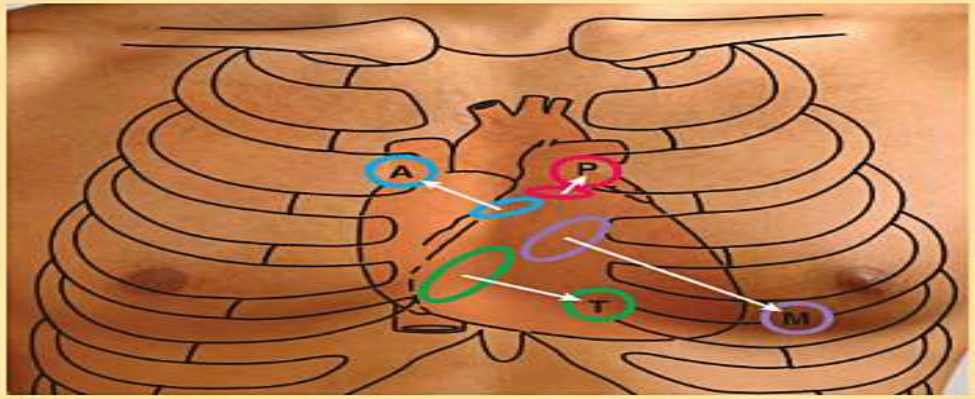
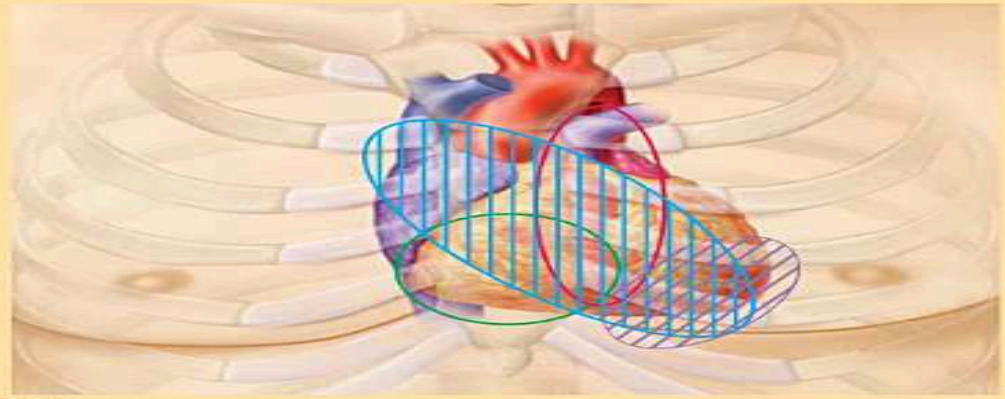


Fig. 3.70 Sulci of the Heart. (**A**) Anterior surface of the heart. (**B**) Diaphragmatic surface and base of the heart.



(A)



(B)

Key

Aortic valve | Pulmonary valve | Tricuspid valve | Mitral valve



(C) Aortic valve



(D) Pulmonary valve



(E) Tricuspid valve



(F) Mitral valve: male



(G) Mitral valve: female

FIGURE SA4.7. Location of valves and areas of auscultation. The location of each valve is indicated by a *colored oval* and the area of auscultation as a *circle* of the same color. Tricuspid valve (*T*) is *green*, mitral valve (*M*) is *purple*, pulmonary valve (*P*) is *pink*, and aortic valve (*A*) is *blue*. The direction of blood flow is indicated by *white arrows*.

Heart

- It is a muscular pump which maintains the circulation of blood.

***Position:**

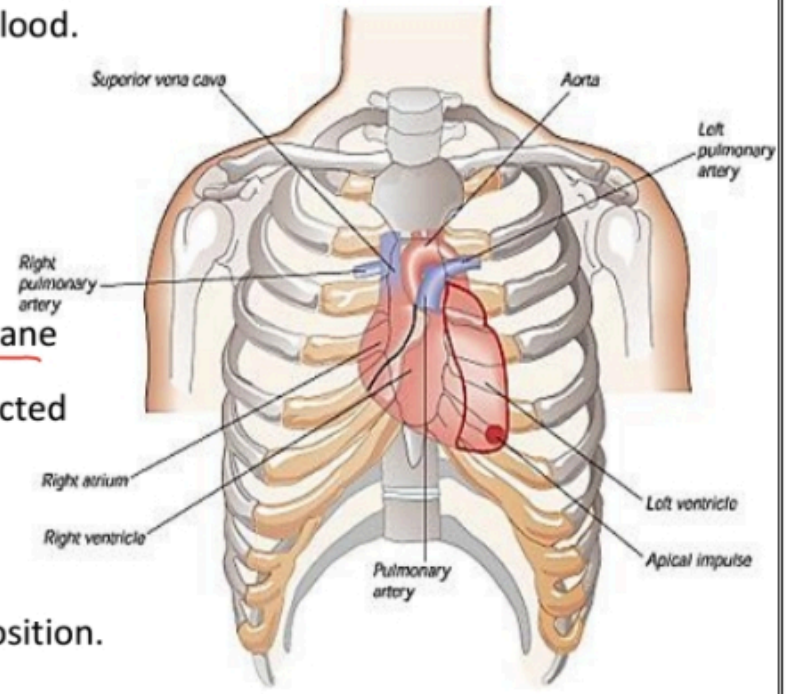
it lies obliquely in the middle mediastinum inside the pericardium so that :

(a) 1/3 of it lies to the Rt. & 2/3 to the Lt. of the median plane

(b) Its **long axis** (the line from the base to the apex) is directed downward, forwards & to the left.

(c) Its **base** (post. surface): lies opposite: -

-The middle 4Th vertebrae (T5-T8) in the recumbent position.



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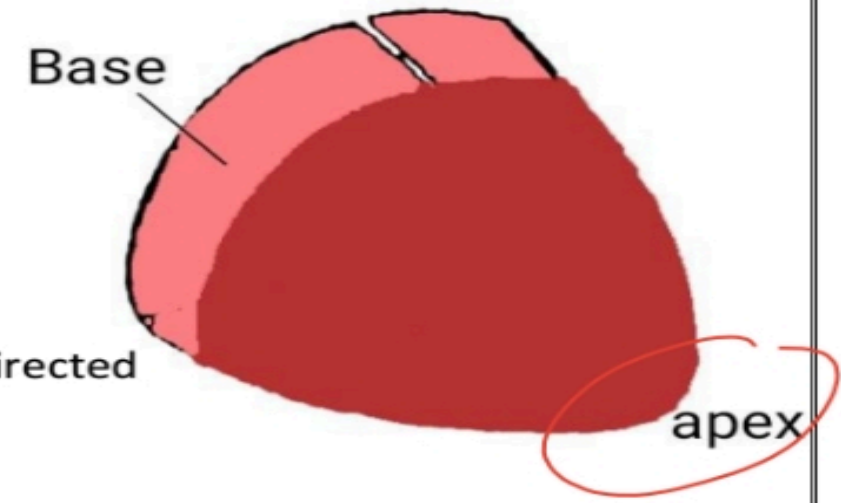
- The vertebrae (T6-T9) in the erect position

(d) Its **apex**: lies opposite the Lt. 5th intercostal space 3½" from the middle line.

*Size: The size of the heart of a person is slightly larger than his closed fist.

- Its long axis is 12 cm.

- Its weight is about 300 gm in the male & 250 gm in the female.



*Shape: cone-shaped having:

(a) **Base** or post. surface :directed backwards & to the right.

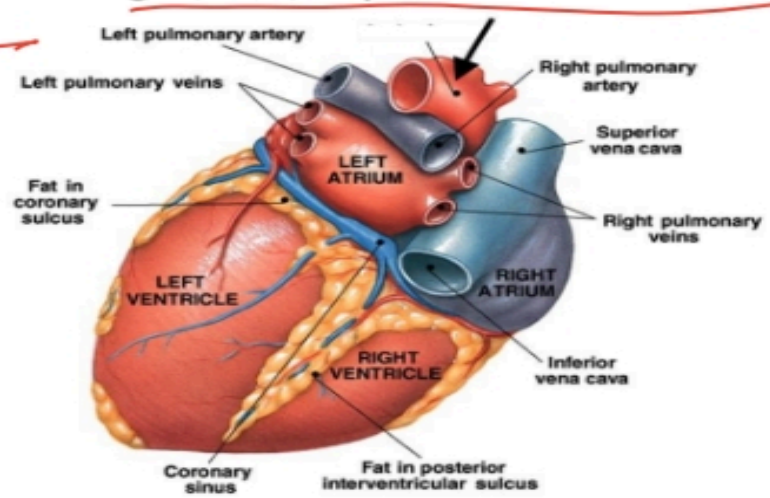
(b) **Apex** is the lower most & left most point of the heart It is directed downwards, forwards & to the left.

(c) **2 Surfaces**:

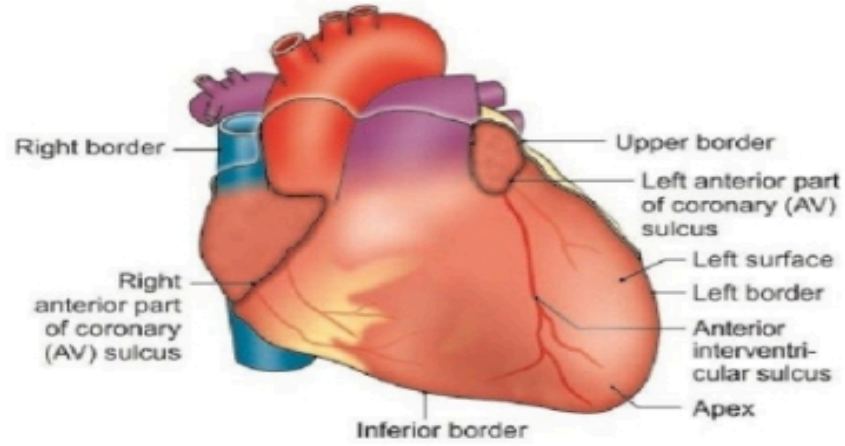
(1) Sternocostal surface (ant aspect).

(2) Inferior or diaphragmatic surface.

(d) 4 Borders: right, left, superior & inferior.



Posterior (diaphragmatic) surface

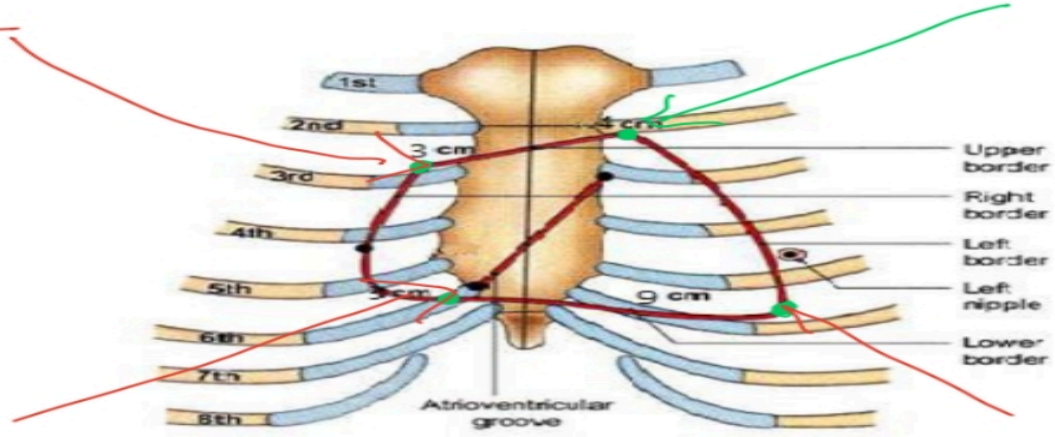


Gross features: Sternocostal surface of heart

Surface anatomy of the heart

*Put the following 4 points on the ant. chest wall:

- **point (a):** on the lower border of the 2nd Lt. costal cartilage 4 cm from the middle line.
- **Point (b):** on the upper border of the 3rd Rt. costal cartilage 3 cm from the middle line.
- **Point (c):** on the Rt. 6th costal cartilage 3 cm from the middle line.
- **Point (d):** on the Lt. 5th intercostal space 9 cm from middle line (apex of the heart)

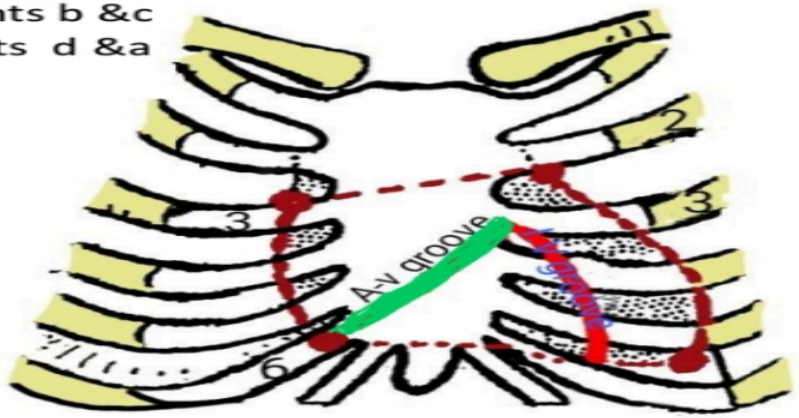


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DR. SAMEH DOSS

(1) **The upper border** is a straight line between the points a & b.
(2) **The lower border:** a straight line between the points c & d.
(3) **The right border**: a line slightly curved Laterally between the points b &c
(4) **The left border** : a line slightly curved Laterally between the points d &a
(5) **Atrioventricular groove** : its ant. part can be represented by a line between 2 points:
    (a) A point on the 3rd Lt. sternocostal junction.
    (b) A point on the 6th Rt. sternocostal junction .
(6) **Interventricular groove**: represented by:
    a line from the 3rd Lt. sternocostal junction drawn parallel to left border of the heart till it meets the inferior border.

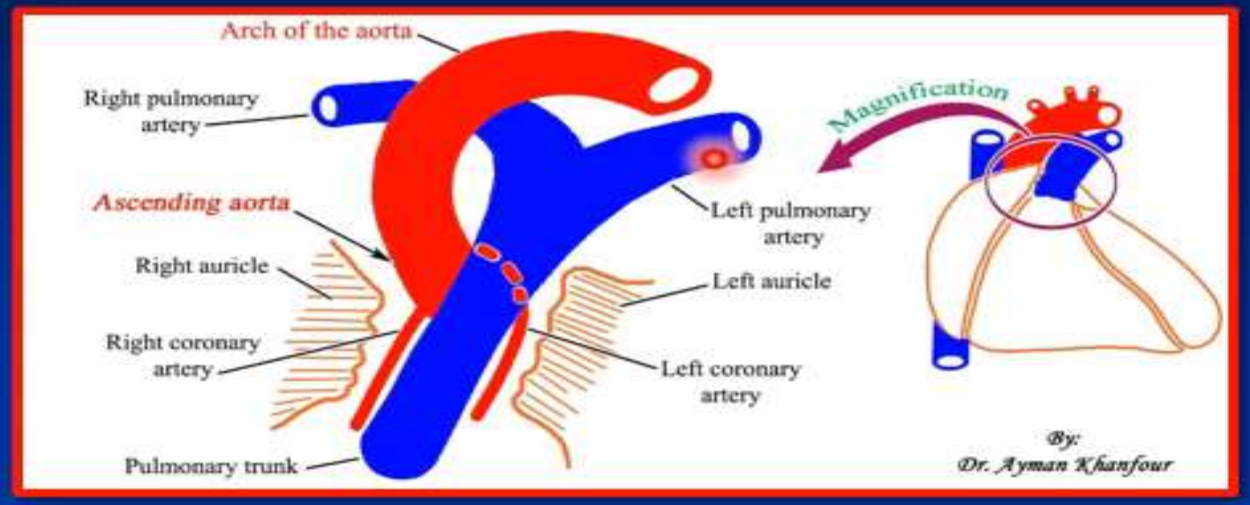


Pulmonary Trunk

Pulmonary Trunk

Begining

- Level of 3rd Lt. intercostal space (From the infundibulum of Rt. ventricle).

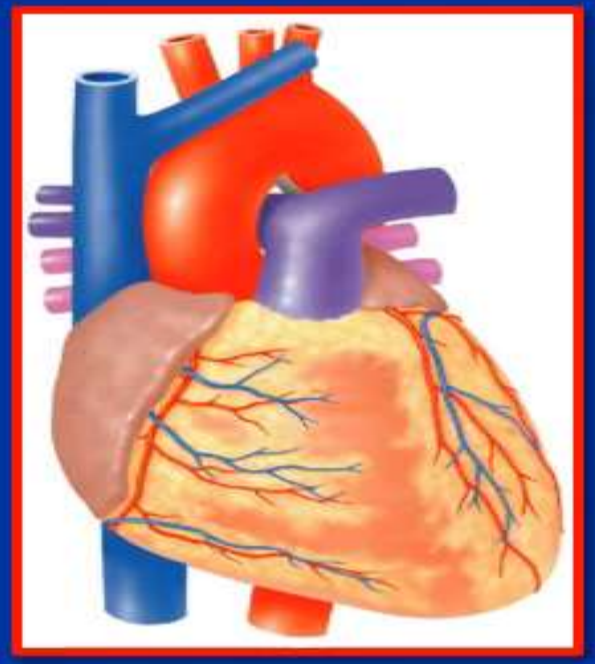


Branches

1. Rt. Pulmonary a.
2. Lt. pulmonary a.

Termination

- Level of T. 4

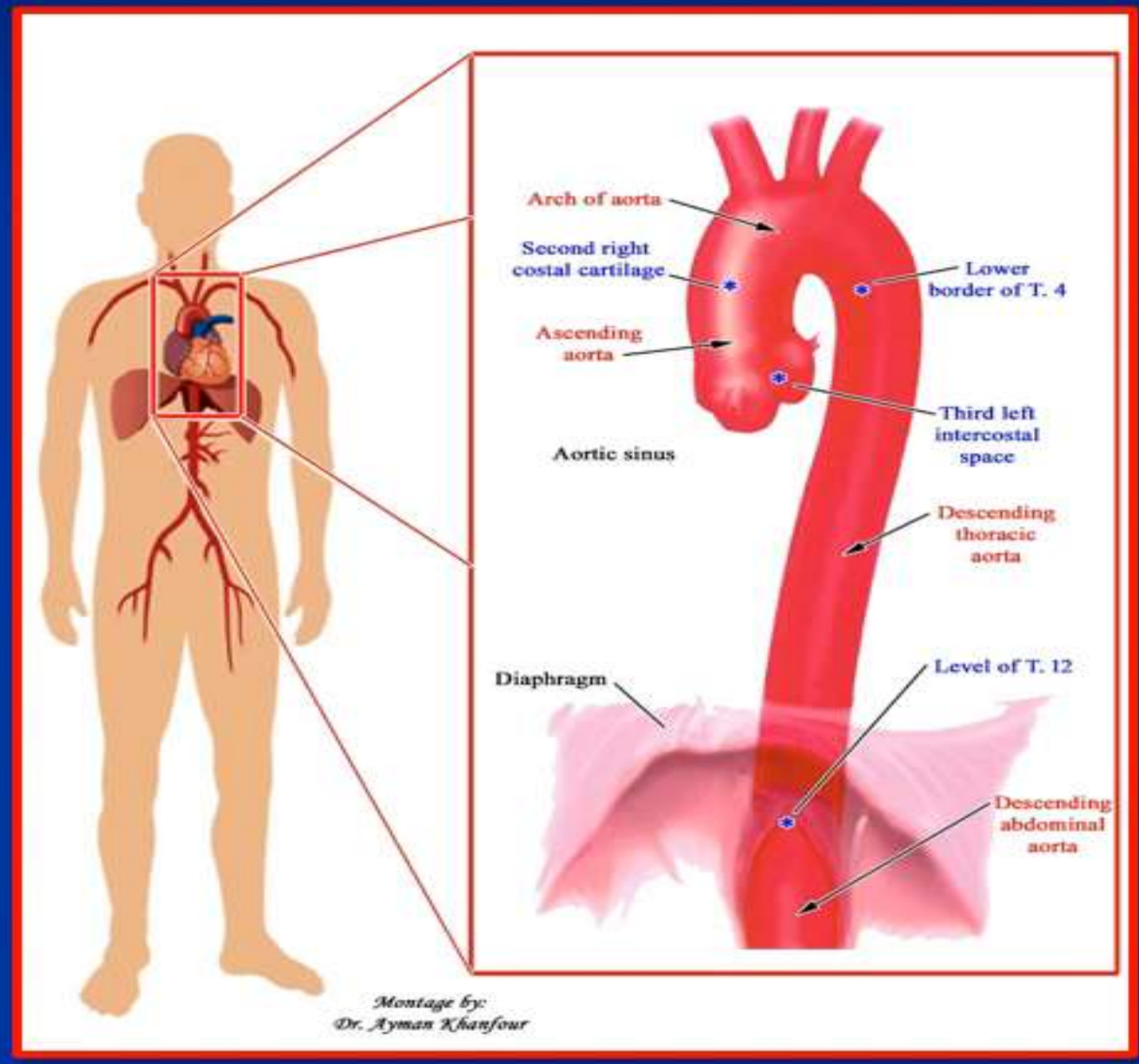


Aorta

Parts of Aorta

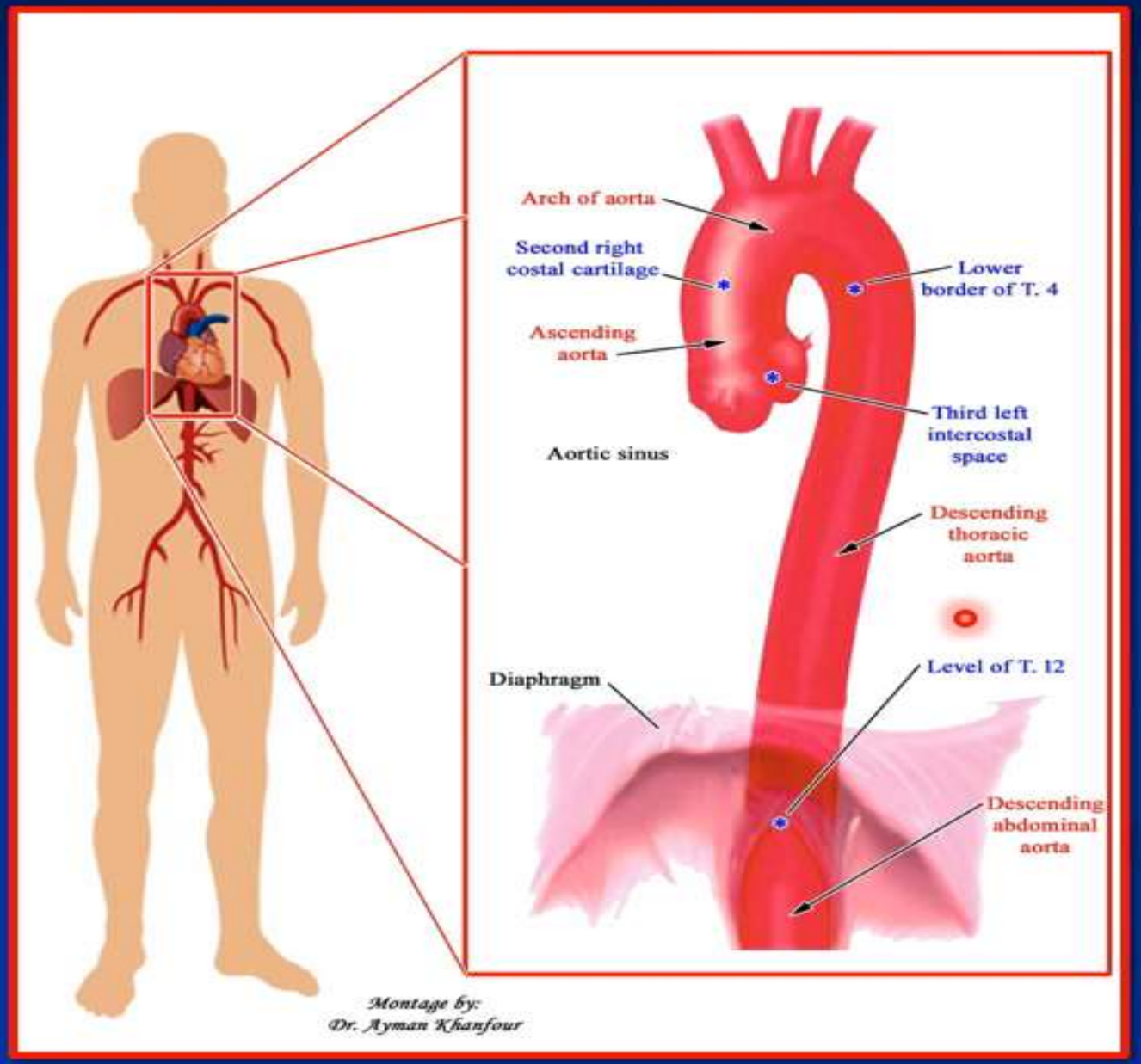
- The aorta is formed of the following parts:

1. Ascending aorta
2. Arch of aorta
3. Descending aorta
   a. Descending thoracic
   b. Descending abdominal

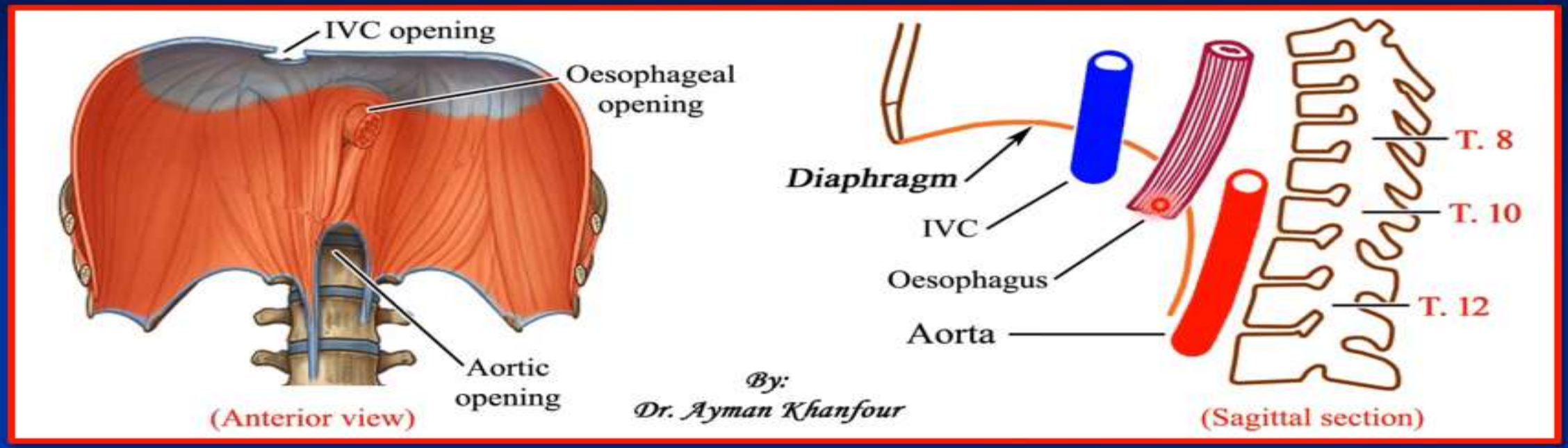


Beginning and Termination





Main Openings in the Diaphragm



Voice Of America

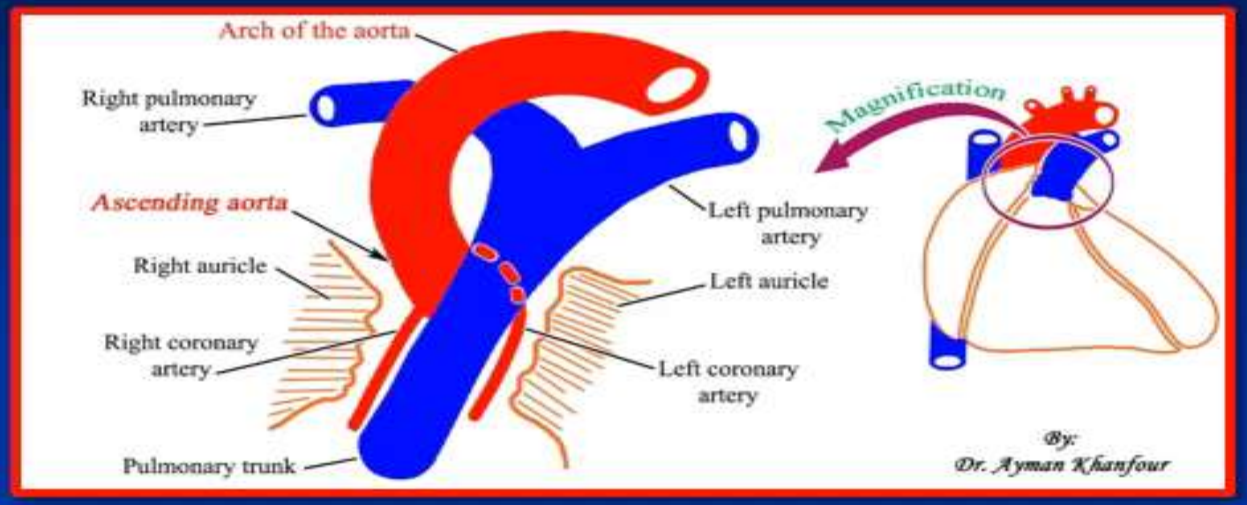
IVC	Oesophagus	Aorta
T 8	T 10	T 12

Ascending Aorta

Ascending Aorta

Begining

- Level of 3rd Lt. intercostal space (From the vestibule of Lt. ventricle).

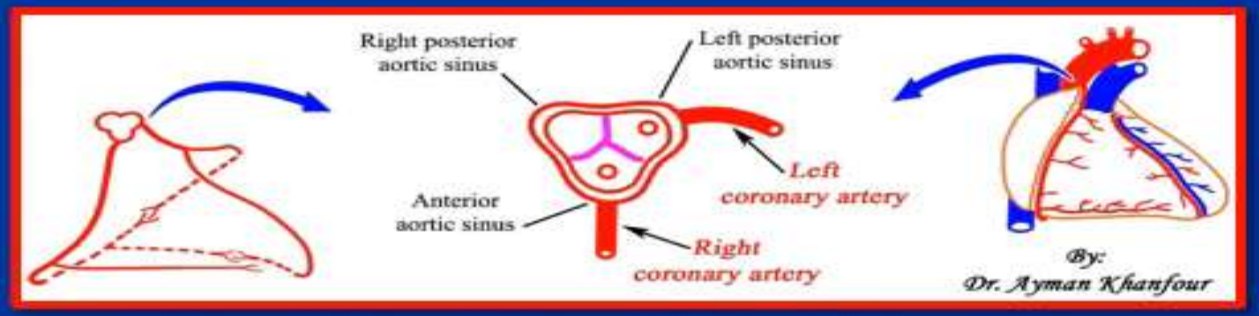


Branches

1. Rt. coronary a.
2. Lt. coronary a.

Termination

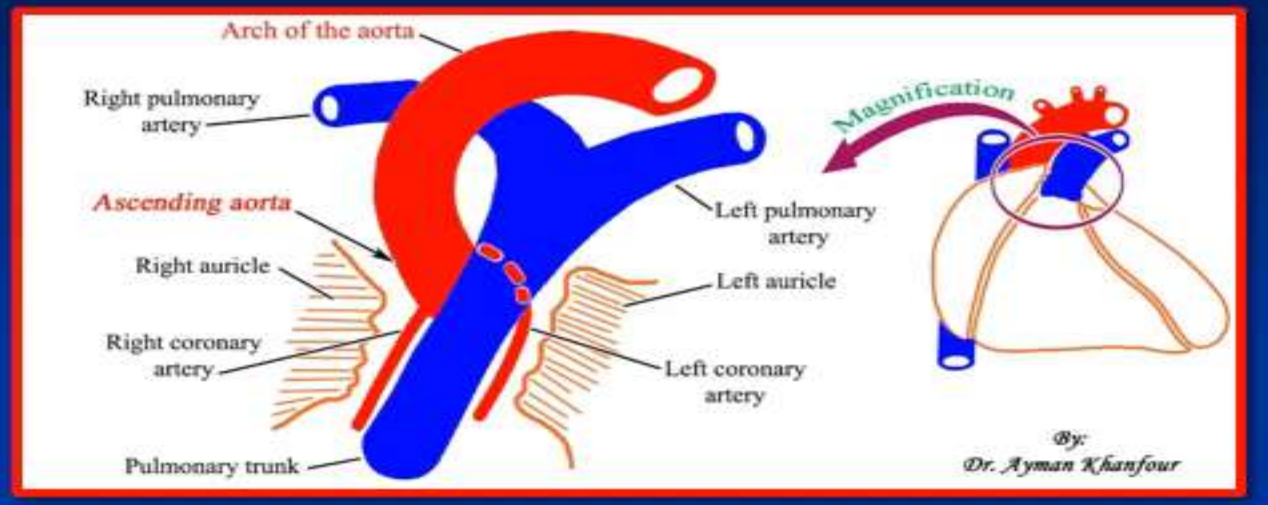
- Level of 2nd Rt. Costal cartilage



Ascending Aorta

Begining

- Level of 3rd Lt. intercostal space (From the vestibule of Lt. ventricle).

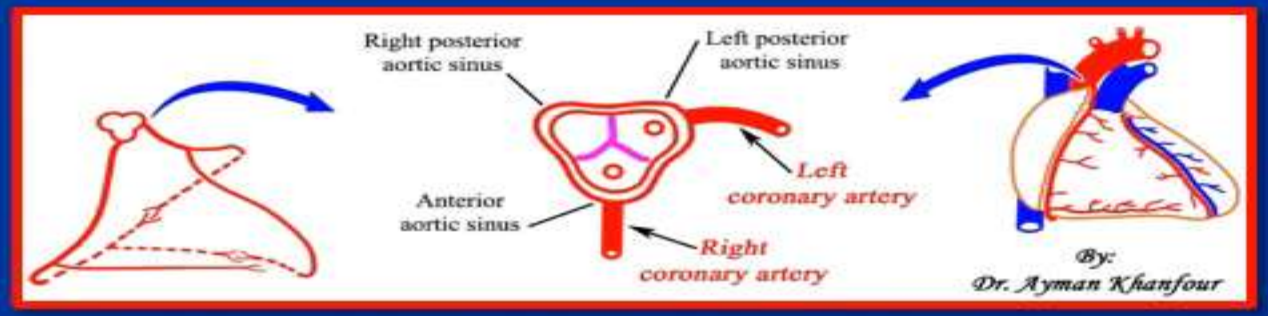


Branches

1. Rt. coronary a.
2. Lt. coronary a.

Termination

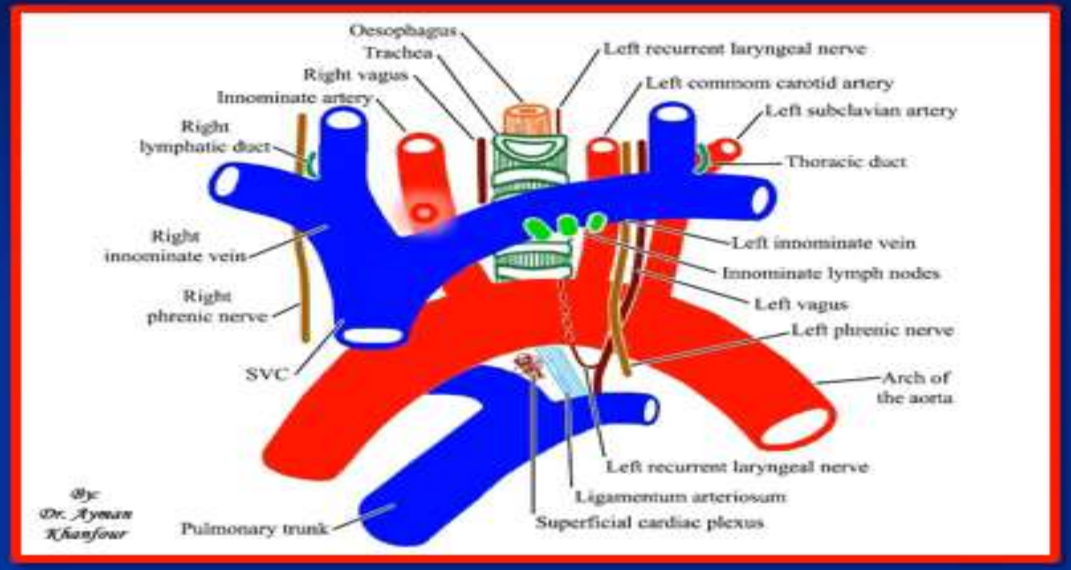
- Level of 2nd Rt. Costal cartilage



Arch of Aorta

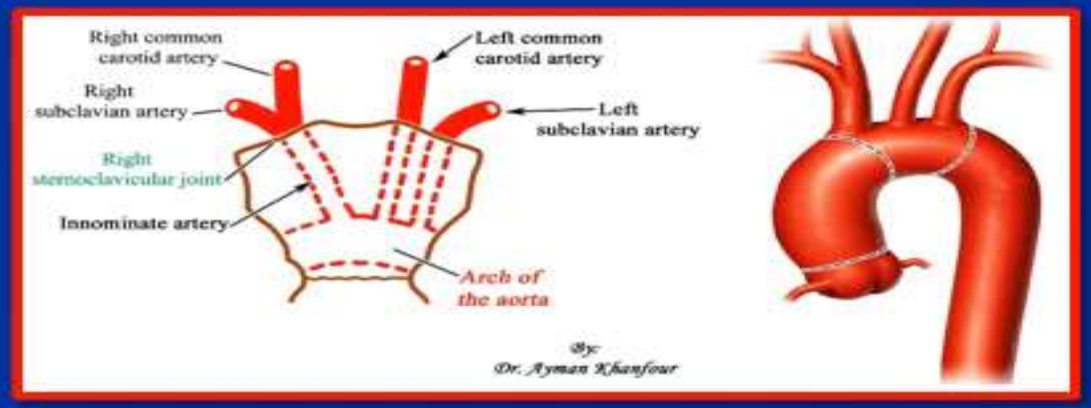
Begining

- Level of 2nd Rt. Costal cartilage



Termination

- Level of T. 4

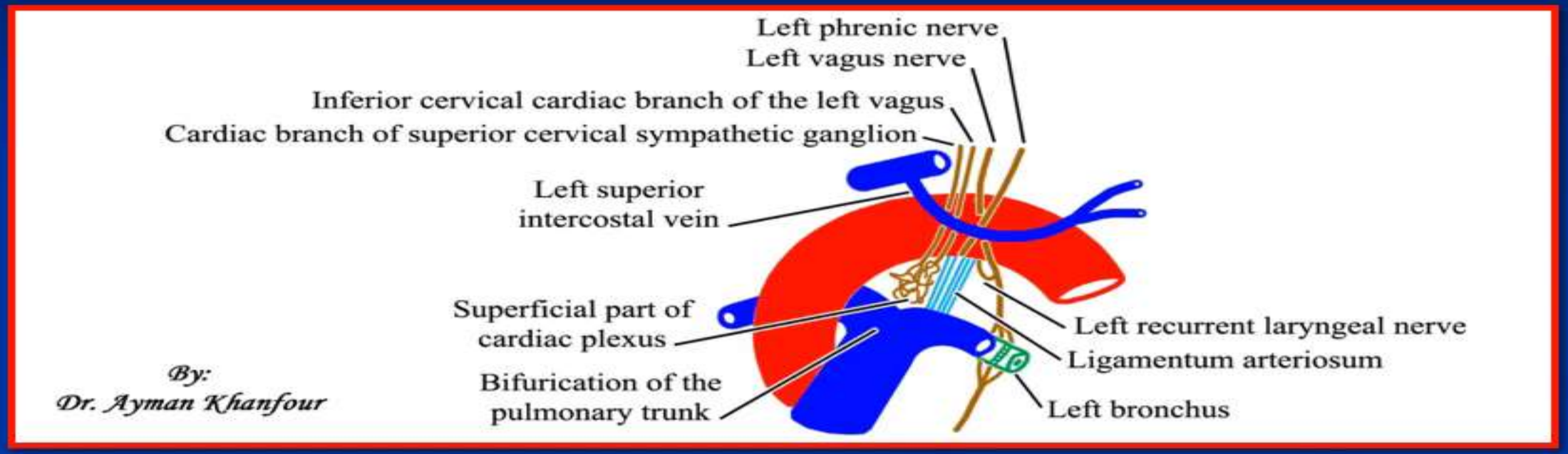


Branches

1. Innominate a.
2. Lt. common carotid a.
3. Lt. subclavian a.

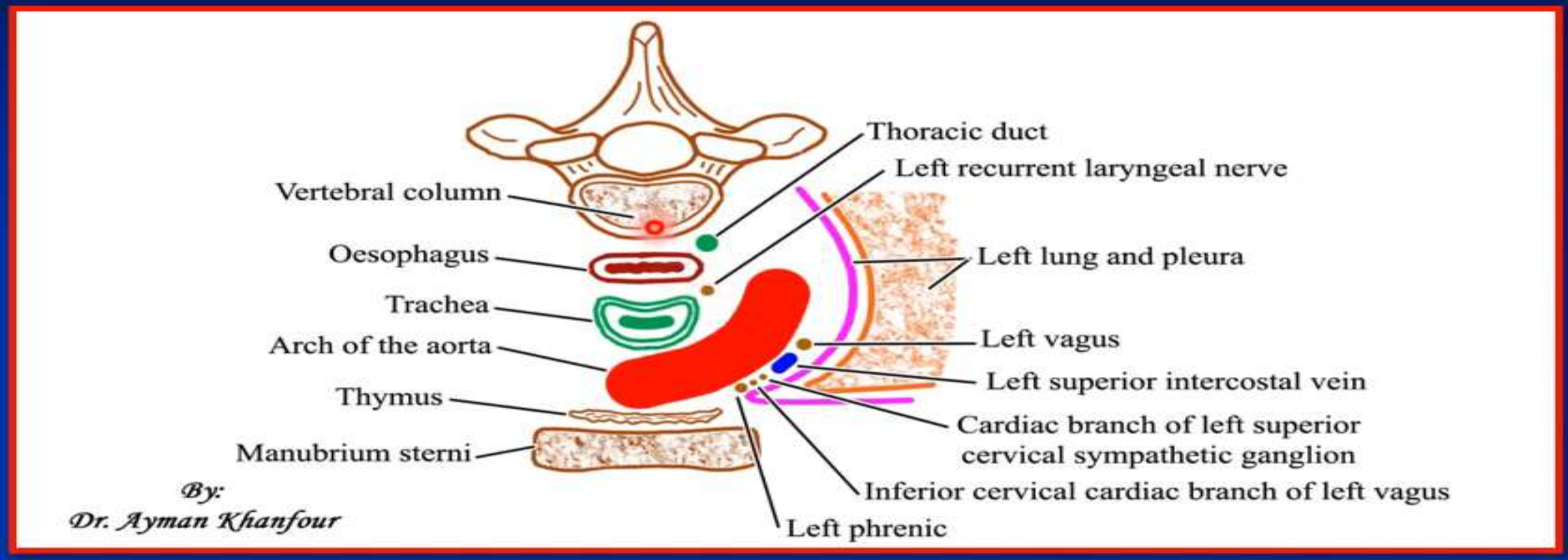
Relations of Aortic Arch

A. Anterior



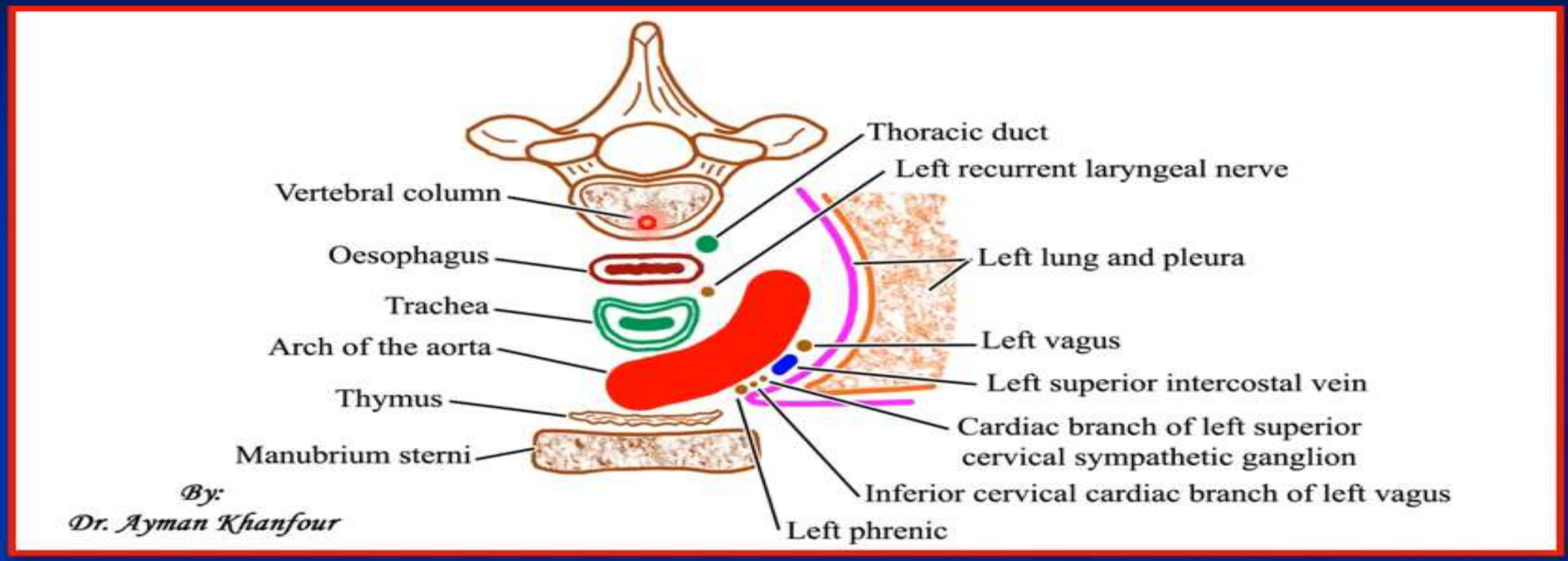
1. **Four nerves (antero-posterior):**
   - **a. Left phrenic.**
   - **b. Left vagus.**
   - **c. Cardiac branch of left vagus**
   - **d. Cardiac branch of sympathetic trunk**
2. **Left superior intercostal vein.**
3. **Left lung and pleura**

B. Posterior (3 +2)



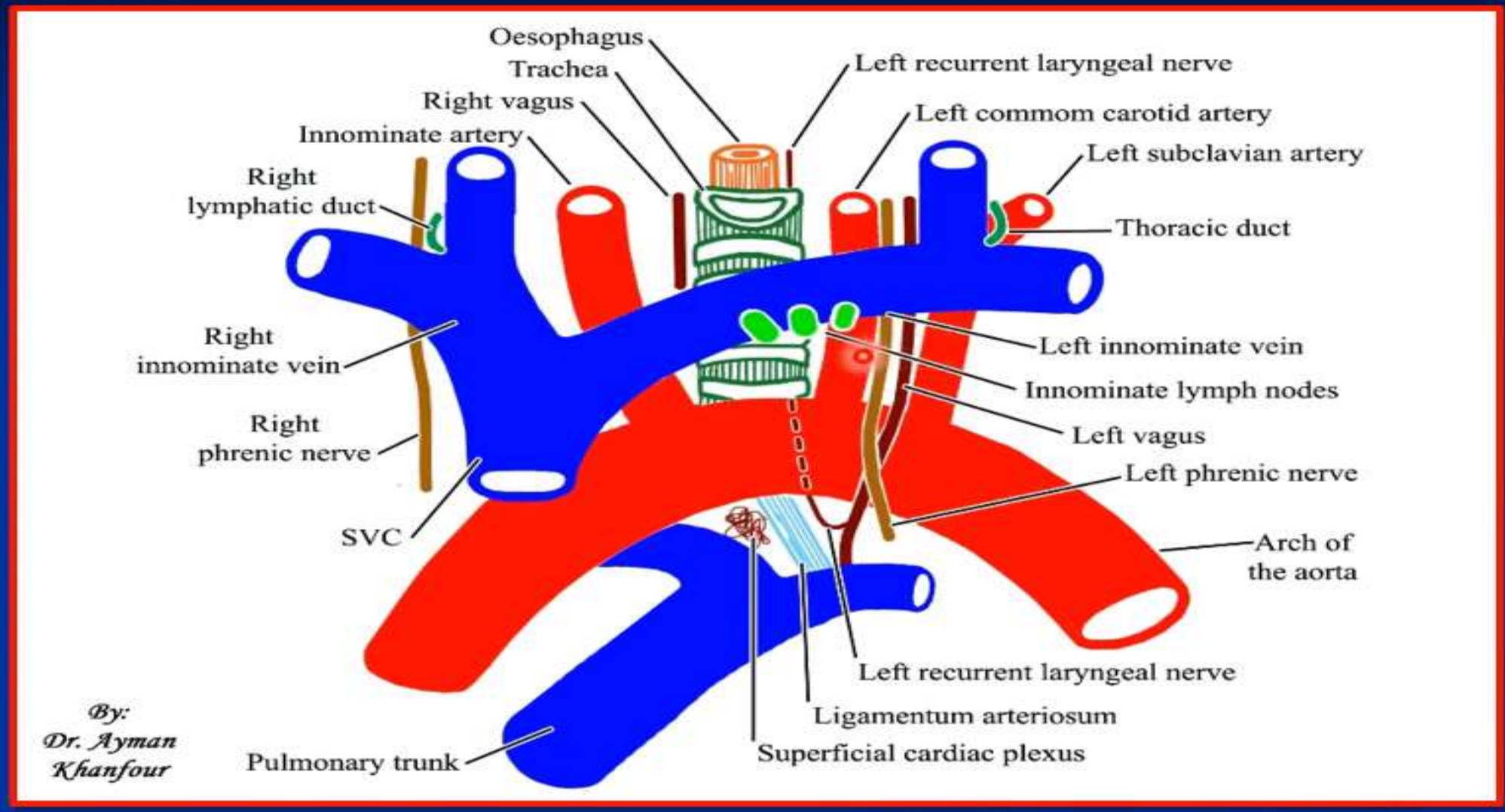
- Trachea.
- Oesophagus.
- Vertebral column.
- Left recurrent laryngeal nerve.
- Thoracic duct.

B. Posterior (3 +2)



- Trachea.
- Oesophagus.
- Vertebral column.
- Left recurrent laryngeal nerve.
- Thoracic duct.

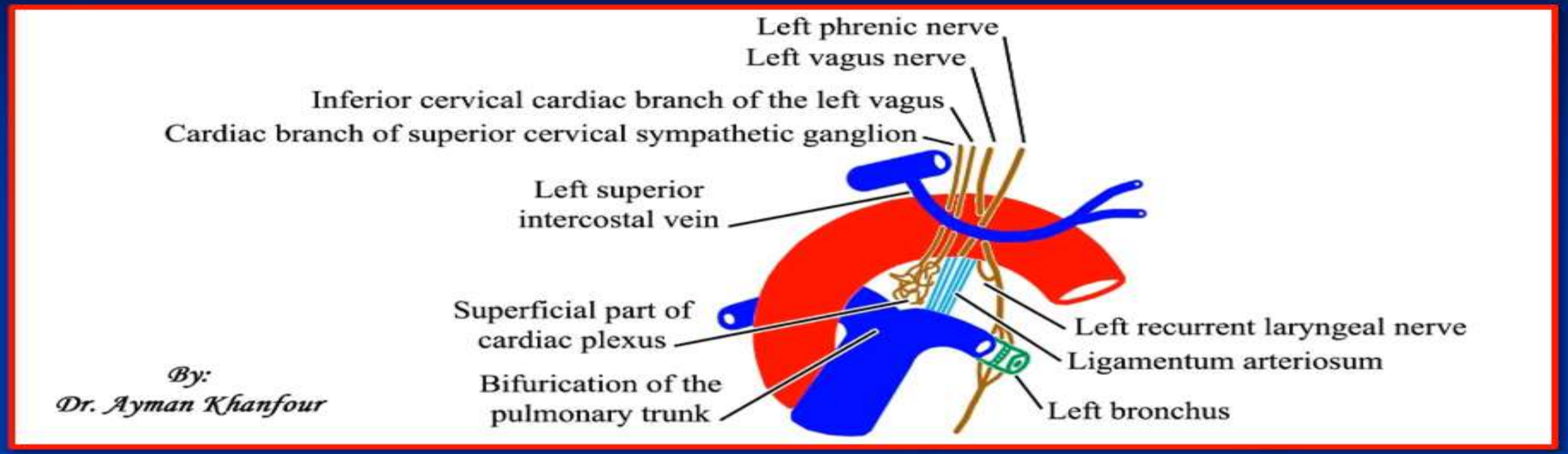
C. Superior



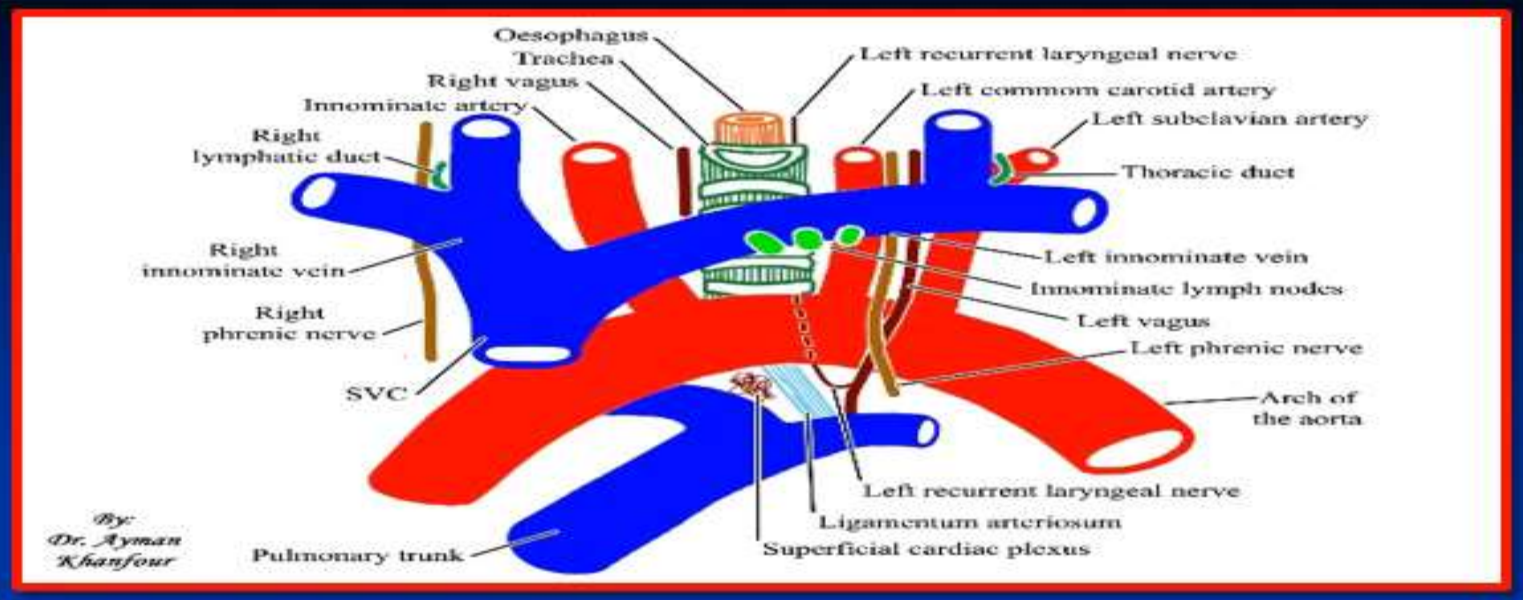
- Great vessels of the aortic arch.

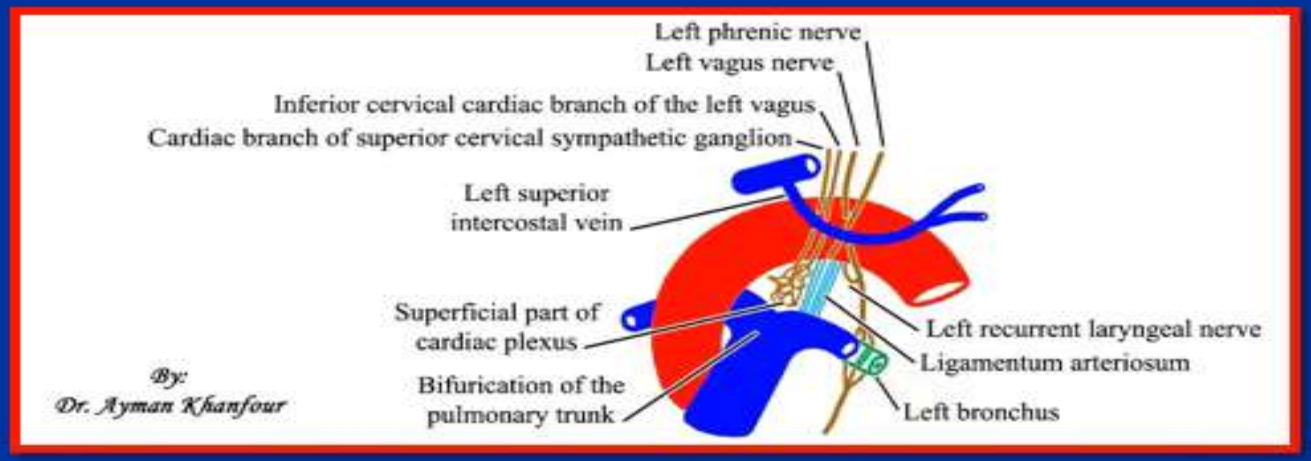
- Left innominate vein.

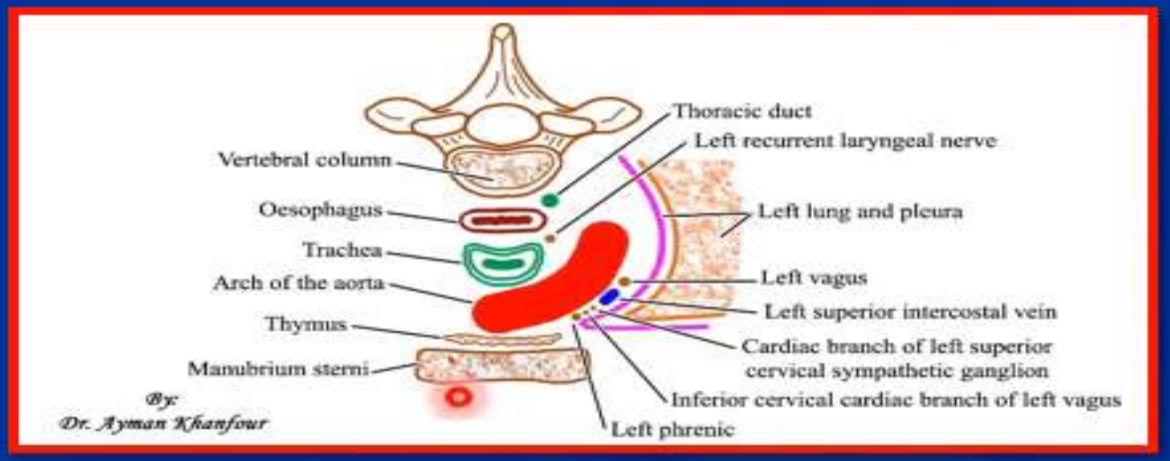
D. Inferior



- **Bifurcation of the pulmonary trunk.**
- **Left bronchus.**
- **Ligamentum arteriosum which is related to:**
  a. **Left recurrent laryngeal nerve (on the left side).**
  b. **Superficial part of the cardiac plexus (on its right side).**

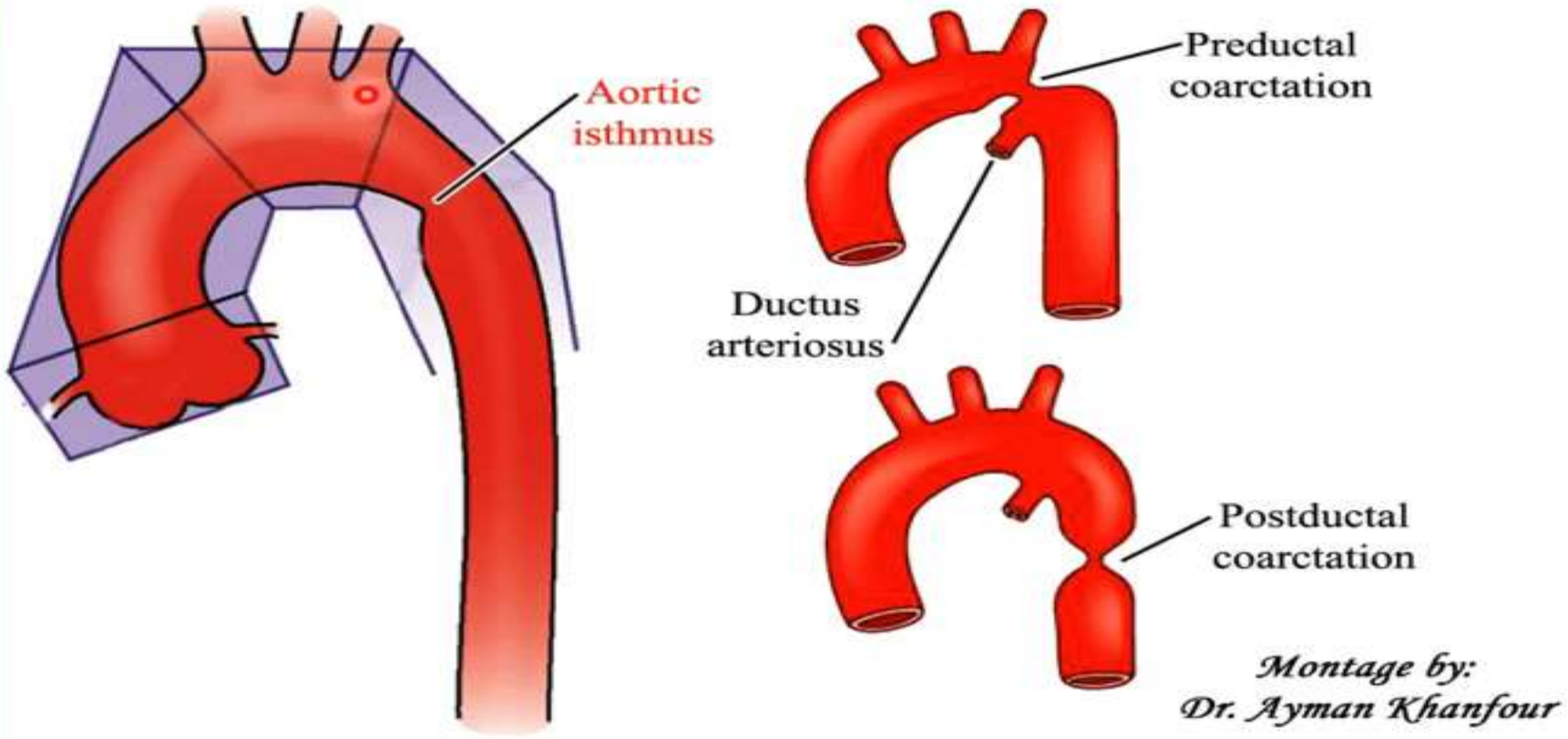




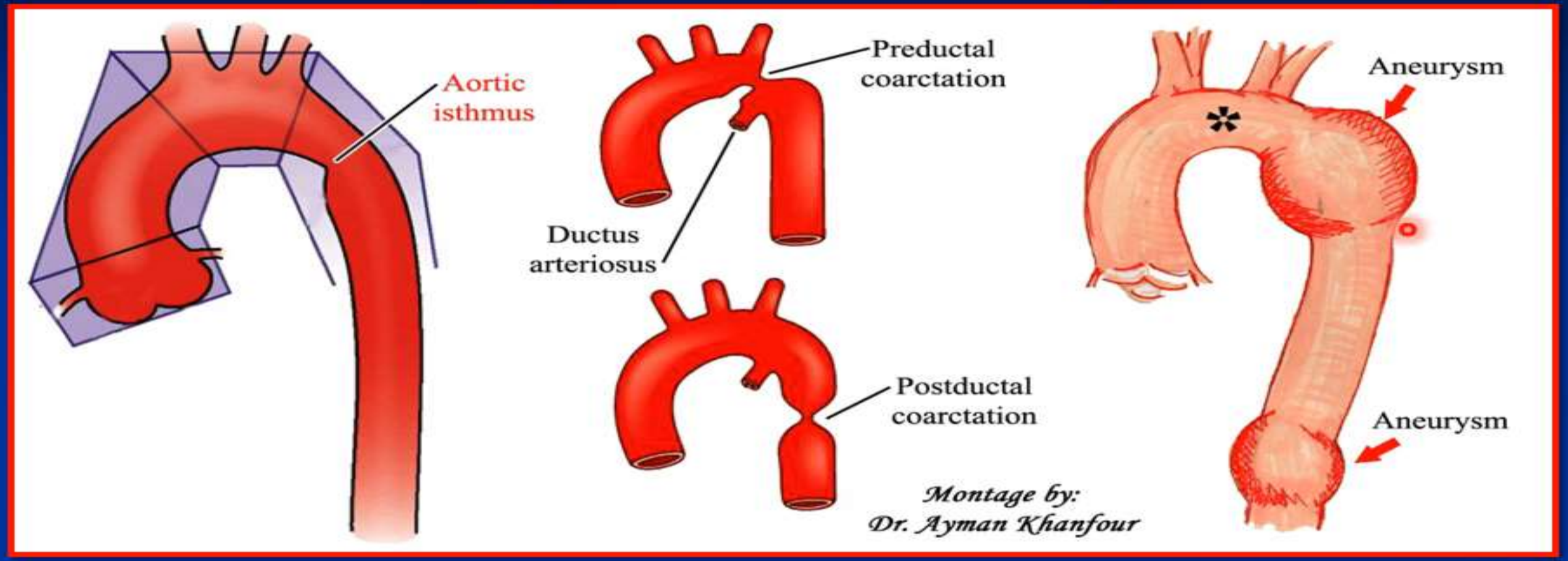


(Relations of the arch of the aorta)

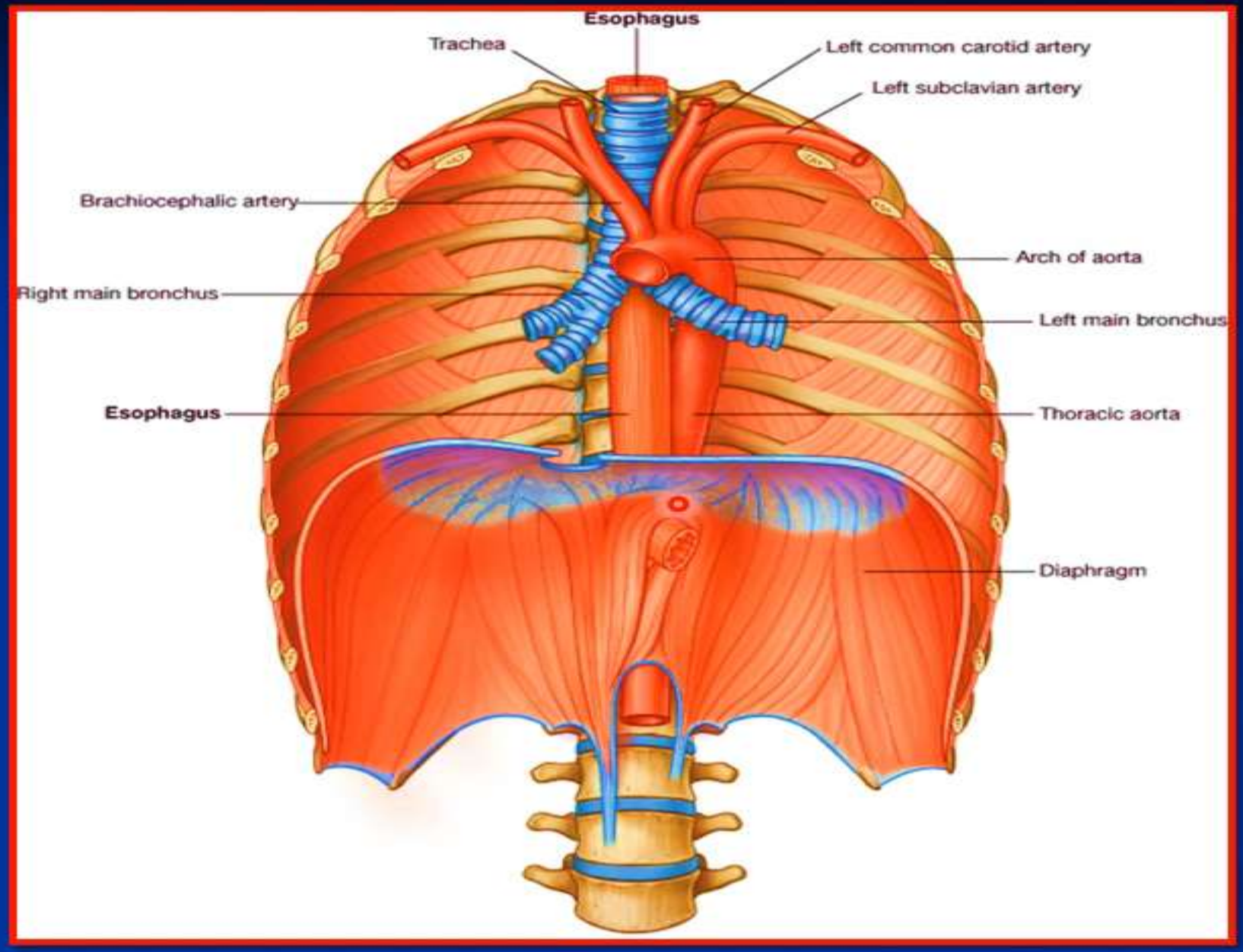
Applied Anatomy



Applied Anatomy



- Immediately distal to origin of left subclavian artery, the arch of aorta narrows forming aortic isthmus.
- It is a common site of coartication of aorta.



(Branches of the arch of the aorta)

Descending Thoracic Aorta

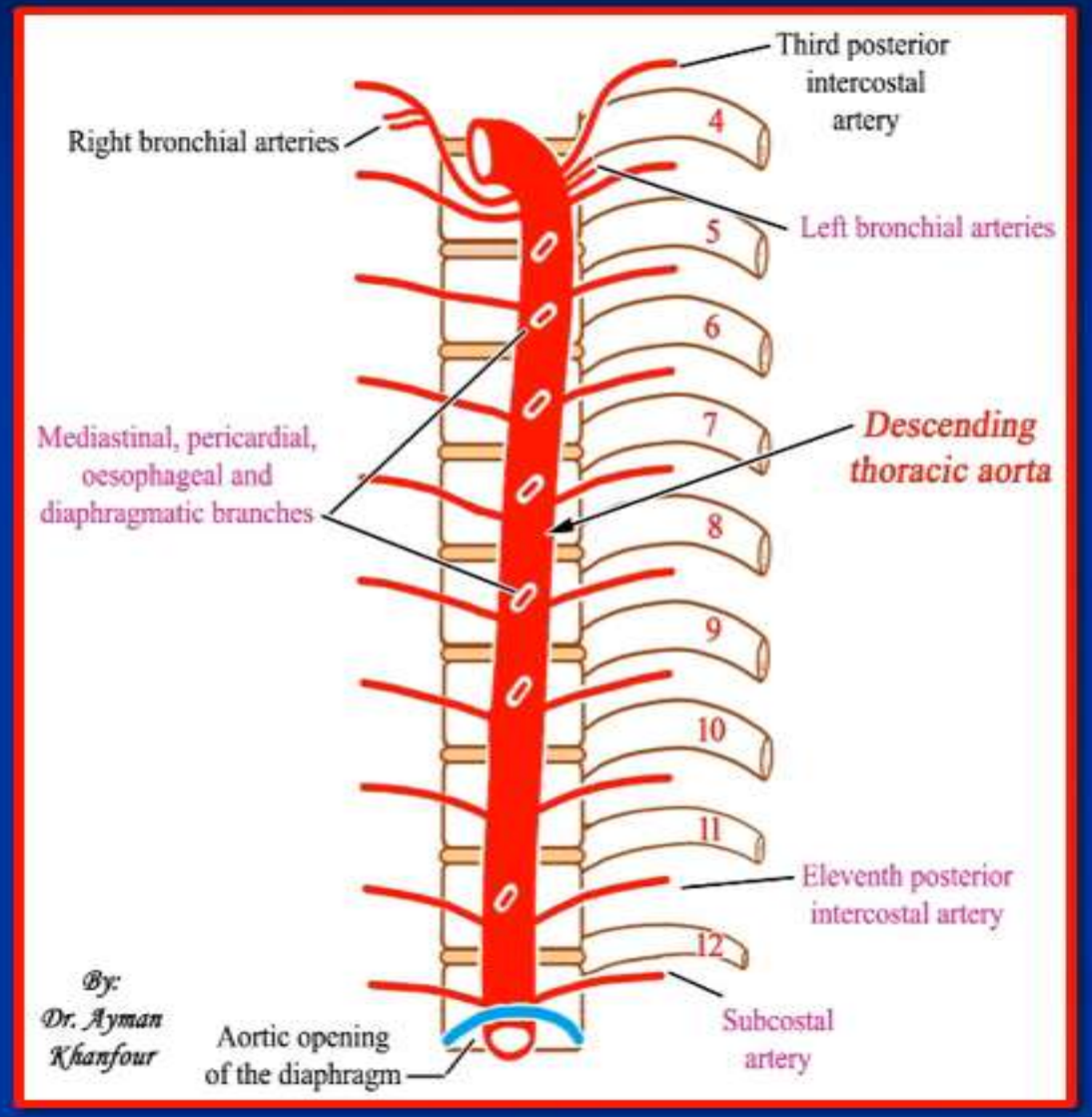
Descending Thoracic Aorta

Begining

- Level of T. 4

Termination

- Level of T. 12



Branches

A. Parietal branches

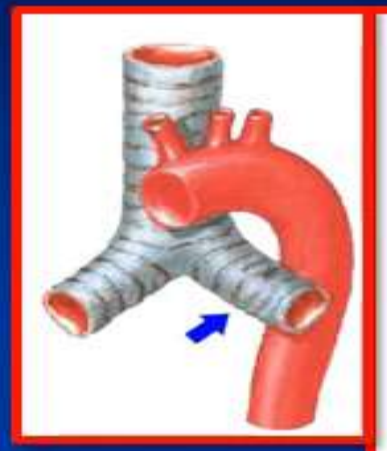
1. Post. intercostal a. (3 - 11).
2. Subcostal a.
3. Phrenic branches.

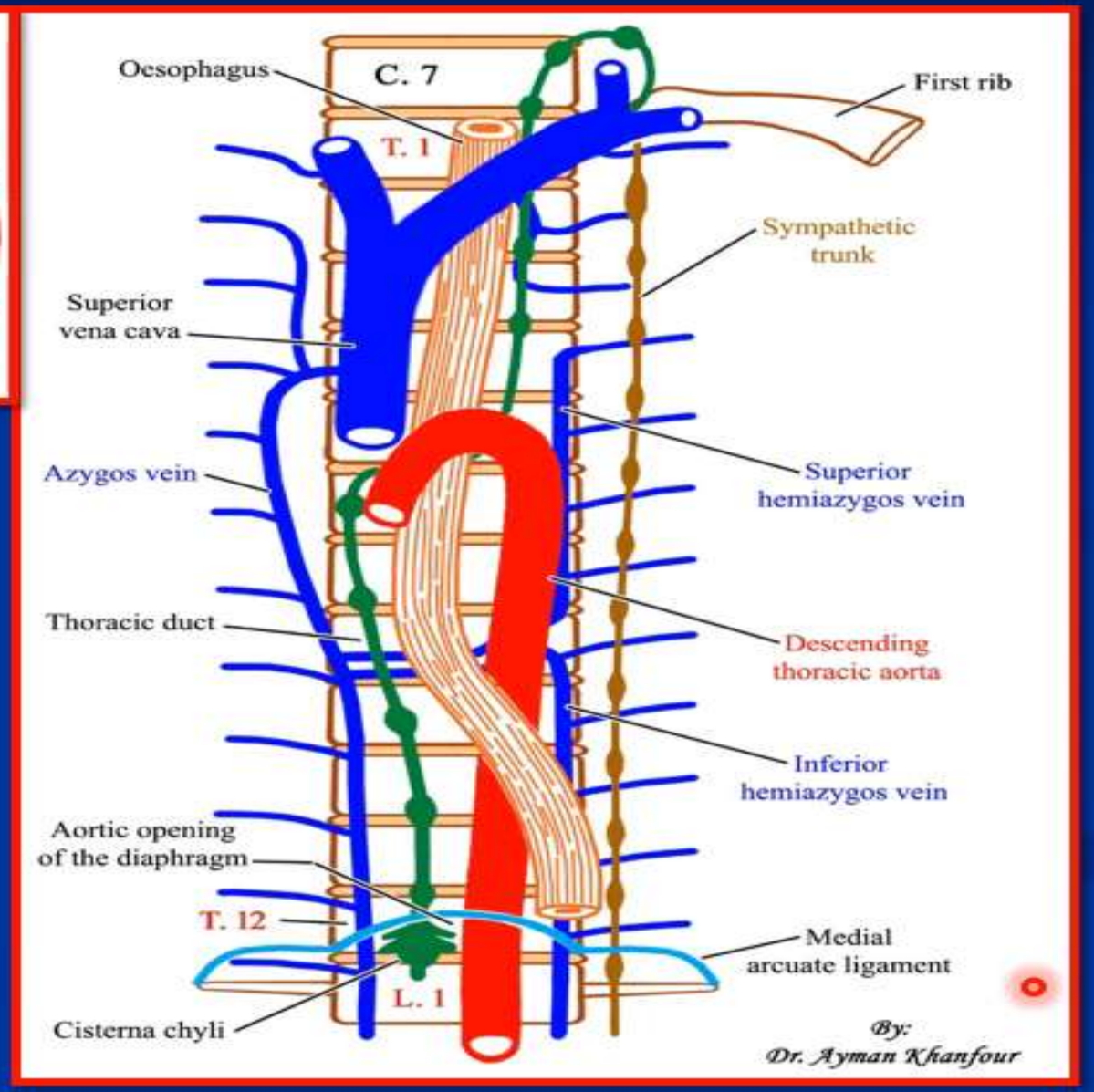
B. Visceral branches

1. Pericardial branches.
2. Bronchial arteries.
3. Mediastinal branches.

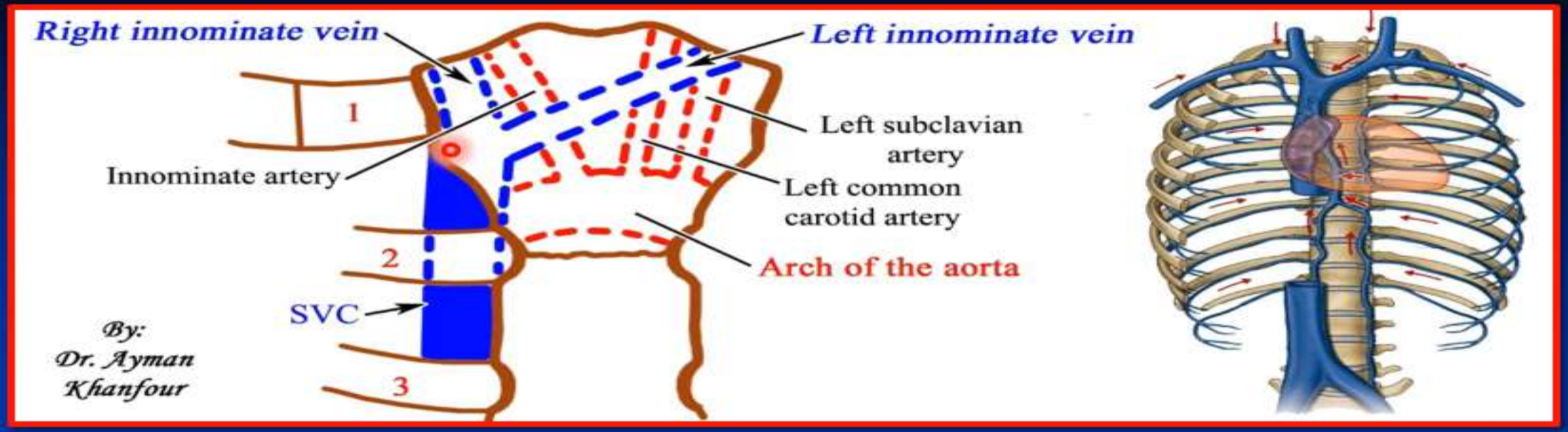
Relations

1. Anterior
   - Oesophagus.
   - Left main bronchus
   - Crura of diaphragm
2. Posterior
   - Thoracic vertebrae
   - Anterior longitudinal lig.
3. Right
   - Thoracic duct
   - Azygos vein.
   - Oesophagus
4. Left
   - Left lung and pleura
   - Oesophagus

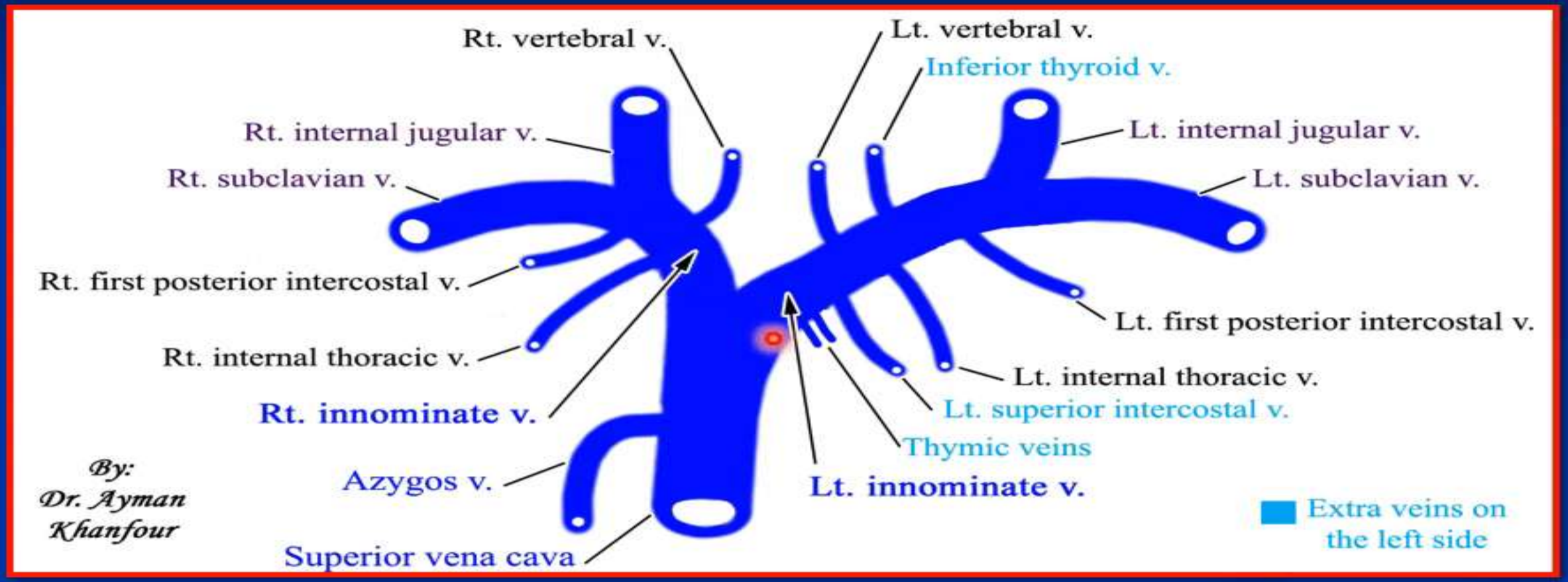




Innominate Veins



	Right Innominate	Left Innominate
Beginning	- Each innominate vein begins behind the corresponding sterno-clavicular joint by the union of the internal jugular vein and the subclavian vein on each side.	
Course	- Vertical.	- Oblique.
Length	- 1 inch. +	- 2 ½ inches. ++₊
Termination	- Both veins unite behind the *lower border of the 1st costal cartilage* to form the S.V.C.	



(Tributaries of the innominate veins)

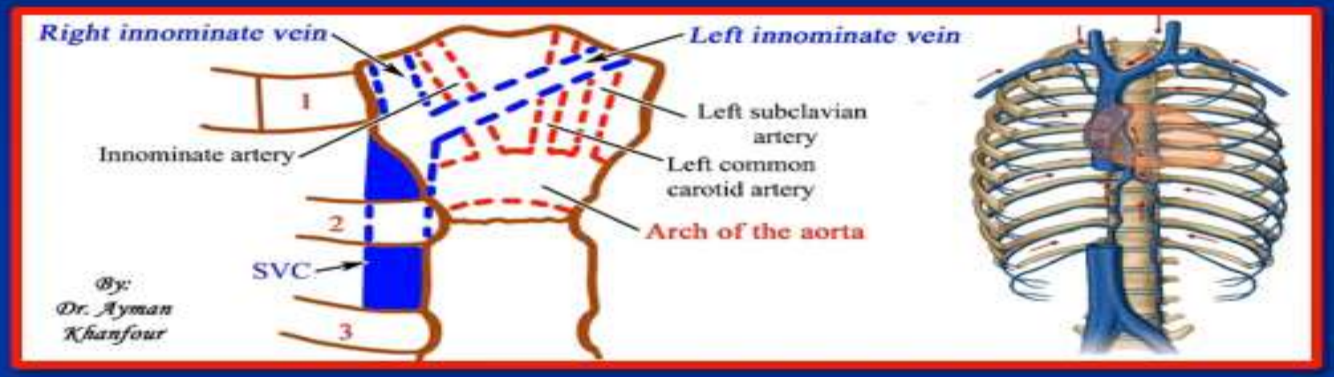
Superior Vena Cava

Superior Vena Cava

S V C
1 2 3

Begining

- 1st Rt. Costal cartilage

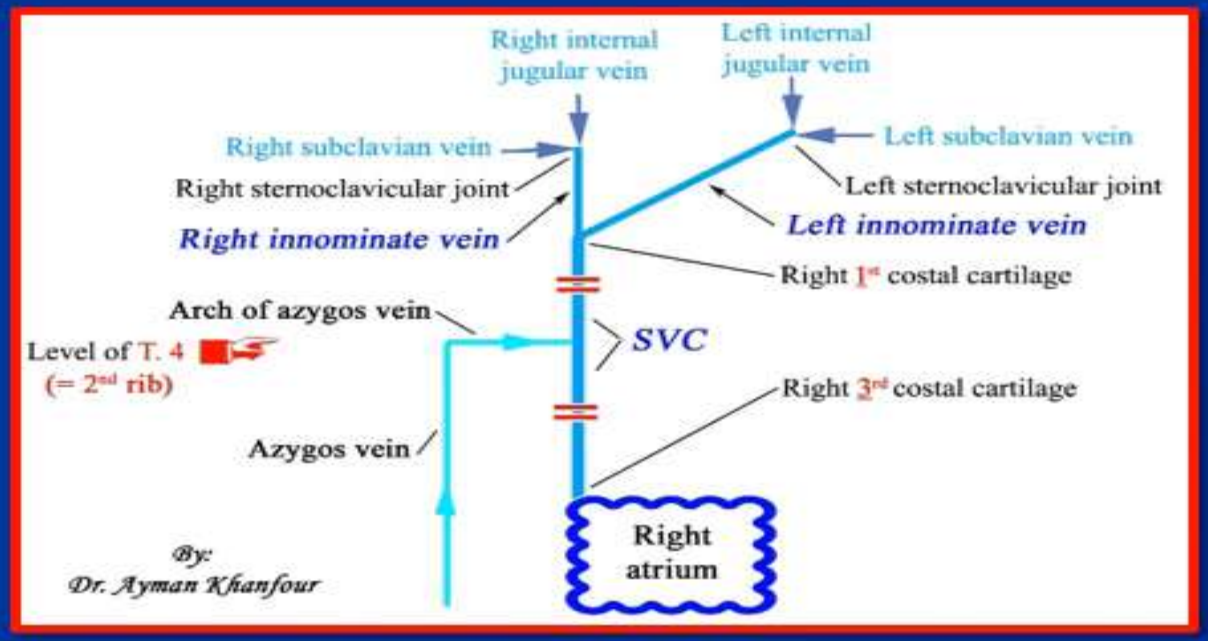


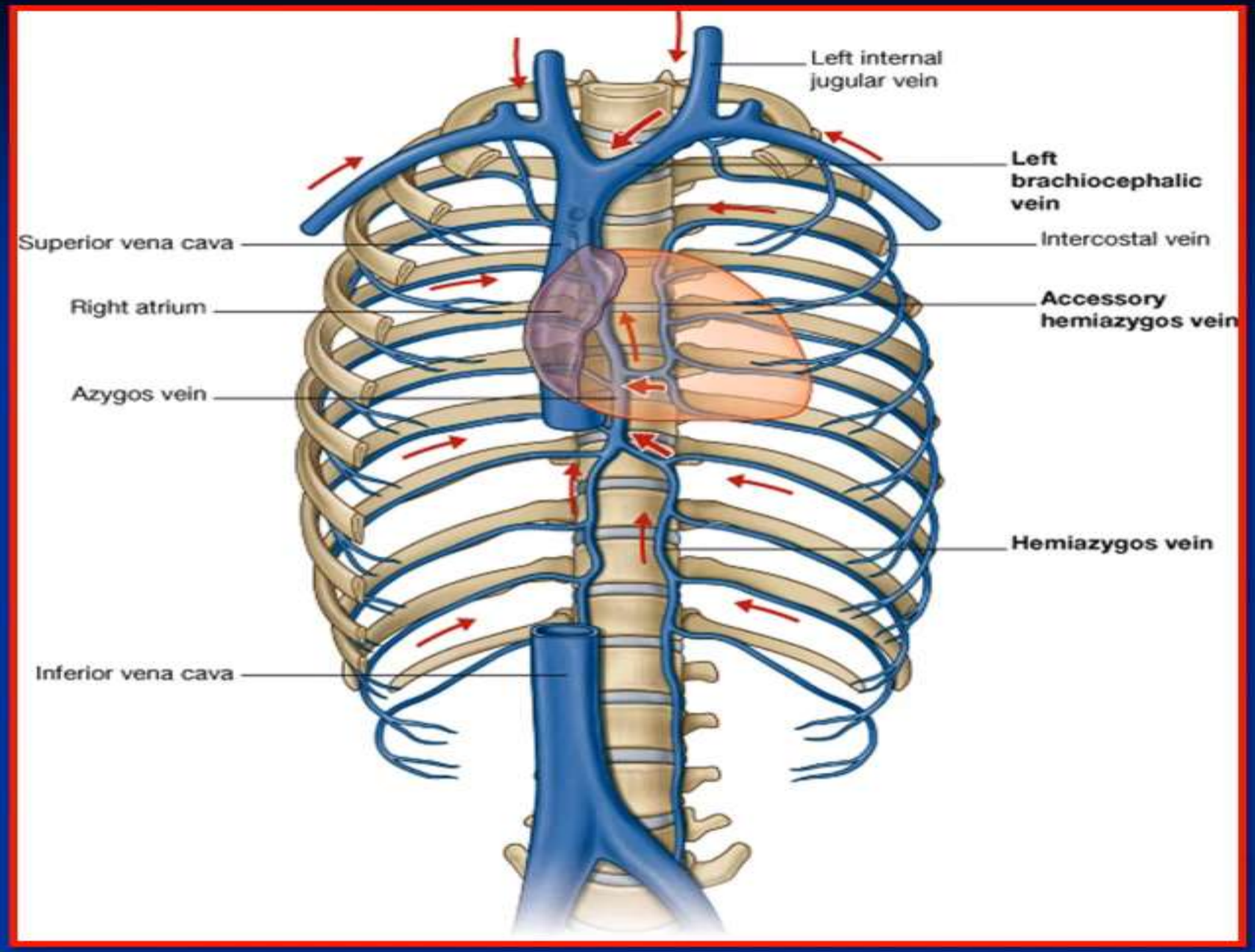
Tributary

- Azygos vein at the level of 2nd Rt. Costal cartilage

Termination

- 3rd Rt. Costal cartilage





(SVC and azygos venous system)

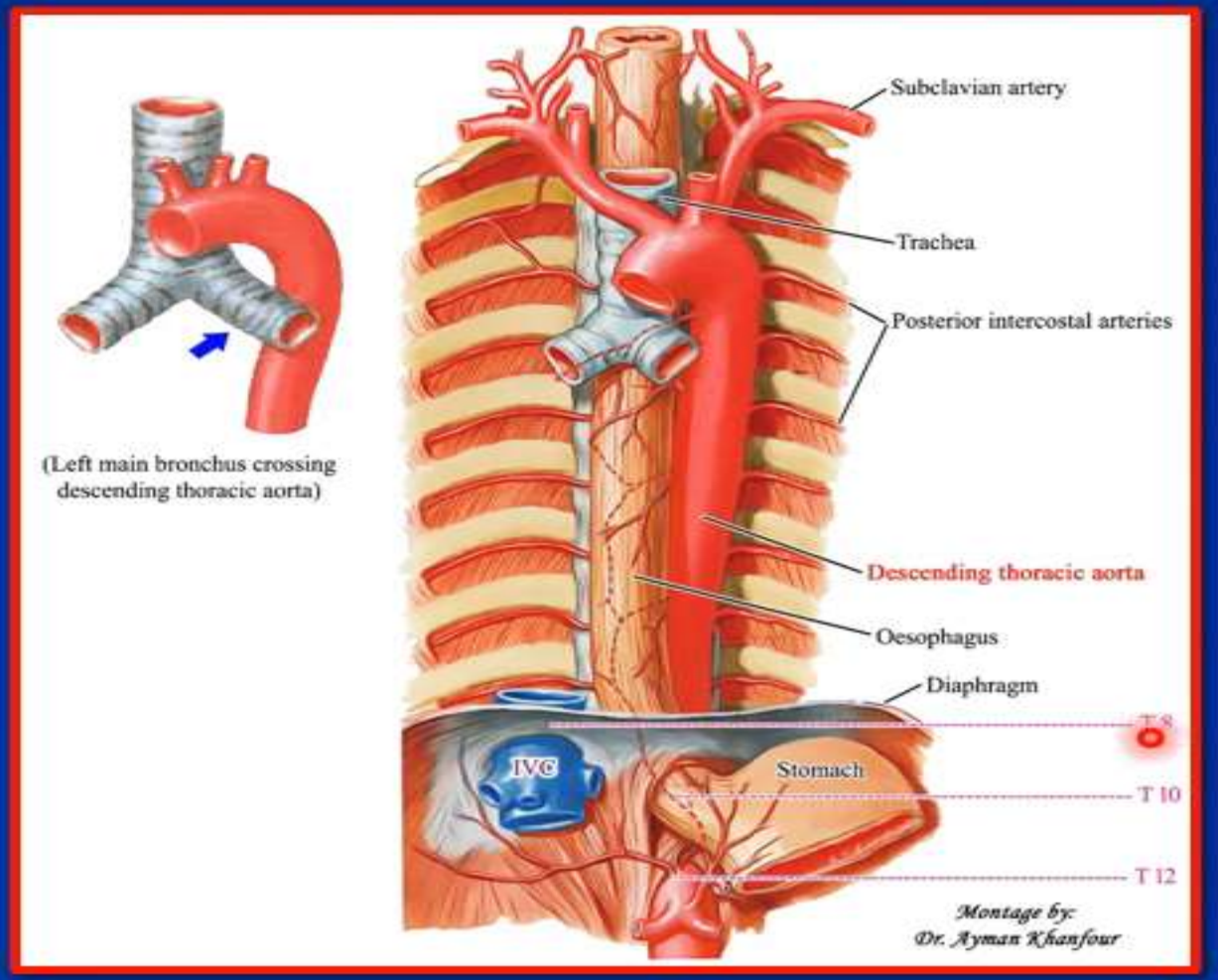
Inferior Vena Cava

Inferior Vena Cava

V O A
8 10 12

Begining

- Level of L. 5

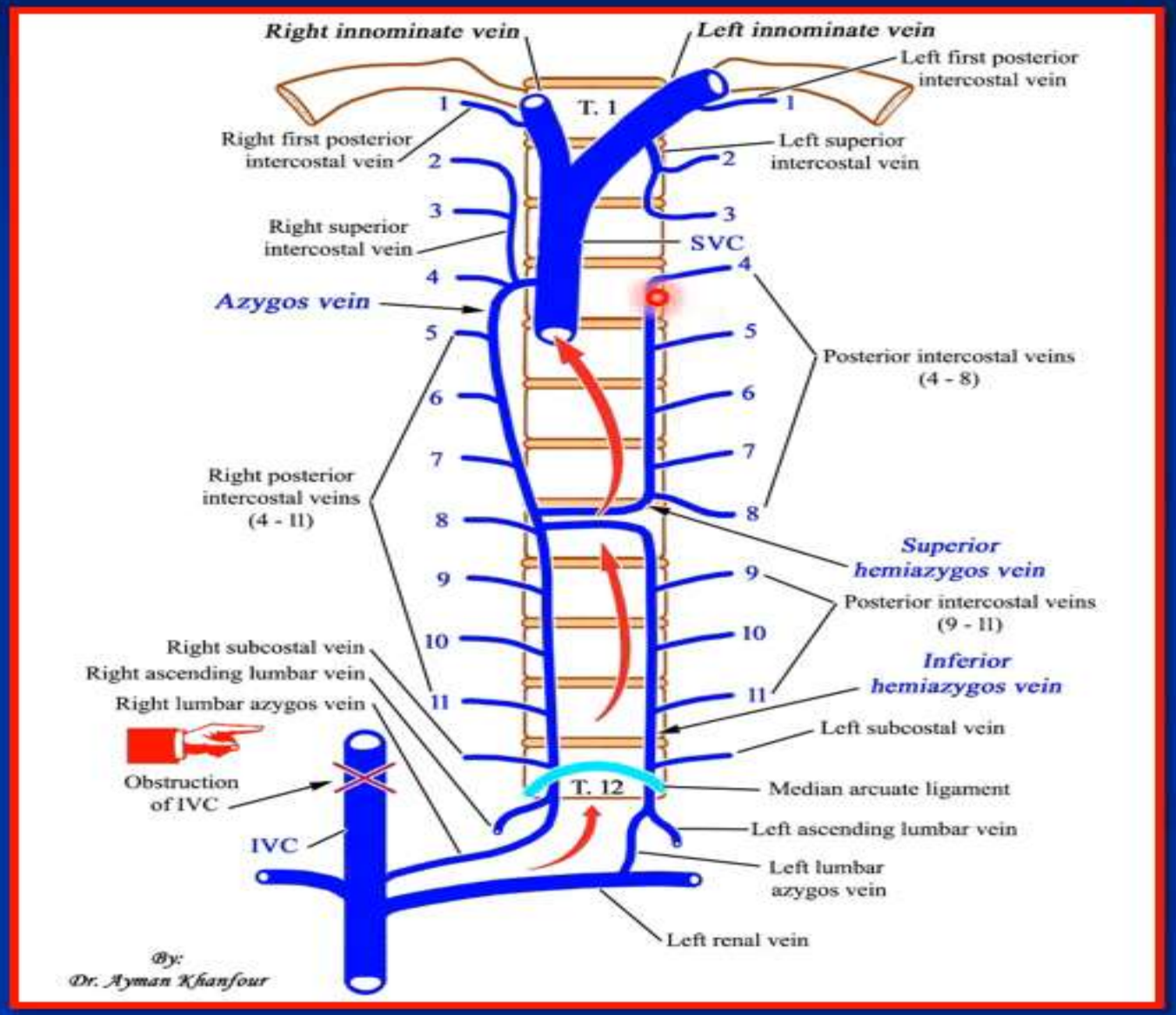


Termination

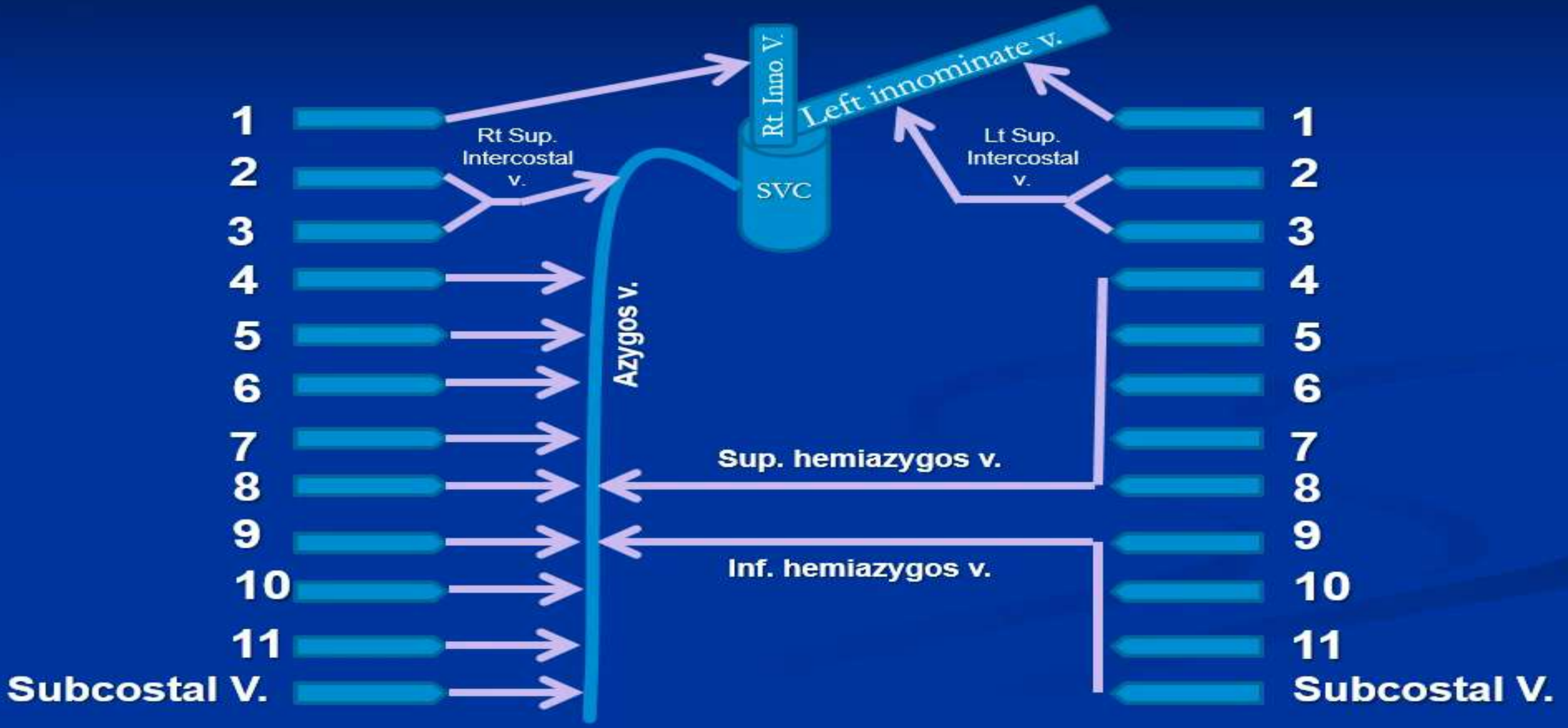
- Level of T. 8 (into Rt. Atrium)

Azygos Veins

Azygos Venous System

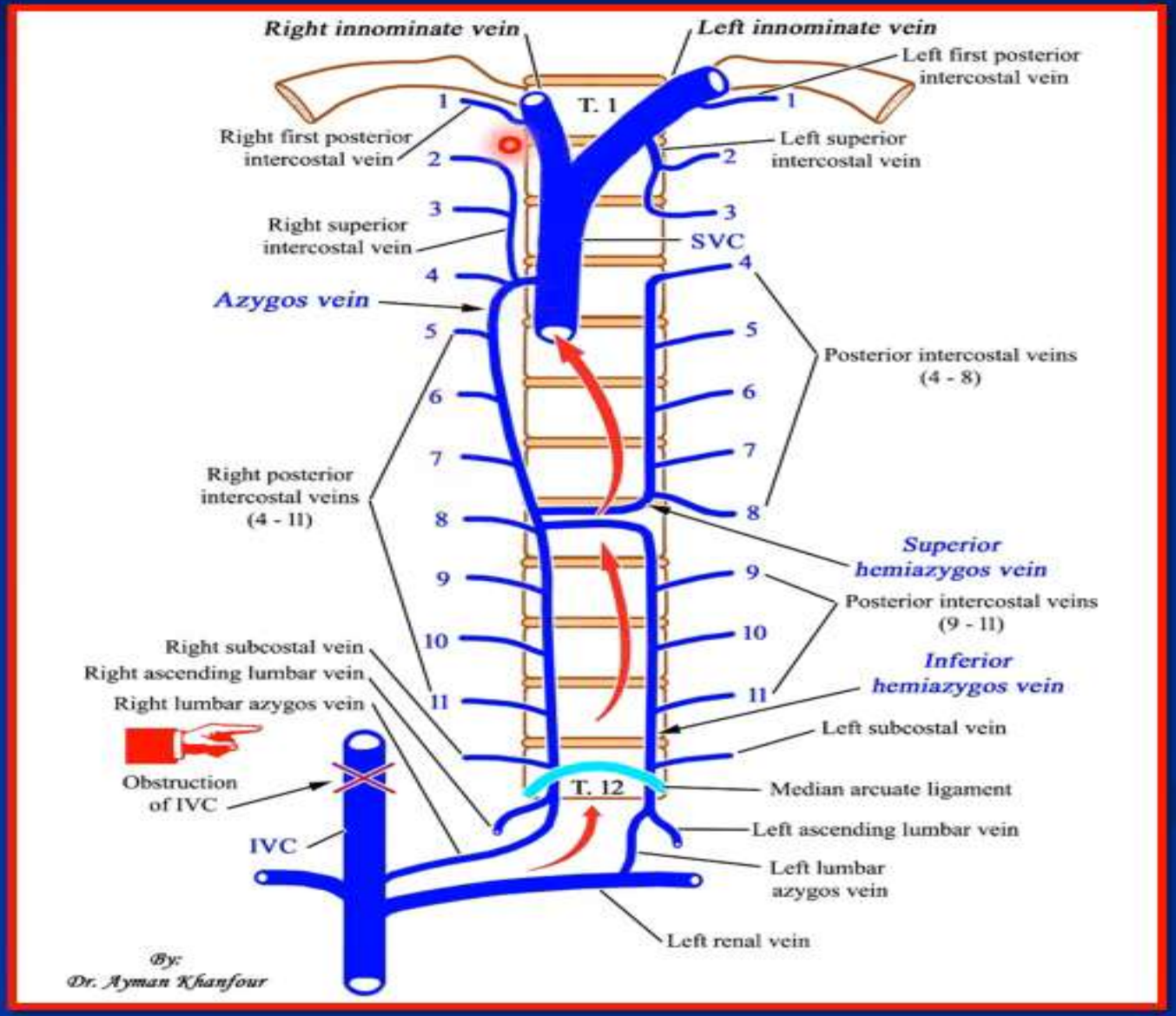


Tributaries of azygos veins:



Applied Anatomy

- In case of obstruction of the IVC, a collateral circulation is developed between the SVC and IVC through the three azygos veins.



Thorathic Duct

Thoracic Duct

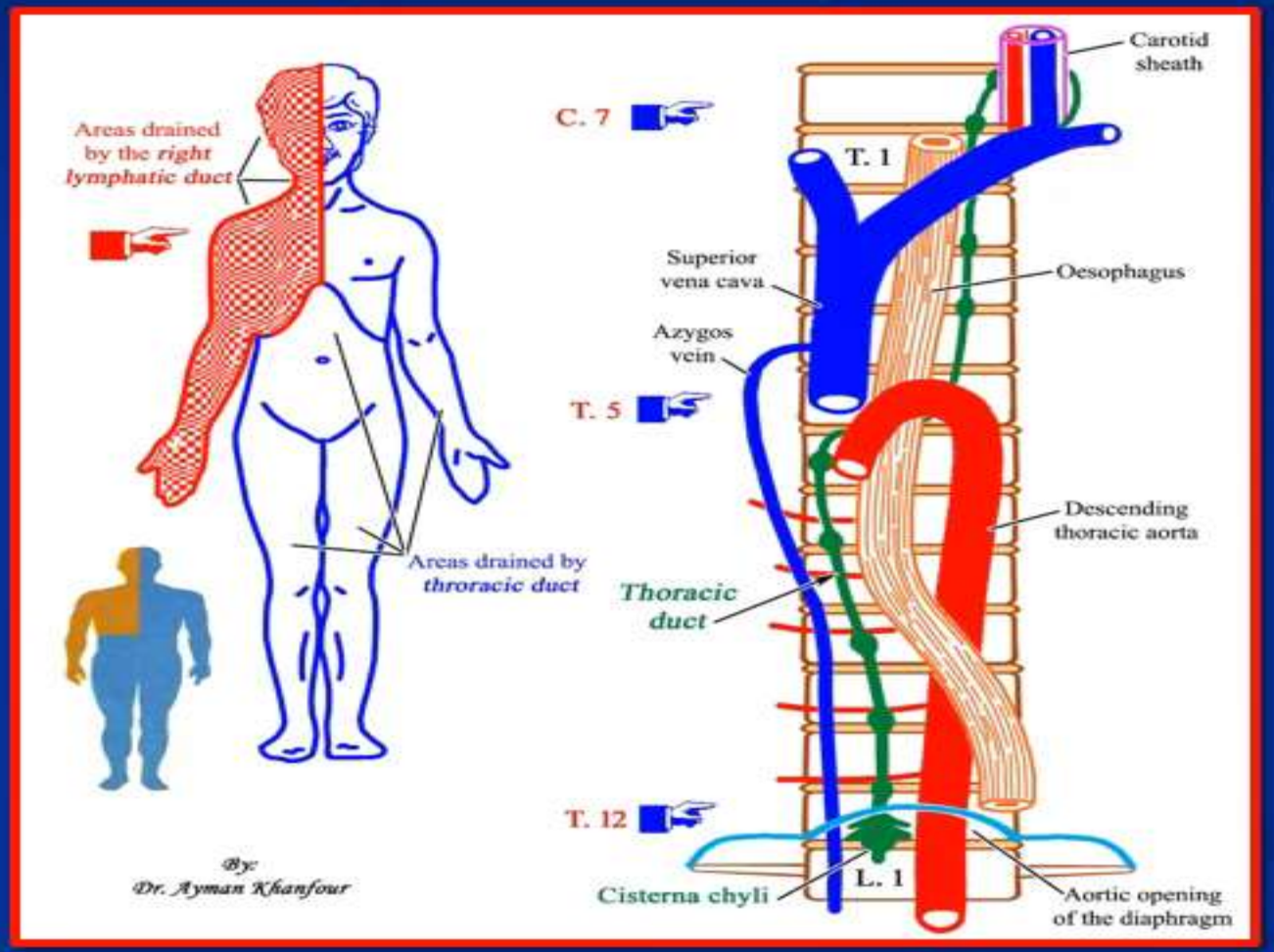
(5 + 7=12)

Begining

- Cisterna chyli at the level T. 12

Course

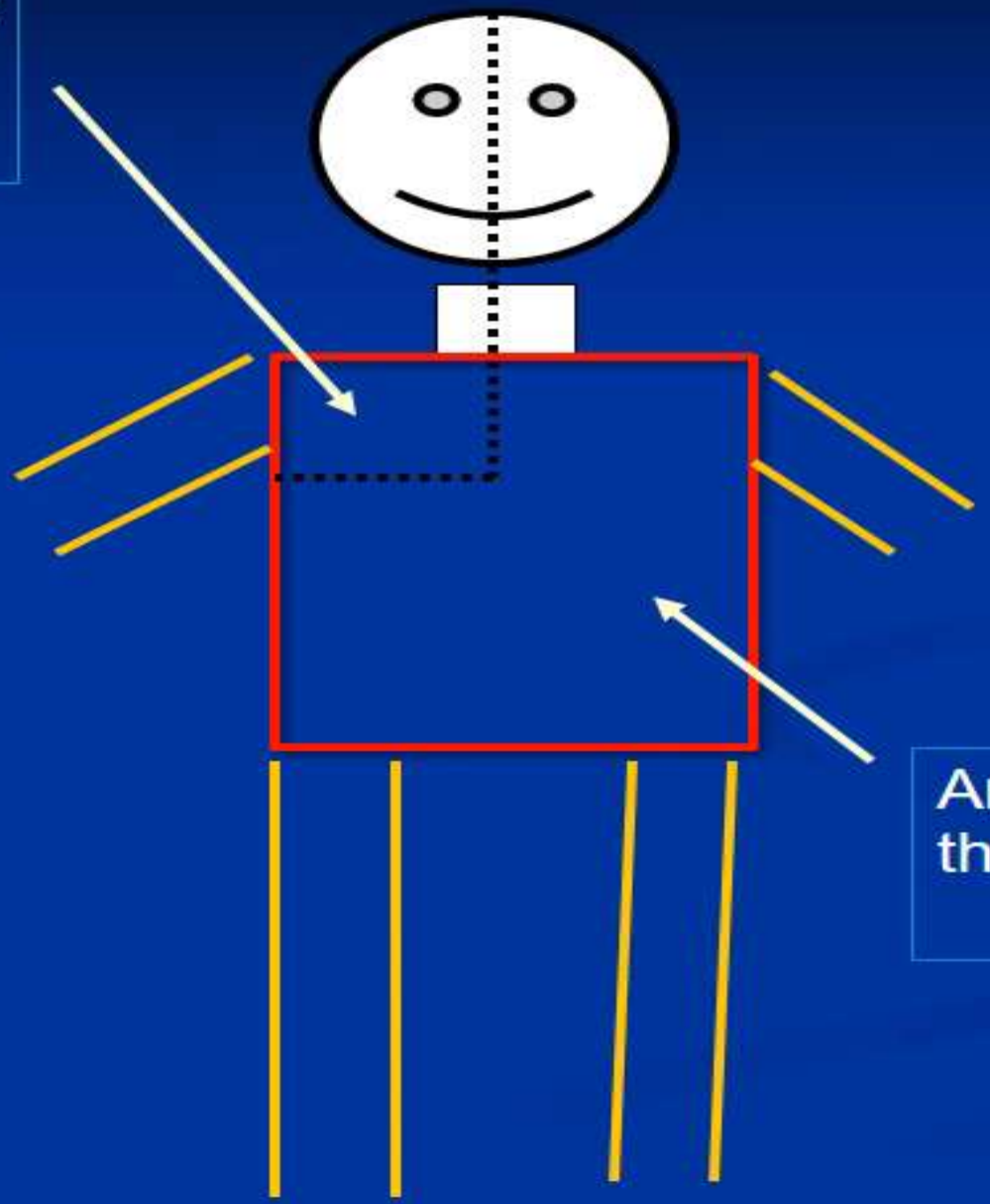
- At level of T. 5, it crosses behind the oesophagus to ascend on its Lt. side.



Termination

- At the level of C. 7, it curves behind carotid sheath to terminate into angle between the left internal jugular & subclavian v.

Area drained by the Rt. Lymphatic duct

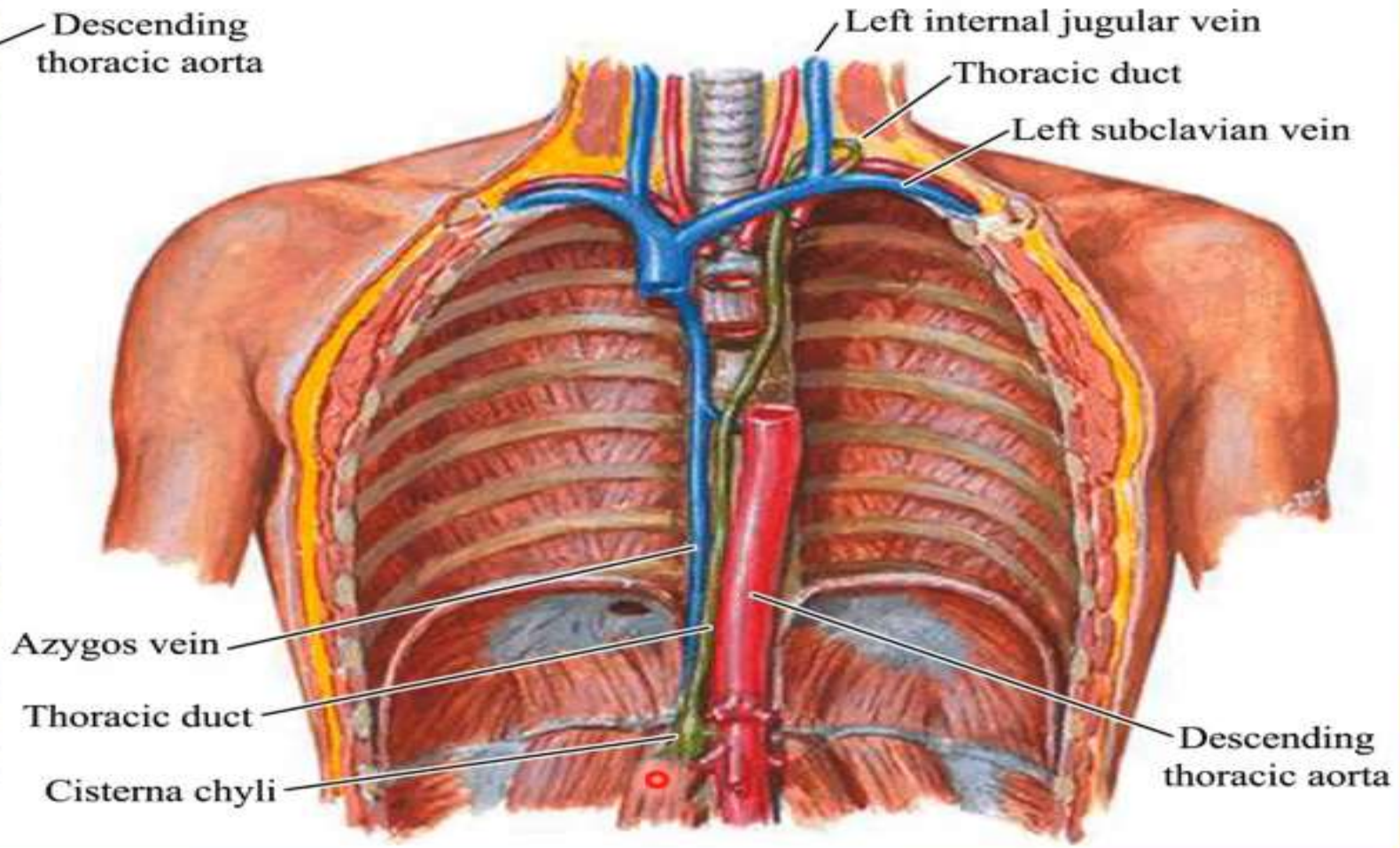


Area drained by the thoracic duct



(Right lateral view)

Montage by:
Dr. Ayman Khanfour



(Thoracic duct)